

IDENTIFICATION

PRODUCT CODE:

MAINDEC-08-DHKLD-A-D
(FORMERLY MAINDEC-08E-DZAC)

PRODUCT TEST:

PDP-8/E TELETYPE AND KLB ASYNCHRONOUS
DATA CONTROL TESTS

DATE:

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MAINTAINER:

DIAGNOSTIC GROUP

AUTHOR:

ED FORTMILLER

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1.

ABSTRACT

THIS PROGRAM CONSISTS OF A PACKAGE OF TEST PROGRAMS FOR TESTING THE KLS LOGIC (EIA OR CURRENT) AND A TELETYPE, ONLY ONE TELETYPE MAY BE TESTED AT A TIME, THE TELETYPE TO BE TESTED CAN BE A KSR33, ASR33, KSR35, ASR35, OR KSR37.

THE TEST PROGRAMS ARE:

PRG0-BASIC TEST OF THE OUTPUT LOGIC (CURRENT AND EIA)
PRG1-BASIC TEST OF THE OUTPUT AND INPUT LOGIC (LOOP AROUND)(EIA)
PRG2-BASIC TEST OF INPUT LOGIC (USES TTY READER)(CURRENT)
PRG3-READER TEST
PRG4-PRINTER TEST
PRG5-PUNCH TEST
PRG6-KEYBOARD TEST
PRG7-COMBINED TEST
PRG10-READER EXERCISER, BINARY COUNT PATTERN
PRG11-PRINTER EXERCISER
PRG12-BINARY COUNT TAPE GENERATOR

2.

REQUIREMENTS

2.1

EQUIPMENT

A. PDP-8/E WITH AT LEAST 4K OF MEMORY
B. FOR EIA A JUMPER TO CONNECT INPUT TO OUTPUT, SEE TEST EQUIPMENT 7.3.
C. KSR33, ASR33, KSR35, ASR35 TO TEST AN 110 BAUD CURRENT OPTION.

2.2

STORAGE

LOCATIONS 0000 THROUGH 7600 ARE USED.

3.

LOADING PROCEDURE

THE BINARY LOADER IS USED TO LOAD THE PROGRAM, REFER TO THE BINARY LOADER DOCUMENTATION IF UNFAMILIAR WITH ITS USE.

4; Use PROCEDURE

4.1 DEVICE CODE SELECTION

BEFORE ANY PROGRAM CAN BE RUN, THE PROGRAM MUST HAVE THE FOLLOWING INFORMATION:

- 1; TYPE OF TELETYPE (33, 35, OR 37) IF TESTING WITH A TELETYPE 33
- 2; DEVICE CODES ASSIGNED,
- 3; BAUD RATE OF DEVICE 110

TO PROVIDE THIS INFORMATION, PROCEED AS FOLLOWS:

A; SET LOCATION 0020 TO 1

- 1; 0000 FOR KSR OR ASR 33 TELETYPE
- 2; 0001 FOR KSR OR ASR 35 TELETYPE
- 3; 0002 FOR KSR 37 TELETYPE

B; SET LOCATION 0021 AS FOLLOWS:

- 1; LOAD ADDRESS 0021.
- 2; SET SR 0 THROUGH 5 TO THE DEVICE CODE OF THE KEY-BOARD/READER TO BE TESTED;
(EG: READER CODE OF 03, SR0-5=03,
- 3; SET SR 6 THROUGH 11 TO THE DEVICE CODE OF THE PRINTER/PUNCH TO BE TESTED;
(EG: PRINTER CODE OF 04, SR6-11=04,
- 4; PRESS DEPOSIT.

C; SET LOCATION 0022 AS FOLLOWS:

- 1; LOAD ADDRESS 0022;
- 2; PLACE THE FOLLOWING IN THE SR:
0110 FOR 110 BAUD; OR
0150 FOR 150 BAUD; OR
0300 FOR 300 BAUD; OR
0600 FOR 600 BAUD; OR
1200 FOR 1200 BAUD; OR
2400 FOR 2400 BAUD;
3. PRESS DEPOSIT.

D; REFER TO INDIVIDUAL PROGRAM USE PROCEDURE.

4.2

PRG0 USE PROCEDURE

- A. PERFORM DEVICE SELECTION IF NOT PREVIOUSLY DONE, REFER TO SECTION 4.1.
- B. INSURE THAT TELETYPE IS ON-LINE IF ON THE KL8 BEING TESTED.
- C. INSURE THAT THERE IS PAPER IN TELEPRINTER.
- D. LOAD ADDRESS 0200.
- E. SET SR TO 0000.
- F. PRESS CLEAR AND CONTINUE.
- G. PROGRAM HALTS AT LOCATION 0236 TO PERMIT SETTING OF SR OPTIONS. SET ANY DESIRED OPTIONS. NORMAL RUN IS WITH SR=0000. PRESS CONTINUE.

PRG0 SR OPTIONS:

- SR0=1 HALT AT END OF ROUTINE. ROUTINE NUMBER IN AC.
- SR1=1 SELECT ROUTINE WHOSE NUMBER IS SET IN SR6 = SR11.
- SR2=1 LOOP PROGRAM.
- SR6 THROUGH SR11 ROUTINE NUMBER TO BE SELECTED.
- H. PROGRAM IS EXECUTED AND HALTS AT LOCATION 0300 PROGRAM END. HALT, IF NO LOOP OPTIONS ARE SET, AND IF NO ERROR OCCURRED.

4.3

PRG1 USE PROCEDURE

- A. PERFORM DEVICE SELECTION IF NOT PREVIOUSLY DONE, REFER TO SECTION 4.1.
- B. CONNECT EIA OUTPUT TO EIA INPUT.
ON THE 40 PIN SIDE CONNECTOR, CONNECT-
PIN E TO PIN M
PIN F TO PIN J
- C. LOAD ADDRESS 0200.
- D. SET SR TO 0001.
- E. PRESS CLEAR AND CONTINUE.

(4,3 CONT'D)

F. PROGRAM HALTS AT LOCATION 0236 TO PERMIT SETTING OF SR
OPTIONS. SET ANY DESIRED OPTIONS. NORMAL RUN IS WITH
SR=0000. PRESS CONTINUE.

PRG1 SR OPTIONS:

SR0=1 HALT AT END OF ROUTINE. ROUTINE NUMBER IN AC.
SR1=1 SELECT ROUTINE WHOSE NUMBER IS SET IN SR6 = SR11.
SR2=1 LOOP PROGRAM.
SR6 THROUGH SR11 ROUTINE NUMBER TO BE SELECTED.

G. PROGRAM IS EXECUTED AND HALTS AT LOCATION 0300 PROGRAM END
HALT, IF NO LOOP OPTIONS ARE SET, AND IF NO ERRORS OCCUR.

4,4 PRG2 USE PROCEDURE

A. PERFORM DEVICE SELECTION IF NOT PREVIOUSLY DONE. REFER TO
SECTION 4.1.

B. INSURE THAT TELETYPE IS ON-LINE.

C. LOAD THE BINARY COUNT PATTERN TEST TAPE IN THE READER.

D. TURN ON READER.

E. LOAD ADDRESS 0200.

F. SET SR TO 0002.

G. PRESS CLEAR AND CONTINUE.

H. PROGRAM HALTS AT LOCATION 0236 TO PERMIT SETTING OF SR
OPTIONS. SET ANY DESIRED OPTIONS. NORMAL RUN IS WITH
SR=0000. PRESS CONTINUE.

PRG2 SR OPTIONS:

SR0=1 HALT AT END OF ROUTINE. ROUTINE NUMBER IN AC.
SR1=1 SELECT ROUTINE WHOSE NUMBER IS SET IN SR6 = SR11.
SR2=1 LOOP PROGRAM.
SR6 THROUGH SR11 ROUTINE NUMBER TO BE SELECTED.

I. PROGRAM IS EXECUTED AND HALTS AT LOCATION 0300. PROGRAM END
HALT, IF NO "LOOP" OPTIONS ARE SET, AND IF NO ERRORS OCCUR.

4.5 PRG3 USE PROCEDURE

- A. PERFORM DEVICE SELECTION IF NOT PREVIOUSLY DONE.
REFER TO SECTION 4.1.
- B. INSURE TELETYPE IS ON-LINE.
- C. LOAD BINARY COUNT PATTERN TEST TAPE IN READER.
- D. TURN ON READER.
- E. LOAD ADDRESS 0200.
- F. SET SR TO 0003.
- G. PRESS CLEAR AND CONTINUE.
- H. PROGRAM HALTS AT LOCATION 0236 TO PERMIT SETTING OF SR
OPTIONS. SET ANY DESIRED OPTIONS. NORMAL RUN IS WITH
SR=0000. PRESS CONTINUE.

PRG3 SR OPTIONS:

- SR0=1 HALT AT END OF ROUTINE. ROUTINE NUMBER IN AC.
- SR1=1 SELECT ROUTINE WHOSE NUMBER IS SET IN SR6 THROUGH
SR11.
- SR2=1 LOOP PROGRAM.
SR6 THROUGH SR11 ROUTINE NUMBER TO BE SELECTED.
- I. PROGRAM IS EXECUTED AND HALTS AT LOCATION 0300 PROGRAM END.
HALT, IF NO "LOOP" OPTIONS ARE SET, AND IF NO ERRORS OCCUR.

4.6 PRG4 USE PROCEDURE

- A. PERFORM DEVICE SELECTION IF NOT PREVIOUSLY DONE.
REFER TO SECTION 4.1.
- B. INSURE TELETYPE IS ON LINE.
- C. LOAD ADDRESS 0200.
- D. SET SR TO 0004.

(4,6 CONT'D)

- E. PRESS CLEAR AND CONTINUE.
- F. PROGRAM HALTS AT LOCATION 0236 TO PERMIT SETTING OF SR OPTIONS. SET ANY DESIRED OPTIONS, NORMAL RUN IS WITH SR=0000, PRESS CONTINUE.
- PRG4 SR OPTIONS
- SR0=1 HALT AT END OF ROUTINE, ROUTINE NUMBER IN AC;
SR1=1 SELECT ROUTINE WHOSE NUMBER IS SET IN SR6 - SR11;
SR2=1 LOOP PROGRAM;
SR6 THROUGH SR11 ROUTINE NUMBER TO BE SELECTED.
- G. PROGRAM IS EXECUTED AND HALTS AT LOCATION 0300, PROGRAM END HALT IF NO "LOOP" OPTIONS ARE SET, AND IF NO ERRORS OCCUR.
- 4,7 PRG5 USE PROCEDURE

- A. PERFORM DEVICE SELECTION IF NOT PREVIOUSLY DONE. REFER TO SECTION 4.1.
- B. TURN ON TELETYPE PUNCH.
- C. WITH TELETYPE OFF-LINE, PUNCH A SECTION OF BLANK LEADER ABOUT 6 INCHES LONG. RETURN TO ON-LINE POSITION.
- D. LOAD LEADER IN READER, LEAVING VERY LITTLE SLACK BETWEEN PUNCH AND READER.
- E. TURN ON READER.
- F. LOAD ADDRESS 0200.
- G. SET SR TO 0005.
- H. PRESS CLEAR AND CONTINUE.
- I. PROGRAM BEGINS EXECUTION. SET SR5 TO A 1 IF YOU WISH TO STOP ON ERROR. SR5 SET TO A 0 WILL CAUSE PROGRAM TO HALT AT END OF DATA BLOCK IF ERRORS OCCURRED. THE AC WILL CONTAIN THE ERROR COUNT.
- J. THE PROGRAM RUNS CONTINUOUSLY, UNTIL STOPPED BY USER.

PRG6 USE PROCEDURE

- A. PERFORM DEVICE SELECTION IF NOT PREVIOUSLY DONE. REFER TO SECTION 4.1.
 - B. INSURE TELETYPE IS ON-LINE.
 - C. MAKE SURE THAT THE TELETYPE "PROCEED" LIGHT IS ON, IF TESTING A KSR37 KEYBOARD.
 - D. LOAD ADDRESS 0200.
 - E. SET SR TO 0006.
 - F. PRESS CLEAR AND CONTINUE.
 - G. PROGRAM TITLE IS TYPED, AND PROGRAM HALTS AT LOC 0236 TO PERMIT SETTING OF SR OPTIONS. SET ANY DESIRED OPTIONS. NORMAL RUN IS WITH SR0000. PRESS CONTINUE.
- PRG5 SR OPTIONS:
- SR0#1 HALT AT END OF ROUTINE. ROUTINE NUMBER IN AC.
 - SR1#1 SELECT ROUTINE WHOSE NUMBER IS SET IN SR6 - SR11.
 - SR2#1 LOOP PROGRAM.
 - SR6 THROUGH SR11 ROUTINE NUMBER TO BE SELECTED.
- H. FOLLOW TYPED INSTRUCTIONS.
 - I. WHEN PROGRAM IS COMPLETED, AND PROVIDED THAT NO SR OPTIONS PREVENT IT, THE PROGRAM STOPS AT PROGRAM END HALT AT LOC 0300.

NOTE

CORRECT OPERATION OF KEYBOARD IS VERIFIED BY USER CHECKING THAT THE PRINTED CHARACTERS MATCH WITH THE CHARACTERS KEYED.

PRG7 USE PROCEDURE -----

- A. PERFORM DEVICE SELECTION IF NOT PREVIOUSLY DONE. REFER TO SECTION 4.1.
- B. TURN ON TELETYPE PUNCH.
- C. THWTH TELETYPE OFF-LINE, PUNCH A SECTION OF BLANK LEADER ABOUT 6 INCHES LONG. RETURN TELETYPE TO ON-LINE POSITION.
- D. LOAD LEADER IN READER, LEAVING VERY LITTLE SLACK BETWEEN PUNCH AND READER.
- E. TURN ON READER.
- F. LOAD ADDRESS 0200.
- G. SET SR TO 0007.
- H. PRESS CLEAR AND CONTINUE.
- I. PROGRAM HALTS AT LOC 0236 TO PERMIT SETTING OF SR OPTIONS; SET ANY DESIRED OPTIONS. NORMAL RUN IS WITH SR=0200, TO HALT ON ERROR, PRESS CONTINUE.

PRG6 SR OPTIONS:

SR0=1 HALT AT END OF ROUTINE; ROUTINE NUMBER IN AC.
SR1=1 SELECT ROUTINE WHOSE NUMBER IS SET IN SR6 THROUGH SR11.
SR2=1 LOOP PROGRAM.
SR3=1 HALT ON ERROR; BAD CHARACTER IN AC.
SR5=0 HALT AT END OF DATA BLOCK IF ERRORS OCCURRED; ERROR COUNT IN AC.
SR6 THROUGH SR11 ROUTINE NUMBER TO BE SELECTED.
- J. PROGRAM IS EXECUTED AND HALTS AT PROGRAM END HALT AT LOC 0300 UNLESS PREVENTED FROM ENDING, BY SR OPTIONS, OR IF ERRORS OCCUR.

4.10

PG10 USE PROCEDURE

- A. PERFORM DEVICE SELECTION IF NOT PREVIOUSLY DONE, REFER TO SECTION 4.1.
- B. INSURE THAT TELETYPE IS ON-LINE.
- C. LOAD BINARY COUNT PATTERN TEST TAPE IN READER.
- D. TURN ON READER.
- E. LOAD ADDRESS 0200.
- F. SET SR TO 0010.
- G. PRESS CLEAR AND CONTINUE.
- H. PROGRAM RUNS CONTINUOUSLY UNTIL STOPPED BY USER. THE FOLLOWING SR OPTIONS MAY BE SET AT ANY TIME.

SR0=1 PROGRAM HALTS WITH ACCUMULATED ERROR COUNT IN AC;
SR3=1 PROGRAM READS TAPE AT FULL SPEED.
SR3=0 PROGRAM READS TAPE WITH RANDOM STALLS BETWEEN CHARACTERS;
SR5=1 HALT ON ERROR. PROGRAM HALTS IF READ ERROR OCCURS;
BAD CHARACTER IS DISPLAYED IN AC.
SR6=0 NO HALT ON ERROR.

4.11

PG11 USE PROCEDURE

- A. PERFORM DEVICE SELECTION IF NOT PREVIOUSLY DONE, REFER TO SECTION 4.1.
- B. MAKE SURE THAT TELETYPE IS ON-LINE, AND IF KSR37, THAT KEYBOARD "PROCEED" LIGHT IS ON.
- C. LOAD ADDRESS 0200.
- D. SET SR TO 0011.
- E. PRESS CLEAR AND CONTINUE.
- F. THE PROGRAM IDENTIFIES ITSELF, AND REQUESTS DATA TO BE TYPED.
- G. TYPE IN DATA AS FOLLOWS:
 1. TYPE THE 3 CHARACTERS TO BE TYPED AND A DELETE CODE (RUBOUT) IF YOUR WISH NOT TO STALL BETWEEN CHARACTERS OR,
 2. TYPE THE 3 CHARACTERS TO BE TYPED AND ANY OTHER CHARACTER OTHER THAN THE DELETE CODE TO STALL BETWEEN CHARACTERS.

(4,11 CONT'D)

H. THE PROGRAM WILL CONTINUOUSLY TYPE LINES CONTAINING THE THREE DESIRED CHARACTERS.

I. TO CHANGE THE CHARACTER TO BE TYPED, SET SR0 TO A 1. THE PROGRAM WILL REQUEST NEW DATA WHEN THE CURRENT LINE IS COMPLETED. TYPE IN THE DATA AS IN STEP G.

4.12 PRG12 USE PROCEDURE

A. PERFORM DEVICE SELECTION IF NOT PREVIOUSLY DONE. REFER TO SECTION 4.1.

B. INSURE TELETYPE IS ON-LINE.

C. TURN OFF TELETYPE READER.

D. LOAD BLANK TAPE IN PUNCH.

E. TURN ON PUNCH.

F. LOAD ADDRESS 0200.

G. SET SR TO 0012.

H. PRESS CLEAR AND CONTINUE.

I. PROGRAM PUNCHES BINARY COUNT PATTERN TEST TAPE UNTIL STOPPED BY USER.

5. PROGRAM AND/OR OPERATOR ACTION

5.1 NORMAL HALTS

LOC 0236

SR SET HALT. OCCURS TO PERMIT SETTING OF DESIRED OPTIONS. PRESS CONTINUE AFTER SETTING DESIRED OPTIONS. (PRG0,PRG1,PRG2).

LOC 0300

PROGRAM END HALT. OCCURS AT END OF PROGRAM, IF NO "LOOP" TYPE OPTION IS SET. SET DESIRED OPTIONS AND PRESS CONTINUE. THIS HALT REOCCURS IF NO OPTIONS ARE SET. (PRG0,PRG1,PRG2,PRG3,PRG4,PRG6,PRG10).

LOC 0324

ROUTINE END HALT. THIS HALT OCCURS AT END OF A TEST ROUTINE IF SR0 IS SET TO A 1. THE AC CONTAINS THE NUMBER OF ROUTINE JUST COMPLETED. (PRG0,PRG1,PRG2,PRG3,PRG4,PRG6,PRG10).

6: ERRORS

6.1 ERROR HALT AND DESCRIPTION

LOC 1526	AN ILLEGAL BAUD RATE WAS SELECTED; RESELECT THE BAUD RATE AND RESTART PROGRAM;
LOC 2103	PRG0, PRG1, AND PRG2 UNEXPECTED INTERRUPT ERROR HALT; A DEVICE OTHER THAN THE ONE BEING TESTED HAS CAUSED AN INTERRUPT. THE AC CONTAINS THE IOT CODE THAT DETECTED THE INTERRUPT (EG, 6031 FOR SYSTEM TELETYPE KEYBOARD). PRESS CONTINUE, THE PROGRAM WILL ATTEMPT TO CLEAR THE UNDESIRABLE FLAG. IF SUCCESSFUL, THIS HALT WILL NOT REOCCUR;
LOC 2237	PRG0, ROUTINE 0, ERROR HALT A; SPF INSTRUCTION FAILED TO SET PRINTER FLAG OR TSF INSTRUCTION FAILED TO SKIP ON PRINTER FLAG SET; PRESSING CONTINUE ENTERS SCOPE LOOP THAT ISSUES SPF AND THEN TSF CONTINUOUSLY; MANUAL RESTART
LOC 2244	PRG0, ROUTINE 0, ERROR HALT B; CAF INSTRUCTION FAILED TO CLEAR PRINTER FLAG OR TSF INSTRUCTION SKIPPED ON NO PRINTER FLAG; PRESSING CONTINUE ENTERS SCOPE LOOP THAT SETS PRINTER FLAG WITH SPF, AND THEN CAF AND TSF ARE ISSUED; MANUAL RESTART
LOC 2253	PRG0, ROUTINE 0, ERROR HALT C; CAF INSTRUCTION FAILED TO CLEAR AC AND/OR LINK; PRESSING CONTINUE ENTERS SCOPE LOOP THAT ISSUES CAF WITH AC AND LINK SET; MANUAL RESTART;
LOC 2262	PRG0, ROUTINE 0, ERROR HALT E; TCF INSTRUCTION FAILED TO CLEAR PRINTER FLAG; PRESSING CONTINUE ENTERS SCOPE LOOP THAT ISSUES TCF WITH THE PRINTER FLAG SET; MANUAL RESTART;
LOC 2315	PRG0, ROUTINE 1, ERROR HALT B; WITH THE PRINTER FLAG SET AND THE INTERRUPT ENABLED, NO INTERRUPT OCCURRED. PRESSING CONTINUE ENTERS SCOPE LOOP THAT TURNS ON INTERRUPT CONTINUOUSLY; MANUAL RESTART;
LOC 2415	PRG0, ROUTINE 2, ERROR HALT A; KIE INSTRUCTION FAILED TO DISABLE THE TELETYPE INTERRUPT ENABLE FLIP-FLOP; PRESSING CONTINUE ENTERS SCOPE LOOP THAT ISSUES KIE CONTINUOUSLY WITH AC 11=0; MANUAL RESTART;
LOC 2427	PRG0, ROUTINE 2, ERROR HALT B; SPI INSTRUCTION SKIPPED WITH FLAG SET AND TELETYPE INTERRUPT ENABLE FLIP-FLOP DISABLED; PRESSING CONTINUE ENTERS SCOPE LOOP THAT ISSUES SPI WITH PRINTER FLAG SET AND TTY INTERRUPT DISABLED; MANUAL RESTART;

LOC 2435

PRG0, ROUTINE 2, ERROR HALT C, SRQ INSTRUCTION
 SKIPPED WITH PRINTER FLAG SET AND TELETYPE INTER-
 RUPT ENABLE FLIP-FLOP DISABLED, PRESSING CONTINUE
 ENTERS SCOPE LOOP THAT ISSUES SRQ WITH PRINTER
 FLAG SET AND TTY INTERRUPT DISABLED, MANUAL RESTART;

LOC 2443

PRG0, ROUTINE 2, ERROR HALT D, KIE INSTRUCTION
 FAILED TO ENABLE TELETYPE INTERRUPT FLIP-FLOP;
 PRESSING CONTINUE ENTERS SCOPE LOOP THAT ISSUES
 KIE CONTINUOUSLY WITH AC11#1, MANUAL RESTART;

LOC 2456

PRG0, ROUTINE 2, ERROR HALT E, SPI INSTRUCTION
 FAILED TO SKIP WITH PRINTER FLAG SET AND TTY INTERRUPT
 ENABLE FLIP-FLOP ENABLED, PRESSING CONTINUE
 ENTERS SCOPE LOOP THAT ISSUES SPI CONTINUOUSLY
 WITH PRINTER FLAG SET AND INTERRUPT ENABLED,
 MANUAL RESTART;

LOC 2465

PRG0, ROUTINE 2, ERROR HALT F, SRQ INSTRUCTION
 FAILED TO SKIP WITH PRINTER FLAG SET AND TTY
 INTERRUPT ENABLE FLIP-FLOP SET, PRESSING CONTINUE
 ENTERS SCOPE LOOP THAT ISSUES SRQ CONTINUOUSLY
 WITH PRINTER FLAG SET AND TTY INTERRUPT ENABLE
 FLIP-FLOP ENABLED, MANUAL RESTART;

LOC 2474

PRG0, ROUTINE 2, ERROR HALT G, CAF INSTRUCTION
 FAILED TO ENABLE TTY INTERRUPT ENABLE FLIP-FLOP,
 PRESSING CONTINUE ENTERS SCOPE LOOP THAT ISSUES
 CAF CONTINUOUSLY, MANUAL RESTART;

LOC 2527

PRG0, ROUTINE 3, ERROR HALT A, TPC INSTRUCTION
 FAILED TO SET PRINTER FLAG IN TWICE THE REQUIRED
 TIME FOR IT TO SET, PRESSING CONTINUE ENTERS
 SCOPE LOOP THAT ISSUES TPC AND DELAYS, CONTINUOUSLY,
 MANUAL RESTART;

LOC 2534

PRG0, ROUTINE 3, ERROR HALT B, TLS FAILED TO
 CLEAR PRINTER FLAG, PRESSING CONTINUE ENTERS
 SCOPE LOOP THAT ISSUES TLS CONTINUOUSLY WITH
 PRINTER FLAG SET, MANUAL RESTART;

LOC 2540

PRG0, ROUTINE 3, ERROR HALT C, TLS INSTRUCTION
 FAILED TO SET PRINTER FLAG IN TWICE THE REQUIRED
 TIME FOR IT TO SET, PRESSING CONTINUE ENTERS
 SCOPE LOOP THAT ISSUES TLS AND DELAYS, CONTINUOUSLY,
 MANUAL RESTART;

LOC 2607
 PRG0, ROUTINE 4, ERROR HALT A, PRINTER FLAG
 SET PRIOR TO 9 BIT TIMES, (EG, 110 BAUD: 9X9.09
 MSEC = 81.81 MSEC AT WHICH TIME THE FLAG MUST BE SET,
 NOT PRIOR TO THIS TIME), EITHER THE PDP-8/E
 TIMING IS TOO SLOW OR THE TTY CLOCK TOO FAST.
 (IS THE SLOW CYCLE JUMPER REMOVED FROM THE PROCES-
 SOR TIMING MODULE AND IS THE CORRECT BAUD RATE
 SELECTED IN LOC 227), PRESSING CONTINUE ENTERS
 SCOPE LOOP THAT ISSUES TLS CONTINUOUSLY. MANUAL
 RESTART.

LOC 2614
 PRG0, ROUTINE 4, ERROR HALT B, PRINTER FLAG NOT
 SET AFTER 9.55 BIT TIMES, (EG, 110 BAUD 9.55X9.09
 MSEC = 86.7 MSEC AT WHICH TIME THE FLAG MUST BE
 SET, NO LATER,) PRESSING CONTINUE ENTERS SCOPE
 LOOP THAT ISSUES TLS CONTINUOUSLY. MANUAL RESTART.

LOC 2654
 PRG0, ROUTINE 5, ERROR HALT A, WHEN ISSUING BACK
 TO BACK TLS/S, FLAG SETTING PRIOR TO 11 BIT TIMES
 FOR 110 BAUD OR 10 BIT TIMES FOR MORE THEN 110
 BAUD. PRESSING CONTINUE ENTERS SCOPE LOOP THAT
 ISSUES TLS CONTINUOUSLY. MANUAL RESTART.

LOC 2656
 PRG0, ROUTINE 5, ERROR HALT B, WHEN ISSUING BACK
 TO BACK TLS/S, FLAG TAKING LONGER THAN 11 BIT TIMES
 TO SET FOR 110 BAUD OR 10 BIT TIMES FOR MORE THAN
 110 BAUD. PRESSING CONTINUE ENTERS SCOPE LOOP THAT
 ISSUES TLS CONTINUOUSLY. MANUAL RESTART.

LOC 2670
 PRG0, ROUTINE 6, ERROR HALT A, WITH LINK, ION,
 AND INT BUS EQUAL TO ZERO, AC DID NOT EQUAL ZERO
 AFTER ISSUING GTF. NO SCOPE LOOP. MANUAL RESTART.

LOC 2675
 PRG0, ROUTINE 6, ERROR HALT B, GTF INSTRUCTION
 CLEARED THE LINK. NO SCOPE LOOP. MANUAL RESTART.

LOC 2701
 PRG0, ROUTINE 6, ERROR HALT C, GTF INSTRUCTION
 FAILED TO BRING LINK INTO AC 0, NO SCOPE LOOP.
 MANUAL RESTART.

LOC 2732
 PRG0, ROUTINE 6, ERROR HALT D, GTF INSTRUCTION FAILED
 TO BRING INT BUS INTO AC 2, NO SCOPE LOOP.
 MANUAL RESTART.

LOC 2724
 PRG0, ROUTINE 6, ERROR HALT E, GTF INSTRUCTION
 CLEARED ION, NO SCOPE LOOP. MANUAL RESTART.

LOC 2731
 PRG0, ROUTINE 6, ERROR HALT F, GTF INSTRUCTION
 FAILED TO BRING ION INTO AC 4, NO SCOPE LOOP.
 MANUAL RESTART.

LOC 2750
 PRG0, ROUTINE 7, ERROR HALT A, RTF INSTRUCTION
 FAILED TO RESET LINK WITH AC 0=0, NO SCOPE
 LOOP. MANUAL RESTART.

(6,1 CONT'D)

LOC 2754	PRG0, ROUTINE 7, ERROR HALT B, RTF INSTRUCTION FAILED TO SET LINK WITH AC0=1, NO SCOPE LOOP, MANUAL RESTART;
LOC 2757	PRG0, ROUTINE 7, ERROR HALT C, RTF INSTRUCTION FAILED TO TURN THE INTERRUPT ON, NO SCOPE LOOP, MANUAL RESTART;
LOC 3025	PRG1, ROUTINE 1, ERROR HALT A, RECEIVER FLAG NOT SETTING UPON COMPLETION OF ISSUING A TLS OR KSF FAILED TO SKIP ON RECEIVER FLAG SET, PRESSING CONTINUE ENTERS SCOPE LOOP THAT CLEARS THE RECEIVER FLAG AND ISSUES A TLS AND WAITS TWICE THE TIME FOR THE FLAG TO SET AND THEN ISSUES A KSF; MANUAL RESTART;
LOC 3053	PRG1, ROUTINE 2, ERROR HALT A, SAME AS PRG1, ROUTINE 1, ERROR HALT A;
LOC 3062	PRG1, ROUTINE 2, ERROR HALT B, KSF INSTRUCTION FAILED TO SKIP ON RECEIVER FLAG, PRESSING CONTINUE ENTERS SCOPE LOOP THAT ISSUES KSF CONTINUOUSLY, MANUAL RESTART;
LOC 3113	PRG1, ROUTINE 3, ERROR HALT A, SAME AS PRG1, ROUTINE 1, ERROR HALT A;
LOC 3122	PRG1, ROUTINE 3, ERROR HALT B, KSF INSTRUCTION SKIPPED ON RECEIVER FLAG NOT SET, PRESSING CONTINUE ENTERS SCOPE LOOP THAT ISSUES KSF WITH NO RECEIVER FLAG SET CONTINUOUSLY, MANUAL RESTART;
LOC 3160	PRG1, ROUTINE 4, ERROR HALT A, THE READER FLAG FAILED TO CAUSE AN INTERRUPT, PRESSING CONTINUE ENTERS SCOPE LOOP THAT TURNS THE INTERRUPT ON CONTINUOUSLY, MANUAL RESTART;
LOC 3230	PRG1, ROUTINE 5, ERROR HALT A, SRQ INSTRUCTION FAILED TO SKIP ON READER FLAG SET AND TELETYPE INTERRUPT ENABLE FLIP-FLOP ENABLED, PRESSING CONTINUE ENTERS SCOPE LOOP THAT ISSUES SRQ CONTINUOUSLY WITH TTY ENABLED AND READER FLAG SET, MANUAL RESTART;
LOC 3235	PRG1, ROUTINE 5, ERROR HALT B, SPI INSTRUCTION FAILED TO SKIP ON READER FLAG SET AND TELETYPE INTERRUPT ENABLE FLIP-FLOP ENABLED, PRESSING CONTINUE ENTERS SCOPE LOOP THAT ISSUES SPI CONTINUOUSLY WITH TTY ENABLED AND READER FLAG SET, MANUAL RESTART;
LOC 3242	PRG1, ROUTINE 5, ERROR HALT C, CAF INSTRUCTION FAILED TO CLEAR THE READER FLAG, PRESSING CONTINUE ENTERS SCOPE LOOP THAT ISSUES CAF CONTINUOUSLY WITH THE RECEIVER FLAG SET, MANUAL RESTART;

LOC 3257
 PRG1, ROUTINE 5, ERROR HALT D, SRQ INSTRUCTION
 SKIPPED WITH NO RECEIVER FLAG SET, PRESSING
 CONTINUE ENTERS SCOPE LOOP THAT ISSUES SRQ
 INSTRUCTION CONTINUOUSLY, MANUAL RESTART.

LOC 3264
 PRG1, ROUTINE 5, ERROR HALT E, SPI INSTRUCTION
 SKIPPED WITH NO RECEIVER FLAG SET, PRESSING
 CONTINUE ENTERS SCOPE LOOP THAT ISSUES SPI
 CONTINUOUSLY WITH NO RECEIVER FLAG SET, MANUAL
 RESTART.

LOC 3310
 PRG1, ROUTINE 6, ERROR HALT A, RECEIVER FLAG
 NOT SETTING AT THE END OF 10 BIT TIMES FOR A
 NON 110 BAUD DEVICE OR 11 BIT TIMES FOR A 110
 BAUD DEVICE, PRESSING CONTINUE ENTERS SCOPE LOOP
 THAT ISSUES T1S CONTINUOUSLY TO SET RECEIVER
 FLAG, MANUAL RESTART.

LOC 3366
 PRG1, ROUTINE 7 OR 10, ERROR HALT, DATA SENT
 DOES NOT COMPARE WITH THE DATA RECEIVED, MQ
 CONTAINS DATA THAT WAS SENT, AC CONTAINS THE
 DATA THAT WAS RECEIVED, PRESSING CONTINUE ENTERS
 SCOPE LOOP THAT SENDS THE DATA IN THE MQ, MANUAL
 RESTART.

LOC 3424
 PRG1, ROUTINE 11, ERROR HALT A, KRS INSTRUCTION
 FAILED TO INCLUSIVE "OR" KBRD BUFFER WITH AC,
 PRESSING CONTINUE ENTERS SCOPE LOOP THAT ISSUES
 KRS CONTINUOUSLY, MANUAL RESTART.

LOC 3464
 PRG1, ROUTINE 12, ERROR HALT A, KRB INSTRUCTION
 FAILED TO "JAM TRANSFER" THE KBRD BUFFER INTO
 THE AC, PRESSING CONTINUE ENTERS SCOPE LOOP THAT
 ISSUES KRB CONTINUOUSLY, MANUAL RESTART.

LOC 3474
 PRG1, ROUTINE 12, ERROR HALT B, KRB INSTRUCTION
 FAILED TO CLEAR THE READER FLAG, PRESSING CONTINUE
 ENTERS SCOPE LOOP THAT ISSUES KRB CONTINUOUSLY
 WITH THE RECEIVER FLAG SET, MANUAL RESTART.

LOC 3524
 PRG1 OR PRG2, ROUTINES 0, ERROR HALT, KCC INSTRUCTION
 FAILED TO CLEAR THE AC, PRESSING CONTINUE ENTERS
 SCOPE LOOP THAT ISSUES KCC CONTINUOUSLY WITH
 AC#7777, MANUAL RESTART.

LOC 3543
 PRG2, ROUTINE 1, ERROR HALT, AFTER ISSUING A KCC
 INSTRUCTION AND WAITING TWICE THE AMOUNT OF
 TIME REQUIRED FOR THE RECEIVER FLAG TO SET, IT
 WAS NOT SET, PRESSING CONTINUE ENTERS A SCOPE
 LOOP THAT REPEATS THE TEST, MANUAL RESTART.

LOC 3562
 PRG2, ROUTINE 2, ERROR HALT A, SAME AS PRG 2,
 ROUTINE 1, ERROR HALT.

LOC 3564 PRG2, ROUTINE 2, ERROR HALT B, WITH RECEIVER FLAG SET, KSF COMMAND FAILED TO SKIP; PRESSING CONTINUE ENTERS SCOPE LOOP THAT SKIPS ON FLAG CONTINUOUSLY, MANUAL RESTART.

LOC 3621 PRG2, ROUTINE 3, ERROR HALT A, SAME AS PRG 2, ROUTINE 1, ERROR HALT.

LOC 3623 PRG2, ROUTINE 3, ERROR HALT B, KCC FAILED TO RESET, OR KSF INSTRUCTION SKIPPED WITH FLAG=0; PRESSING CONTINUE ENTERS SCOPE LOOP THAT CLEARS THE FLAG AND SKIPS ON THE FLAG CONTINUOUSLY, MANUAL RESTART.

LOC 3657 PRG2, ROUTINE 4, ERROR HALT, WITH READER FLAG=1 AND INTERRUPT ENABLED, NO INTERRUPT OCCURRED; PRESSING CONTINUE ENTERS SCOPE LOOP THAT TURNS INTERRUPT ON CONTINUOUSLY, MANUAL RESTART.

LOC 3706 PRG2, ROUTINE 5, ERROR HALT, TIMING ERROR, FLAG NOT=1 103 MSEC AFTER KCC INSTRUCTION; PRESSING CONTINUE ENTERS SCOPE LOOP THAT READS TAPE CONTINUOUSLY, MANUAL RESTART.

LOC 3747 PRG2, ROUTINE 6, ERROR HALT A, REREAD ERROR; A REREAD OF THE RBRD BUFFER DID NOT MATCH WITH THE ORIGINAL READ; NEW CHARACTER IS DISPLAYED IN AC, PRESS CONTINUE.

LOC 3752 PRG2, ROUTINE 6, ERROR HALT B; FOLLOW UP HALT, TO PRG2, ROUTINE 6, ERROR HALT A; THE "HOLD" CHARACTER IS DISPLAYED IN THE AC; PRESSING CONTINUE ENTERS SCOPE LOOP THAT READS THE TELETYPE BUFFER CONTINUOUSLY, MANUAL RESTART.

LOC 3756 PRG2, ROUTINE 6, ERROR HALT C, KRS INSTRUCTION FAILED TO "INCLUSIVE OR" KBRD BUFFER WITH AC; PRESSING CONTINUE ENTERS SCOPE LOOP THAT ISSUES KRS CONTINUOUSLY WITH AC=7777; MANUAL RESTART.

LOC 4015 PRG2, ROUTINE 7, ERROR HALT A, KCR INSTRUCTION CLEARED THE AC; PRESSING CONTINUE ENTERS SCOPE LOOP THAT ISSUES KCR CONTINUOUSLY WITH AC=7777; MANUAL RESTART.

LOC 4021 PRG2, ROUTINE 7, ERROR HALT B, KCR INSTRUCTION FAILED TO CLEAR READER RUN; PRESSING CONTINUE ENTERS SCOPE LOOP THAT ISSUES KCR CONTINUOUSLY WITH READER RUN SET, MANUAL RESTART.

LOC 4073 PRG2, ROUTINE 10, ERROR HALT A, KIE INSTRUCTION FAILED TO DISABLE TELETYPE INTERRUPT ENABLE FLIP-FLOP; PRESSING CONTINUE ENTERS A SCOPE LOOP THAT ISSUES KIE WITH AC=0 CONTINUOUSLY, MANUAL RESTART.

LOC 4107

PRG2, ROUTINE 10, ERROR HALT B, SRQ INSTRUCTION SKIPPED WITH THE TELETYPE INTERRUPT ENABLE FLIP-FLOP DISABLED AND READER FLAG SET, PRESSING CONTINUE ENTERS A SCOPE LOOP THAT ISSUES SRQ CONTINUOUSLY WITH READER FLAG SET AND TTY DISABLED, MANUAL RESTART.

LOC 4115

PRG2, ROUTINE 10, ERROR HALT C, SPI INSTRUCTION SKIPPED WITH THE TELETYPE INTERRUPT ENABLE FLIP-FLOP DISABLED AND READER FLAG SET, PRESSING CONTINUE ENTERS A SCOPE LOOP THAT ISSUES SPI CONTINUOUSLY WITH READER FLAG SET AND TTY DISABLED, MANUAL RESTART.

LOC 4123

PRG2, ROUTINE 10, ERROR HALT D, KIE INSTRUCTION FAILED TO ENABLE THE TELETYPE INTERRUPT ENABLE FLIP-FLOP WITH AC11#1, PRESSING CONTINUE ENTERS A SCOPE LOOP THAT ISSUES KIE WITH AC11#1 CONTINUOUSLY, MANUAL RESTART.

LOC 4135

PRG2, ROUTINE 10, ERROR HALT E, SRQ INSTRUCTION FAILED TO SKIP WITH THE READER FLAG SET AND TTY INTERRUPT ENABLED, PRESSING CONTINUE ENTERS A SCOPE LOOP THAT ISSUES SRQ WITH READER FLAG SET AND TTY ENABLED, MANUAL RESTART.

LOC 4144

PRG2, ROUTINE 10, ERROR HALT F, SPI INSTRUCTION FAILED TO SKIP WITH THE READER FLAG SET AND TTY INTERRUPT ENABLED, PRESSING CONTINUE ENTERS A SCOPE LOOP THAT ISSUES SPI WITH READER FLAG SET AND TTY ENABLED, MANUAL RESTART.

LOC 4206

PRG2, ROUTINE 11, ERROR HALT A, CAF INSTRUCTION FAILED TO CLEAR AC AND/OR LINK, PRESSING CONTINUE ENTERS A SCOPE LOOP THAT ISSUES CAF CONTINUOUSLY WITH AC AND LINK SET, MANUAL RESTART.

LOC 4215

PRG2, ROUTINE 11, ERROR HALT B, CAF INSTRUCTION FAILED TO CLEAR THE READER FLAG, PRESSING CONTINUE ENTERS A SCOPE LOOP THAT ISSUES CAF CONTINUOUSLY WITH READER FLAG SET, MANUAL RESTART.

LOC 4224

PRG2, ROUTINE 11, ERROR HALT C, CAF INSTRUCTION FAILED TO ENABLE THE TELETYPE INTERRUPT ENABLE FLIP-FLOP, PRESSING CONTINUE ENTERS A SCOPE LOOP THAT ISSUES CAF CONTINUOUSLY, MANUAL RESTART.

LOC 4276

PRG2, ROUTINE 12, ERROR HALT A, KRB INSTRUCTION FAILED TO CLEAR THE READER FLAG, PRESSING CONTINUE ENTERS A SCOPE LOOP THAT ISSUES KRB CONTINUOUSLY WITH READER FLAG SET, MANUAL RESTART.

LOC 4303

PRG2, ROUTINE 12, ERROR HALT B, KRB INSTRUCTION FAILED, SET READER FLAG BY 103 MSEC AFTER KRB WAS ISSUED, PRESSING CONTINUE ENTERS A SCOPE LOOP THAT ISSUES KRB CONTINUOUSLY, MANUAL RESTART.

(6,1 CONT'D)

LOC 4305	PRG2, ROUTINE 12, ERROR HALT C, KRB INSTRUCTION FAILED TO READ THE CORRECT DATA OFF OF TAPE, PRESS CONTINUE TO TRY TEST AGAIN, MANUAL RESTART.
LOC 4337	PRG3, ROUTINE 0, ERROR HALT A, READ ERROR, BAD CHARACTER IN AC, PRESS CONTINUE.
LOC 4342	PRG3, ROUTINE 0, ERROR HALT B, FOLLOW UP HALT; EXPECTED CHARACTER IN AC, PRESSING CONTINUE RESUMES TEST.
LOC 4371	PRG3, ROUTINE 1, ERROR HALT A, READ ERROR; BAD CHARACTER IN AC, PRESS CONTINUE.
LOC 4374	PRG3, ROUTINE 1, ERROR HALT B, FOLLOW UP HALT; EXPECTED CHARACTER IN AC, PRESSING CONTINUE RESUMES TEST.
LOC 4427	PRG3, ROUTINE 2, ERROR HALT A, READ ERROR; BAD CHARACTER IN AC, PRESS CONTINUE.
LOC 4432	PRG3, ROUTINE 2, ERROR HALT B, FOLLOW UP HALT; EXPECTED CHARACTER IN AC, PRESSING CONTINUE RESUMES TEST.
LOC 5415	PRG6, ROUTINE 0, KSF COMMAND FAILED TO SKIP ON KEYBOARD FLAG, PRESS CONTINUE TO ENTER SCOPE LOOP THAT SKIPS ON FLAG CONTINUOUSLY.
LOC 5707	PRG10, READ ERROR HALT A, BAD CHARACTER IN AC; PRESS CONTINUE, HALT OCCURS IF SR5#1.
LOC 5712	PRG10, READ ERROR HALT B, FOLLOW UP HALT TO PRG10 READ ERROR HALT A, EXPECTED CHARACTER IS DISPLAYED IN AC, TO PROCEED, PRESS CONTINUE.
LOC 5717	PRG10, ERROR COUNT HALT, HALT OCCURS WHENEVER SR0 IS SET TO A 1, THE AC THEN CONTAINS THE ACCUMULATED ERROR COUNT, IF ANY, TO PROCEED, PRESS CONTINUE.

7. MISCELLANEOUS

7.1 EXECUTION TIME (MINUTES:SECONDS)

	110 CURRENT	110 EIA	150 EIA	300 EIA	600 EIA	1200 EIA	2400 EIA
PRG01	1132	1132	1103	0132	0117	019	015
PRG11	N/A	4130	3125	1120	0144	0123	0112
PRG21	2147	N/A	N/A	N/A	N/A	N/A	N/A
PRG31	10100	N/A	N/A	N/A	N/A	N/A	N/A
PRG41	20100	N/A	N/A	N/A	N/A	N/A	N/A
PRG51	CONTINUOUS	N/A	N/A	N/A	N/A	N/A	N/A
PRG61	USER DEP.	N/A	N/A	N/A	N/A	N/A	N/A
PRG71	40100	N/A	N/A	N/A	N/A	N/A	N/A
PRG101	CONTINUOUS	N/A	N/A	N/A	N/A	N/A	N/A
PRG111	USER DEP.	N/A	N/A	N/A	N/A	N/A	N/A
PRG121	CONTINUOUS	N/A	N/A	N/A	N/A	N/A	N/A

7.2 TEST TAPES

IF A BINARY COUNT TEST TAPE IS NOT AVAILABLE ON SITE, USE PRG12 TO GENERATE A BINARY COUNT TEST TAPE. FOR CONVENIENCE OF USE, A TAPE LOOP SHOULD BE MADE, MAKING SURE THAT THE PATTERN IS MATCHED AT THE SPICE POINT.

7.3 TEST EQUIPMENT

FOR TESTING OF THE EIA LOGIC THE INPUT MUST BE CONNECTED TO THE OUTPUT ON THE 40 PIN SIDE CONNECTOR WITH JUMPERS.
PIN E TO PIN M
PIN F TO PIN J

8. PROGRAM DESCRIPTION -----

8.1 PRG0 - BASIC OUTPUT LOGIC TESTS (EIA AND CURRENT)

THIS PROGRAM CONTAINS 8 ROUTINES NUMBERED FROM 0-7 (OCIAL)

RTN01 CHECKS THE ABILITY OF
SPF TO SET PRINTER FLAG.
TSF TO SKIP ON PRINTER FLAG SET.
CAF TO CLEAR PRINTER FLAG, AC, AND LINK.
TCF TO CLEAR PRINTER FLAG.
TSF TO NOT SKIP ON PRINTER FLAG 0.
TEST IS DONE 100 TIMES.

RTN11 CHECKS THAT NO OTHER DEVICE CAN CAUSE AN INTERRUPT
AND THEN CHECKS THAT THE PRINTER FLAG IS CAPABLE OF
INTERRUPTING. TEST IS DONE 4000 TIMES.

RTN21 CHECKS THE ABILITY OF
KIE TO DISABLE TTY INTERRUPT ENABLE FLIP-FLOP.
SPI TO NOT SKIP WITH NO TTY INTERRUPT REQUEST.
SRQ TO NOT SKIP WITH NO TTY INTERRUPT REQUEST.
KIE TO ENABLE TTY INTERRUPT ENABLE FLIP-FLOP.
SPI TO SKIP ON A TTY INTERRUPT REQUEST.
SRQ TO SKIP ON A TTY INTERRUPT REQUEST.
CAF TO ENABLE TTY INTERRUPT ENABLE FLIP-FLOP.
TEST IS DONE 4000 TIMES.

RTN31 CHECKS THE ABILITY OF
TPC TO SET THE PRINTER FLAG.
TLS TO CLEAR THE PRINTER FLAG.
TLS TO SET THE PRINTER FLAG.
TEST IS DONE 100 TIMES.

RTN41 PRINTER TIMING TEST
CHECKS THAT THE FLAG IS NOT SET JUST PRIOR TO
9 BIT TIMES AND THAT THE FLAG IS SET AT 9.5 BIT TIMES.
TEST IS DONE 100 TIMES.

RTN51 PRINTER TIMING TEST
AFTER ISSUING A TLS AND WAITING FOR THE FLAG
TO SET ANOTHER TLS IS ISSUED AND THE FLAG IS
CHECKED JUST PRIOR TO 11 BIT TIMES FOR 110 BAND
AND 10 BIT TIMES FOR NON 110 BAND - THE FLAG
SHOULD NOT BE SET. THE FLAG IS CHECKED AGAIN 1/2
BIT TIME LATER AND THE FLAG SHOULD BE SET AT THIS
TIME. TEST IS DONE 100 TIMES.

RTN61 TEST OF GTF INSTRUCTION. TEST IS DONE 4000 TIMES.

RTN71 TEST OF RTF INSTRUCTION. TEST IS DONE 4000 TIMES.

PRG1 - BASIC EIA INPUT AND OUTPUT LOGIC TESTS

NOTE: ON THE 40 PIN SIDE CONNECTOR: PIN E MUST BE CONNECTED TO PIN M; PIN F MUST BE CONNECTED TO PIN J.

RTN0: CHECKS THAT KCC WILL CLEAR THE AC. TEST IS DONE 100 TIMES.

RTN1: T1S IS USED TO SEND DATA AND KSF CHECKS TO SEE IF THE RECEIVER FLAG SET UPON COMPLETION OF RECEIVING THE DATA. TEST IS DONE 100 TIMES.

RTN2: TEST OF KSF TO SKIP ON RECEIVER FLAG CONSISTENTLY. TEST IS DONE 4000 TIMES.

RTN3: TEST OF KSF TO NOT SKIP ON NO RECEIVER FLAG. TEST IS DONE 500 TIMES.

RTN4: CHECKS THAT NO OTHER DEVICE CAN CAUSE AN INTERRUPT AND THAT THE READER FLAG WILL INTERRUPT. TEST IS DONE 1000 TIMES.

RTN5: CHECKS THE ABILITY OF

SRQ TO SKIP ON A READER INTERRUPT
SPI TO SKIP ON A READER INTERRUPT
CAF TO CLEAR KBRD/READER FLAG.
SRQ TO NOT SKIP ON NO READER FLAG
SPI TO NOT SKIP ON NO READER FLAG
TEST IS DONE 100 TIMES.

RTN6: CHECKS THAT THE READER FLAG SETS NO LATER THAN THE REQUIRED TIME FOR IT TO SET.

110 BAUD - 100 MSEC.
150 BAUD - 66.7 MSEC.
300 BAUD - 33.3 MSEC.
600 BAUD - 16.7 MSEC.
1200 BAUD - 8.33 MSEC.
2400 BAUD - 4.16 MSEC.
TEST IS DONE 100 TIMES.

RTN7: CHECKS DATA HANDLING CAPABILITIES BY SENDING A NUMBER FOLLOWED BY ITS COMPLEMENT. TEST IS DONE 512 TIMES.

RT10: CHECKS DATA HANDLING CAPABILITIES BY SENDING RANDOM NUMBERS. TEST IS DONE 512 TIMES.

RTN11: CHECKS THAT KRS CAN "INCLUSIVE OR" READER BUFFER WITH AC. TEST IS DONE 500 TIMES.

RTN12: CHECKS THAT KRB WILL "JAM TRANSFER" RECEIVER BUFFER TO AC, AND THAT KRB WILL CLEAR READER FLAG. TEST IS DONE 500 TIMES.

PRG2 - BASIC INPUT LOGIC TESTS (CURRENT)

 THIS PROGRAM CONTAINS 11 ROUTINES NUMBERED FROM 0 TO 12 (OCTAL).

RTN0: CHECKS THAT KCC COMMAND IS ABLE TO CLEAR THE AC. TEST
 IS DONE 1000 TIMES.

RTN1: ISSUES KCC, WAITS 200MS AND CHECKS THAT FLAG IS SET.
 A FAILURE TO SKIP INDICATES THAT THE FLAG IS NOT
 SET, OR THAT KSF COMMAND FAILED TO SKIP.

RTN2: WITH FLAG SET, CHECKS THAT KSF COMMAND SKIPS RELIABLY,
 DONE 500 TIMES.

RTN3: CHECKS THAT KSF COMMAND DOES NOT SKIP WITH FLAG RESET;
 DONE 500 TIMES.

RTN4: CHECKS THAT NO OTHER DEVICE CAN CAUSE AN INTERRUPT,
 AND THEN CHECKS THAT READER IS CAPABLE OF INTERRUPTING;

RTN5: TIMING TEST.

RTN6: READS A CHARACTER FROM TAPE AND SAVES IT. IT THEN
 REREADS THE TTI STATISTICALLY 1000 TIMES TO CHECK FOR
 CONSISTENT READING FROM TTI. 256 CHARACTERS ARE READ
 IN THIS MANNER.

RTN7: CHECKS THAT KCR DOES NOT CLEAR AC AND SETS READER
 FLAG, BIT DOES NOT SET READER RUN; DONE 100 TIMES.

RTN10: CHECKS THAT KIE WILL ENABLE AND DISABLE TTY INTERRUPT
 F.F. AND THAT SRQ AND SPI WILL AND WILL NOT SKIP;
 DONE 1000 TIMES

RTN11: CHECKS THAT CAP WILL ENABLE TTY INTERRUPT F.F. AND
 THAT IT WILL CLEAR AC, LINK, AND READER FLAG; DONE
 100 TIMES.

RTN12: CHECKS THAT KRB CAN CLEAR THE READER FLAG AND THAT
 KRB CAN SET THE FLAG. ALSO KRB IS CHECKED FOR READING
 DATA. TEST IS DONE 256 TIMES.

8.4

PRG3 - READER TEST

THIS PROGRAM CONTAINS 3 ROUTINES NUMBERED FROM 0 TO 2,

RTN0: READS 4095 CHARACTERS OF BINARY COUNT PATTERN, FULL
SPEED,

RTN1: READS 2000 CHARACTERS OF BINARY COUNT PATTERN WITH
RANDOM STALLS BETWEEN CHARACTERS,

RTN2: READS 100 RANDOM LENGTH CHARACTER BLOCKS, FIXED STALL
BETWEEN CHARACTERS IN A BLOCK. THE STALL CHANGES FOR
EACH BLOCK AND IS DETERMINED AT RANDOM,

PRG4 - PRINTER TEST

8.5

THIS PROGRAM CONTAINS 41 ROUTINES NUMBERED FROM 0 TO 50 (OCTAL),

RTN0: CARRIAGE RETURN TEST, CHECKS ABILITY OF CARRIAGE
RETURN TO PRINT POSITION 1 FROM ALL OTHER PRINT
POSITIONS, NO PRINTING SHOULD OCCUR IN ANY PRINT
POSITION OTHER THAN POSITION 1,

RTN1: RIGHT MARGIN TEST, THIS TEST SHOWS WHEN THE RIGHT
MARGIN IS NOT CORRECTLY ADJUSTED, THE TEST PRINTS
16 GROUPS OF ---I FOLLOWED BY CHARACTERS I-.

RTN2: SPACE TEST, THE TEST PRINTS / IN ALTERNATE POSITIONS
OF THE LINE, AFTER A DOUBLE CARRIAGE RETURN IT SCAPES
TO THE BLANK POSITIONS AND PRINTS A LEFT SLANT SLASH.
A DOUBLE CARRIAGE RETURN IS ISSUED AFTER PRINTING EACH
LEFT SLANT SLASH.

RTN3: LINE FEED TEST, THE TEST PRINTS A LEFT SLANT SLASH
FOLLOWED BY A LINE FEED, FOLLOWED BY A RANDOM DELAY
UNTIL 81 SLASHES HAVE BEEN PRINTED, THE RESULT SHOULD
APPEAR TO BE A LEFT SLANTED LINE FROM POSITION 1 TO 81.
VERTICAL SPACING VARIATIONS SHOULD BE APPARENT IF
ADJUSTMENT IS REQUIRED.

(8.5 CONT'D)

ROUTINES 4 THROUGH 41 TYPES LINES CONTAINING 3 CHARACTERS AT FULL SPEED AS FOLLOWS:

RTN41	ABC (CAPITALS)
RTN51	DEF " "
RTN61	GHI " "
RTN71	JKL " "
RTN81	MNO " "
RTN91	PQR " "
RTN101	STU " "
RTN111	VWX " "
RTN121	YZ " "
RTN131	123 " "
RTN141	456 " "
RTN151	789 " "
RTN161	! " #
RTN171	\$ % &
RTN181	' ()
RTN191	* + ,
RTN201	- . /
RTN211	: ; <
RTN221	= > ?
RTN231	@ [\
RTN241	^ _ AND LEFT ARROW
RTN251	ABC (LOWER CASE) (KSR37 ONLY)
RTN261	DEF " "
RTN271	GHI " "
RTN281	JKL " "
RTN291	MNO " "
RTN301	PQR " "
RTN311	STU " "
RTN321	VWX " "
RTN331	YZ AND CODE 340
RTN341	" "
RTN351	" "
RTN361	" "
RTN371	" "
RTN381	" "
RTN391	" "
RTN401	" "
RTN411	" "

RTN421 TYPES LINE OF 4 CHARACTERS WHOSE CODE IS 373, 374, 375, AND 376 (KSR37 ONLY).

RTN431 TYPES 2 LINES OF ALL CHARACTERS; FIRST LINE IS TYPED AT FULL SPEED, AND THE 2ND LINE WITH RANDOM STALLS BETWEEN CHARACTERS.

RTN441 TYPES 12 LINES OF ASR33 PRINTER WORST CASE PATTERN; ALTERNATE LINES ARE TYPED WITH RANDOM STALLS BETWEEN CHARACTERS; ROUTINE RUNS ONLY IF KSR33 OR ASR33 IS PRESENT.

THE ASR33 WORST CASE PATTERN USED IS 'LEFT ARROW W7W LEFT ARROW;

RTN451 TYPES 12 LINES OF ASR35 PRINTER WORST CASE PATTERN;
ALTERNATE LINES ARE TYPED WITH RANDOM STALLS BETWEEN
CHARACTERS. ROUTINE RUNS ONLY IF KSR35 OR ASR35 IS
PRESENT.

THE AST35 WORST CASE PATTERN USED IS 'C?C?C

RTN461 TYPES 12 LINES OF KSR37 PRINTER WORST CASE PATTERN;
ALTERNATE LINES ARE TYPED WITH RANDOM STALLS BETWEEN
CHARACTERS. ROUTINE RUNS ONLY IF KSR37 IS PRESENT.

THE KSR 37 WORST CASE PATTERN USED IS:

CAPITAL N, LOWER CASE Q, CAPITAL A, SWING DASH,
CAPITAL A, LOWER CASE Q.

RTN471 TAB TEST, EXECUTED FOR 37 OR 35 TELETYPE ONLY, THE
TEST IS RUN AFTER ROUTINE 3.

RTN501 BACKSPACE TEST, EXECUTED FOR KSR37 TELETYPE ONLY.
THIS TEST IS RUN AFTER ROUTINE 47.

PRG5 - PUNCH TEST

8,6

THIS PROGRAM TESTS THE PUNCH WITH A SPECIAL BINARY COUNT
PATTERN. EVERY BINARY COUNT CHARACTER PUNCHED IS FOLLOWED
BY ITS 1'S COMPLEMENT CHARACTER.

THE TEST SEQUENCE IS AS FOLLOWS:

- A) PUNCH LEADER (CODE 376)
- B) PUNCH SYNC CHARACTER (CODE 377)
- C) PUNCH DATA BLOCK AT FULL SPEED (512 CHARACTERS)
- D) PUNCH TRAILER (CODE 376)
- E) SYNC THE READER
- F) READ AND CHECK DATA BLOCK
- G) PUNCH LEADER (CODE 376)
- H) PUNCH SYNC CHARACTER (CODE 377)
- I) PUNCH DATA BLOCK WITH STALLS. (512 CHARACTERS)
- J) PUNCH TRAILER (CODE 376)
- K) SYNC THE READER
- L) READ AND CHECK DATA BLOCK
- M) REPEAT. (GO TO STEP A)

8.7

PRG6 - KEYBOARD TEST

THIS PROGRAM CONTAINS 3 ROUTINES NUMBERED FROM 0 TO 2.

RTN0: CHECKS THAT KSF COMMAND SKIPS WHEN FLAG=1. TEST IS DONE 1000 TIMES.

RTN1: ECHO TEST, ANY CHARACTERS READ FROM KEYBOARD ARE TYPED, CORRECT OPERATION VERIFICATION IS DONE VISUALLY BY USER, READING A RUBOUT CHARACTER ENDS THE TEST.

RTN2: OCTAL EQUIVALENCE TEST, THE OCTAL EQUIVALENT OF ANY CHARACTERS KEYED IS TYPED, READ A RUBOUT ENDS THE TEST.

8.8

PRG7 - COMBINED READER, PRINT, PUNCH TEST

THIS PROGRAM CONTAINS 25 ROUTINES NUMBERED FROM 0 TO 32 (OCTAL), ALL ROUTINES USE THE FOLLOWING TEST SEQUENCE!

- A) FILL CORE WITH DATA TO BE PUNCHED/PRINTED.
- B) PUNCH LEADER.
- C) PUNCH SYNC CHARACTER.
- D) PUNCH DATA BLOCK (NO DELAY BETWEEN CHARACTERS.)
- E) SYNC THE READER.
- F) READ/CHECK DATA BLOCK (RANDOM DELAY BETWEEN CHARACTERS).
- G) PUNCH DATA BLOCK (RANDOM DELAY BETWEEN CHARACTERS).
- H) READ DATA BLOCK (NO DELAY BETWEEN CHARACTERS).
- I) PUNCH TRAILER.
- J) WAIT FOR READER TO COMPLETE READING DATA BLOCK.
- K) END OF TEST SEQUENCE.

(8,8 CONT/D)

```
RTN0: PUNCH/PRINT AND READ CHECK BLOCK OF ABC
RTN1: PUNCH/PRINT AND READ CHECK BLOCK OF DEF
RTN2: PUNCH/PRINT AND READ CHECK BLOCK OF GHI
RTN3: PUNCH/PRINT AND READ CHECK BLOCK OF JKL
RTN4: PUNCH/PRINT AND READ CHECK BLOCK OF MNO
RTN5: PUNCH/PRINT AND READ CHECK BLOCK OF PQR
RTN6: PUNCH/PRINT AND READ CHECK BLOCK OF STU
RTN7: PUNCH/PRINT AND READ CHECK BLOCK OF VWX
RTN8: PUNCH/PRINT AND READ CHECK BLOCK OF YZ0
RTN9: PUNCH/PRINT AND READ CHECK BLOCK OF 123
RTN10: PUNCH/PRINT AND READ CHECK BLOCK OF 456
RTN11: PUNCH/PRINT AND READ CHECK BLOCK OF 789
RTN12: PUNCH/PRINT AND READ CHECK BLOCK OF !"#
RTN13: PUNCH/PRINT AND READ CHECK BLOCK OF $%&
RTN14: PUNCH/PRINT AND READ CHECK BLOCK OF '()
RTN15: PUNCH/PRINT AND READ CHECK BLOCK OF *+,-./
RTN16: PUNCH/PRINT AND READ CHECK BLOCK OF :;<=>?
RTN17: PUNCH/PRINT AND READ CHECK BLOCK OF @[\]^_
RTN18: PUNCH/PRINT AND READ CHECK BLOCK OF `{|}~
RTN19: PUNCH/PRINT AND READ CHECK BLOCK OF ALL PRINTABLE CHARACTERS
RTN20: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN21: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN22: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN23: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN24: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN25: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN26: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN27: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN28: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN29: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN30: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN31: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
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RTN39: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN40: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN41: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN42: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN43: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN44: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
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RTN46: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
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RTN57: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN58: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN59: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN60: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN61: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN62: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN63: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN64: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN65: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN66: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN67: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN68: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN69: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN70: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN71: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN72: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN73: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN74: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN75: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN76: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN77: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN78: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN79: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN80: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN81: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN82: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN83: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN84: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN85: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN86: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN87: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN88: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN89: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN90: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN91: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN92: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN93: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN94: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN95: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN96: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN97: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN98: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
RTN99: PUNCH/PRINT AND READ CHECK BLOCK OF ASR33 PRINTER
RTN100: PUNCH/PRINT AND READ CHECK BLOCK OF ASR35 PRINTER
```

8,9

```
PRG10 - READER EXERCISER, BINARY COUNT PATTERN
-----
```

THE PROGRAM READS AND CHECKS A BINARY COUNT PATTERN TEST TAPE;
WITH PROGRAM RUNNING SETTING SR0 TO A 1 CAUSES PROGRAM TO HALT
AND DISPLAY THE ACCUMULATED ERROR COUNT IN AC. SR3 SET TO
A 1 GIVES FULL SPEED READING, SR3 SET TO A 0 CAUSES STALLS
BETWEEN CHARACTERS. SR5 SET TO A 1 WILL HALT THE PROGRAM WHEN
AN ERROR OCCURS. THE BAD CHARACTER IS THEN DISPLAYED IN THE
AC. PRESSING CONTINUE DISPLAYS THE EXPECTED CHARACTER.

8,10

```
PRG11 - PRINTER EXERCISER
-----
```

THIS PROGRAM CONTINUOUSLY TYPES LINES OF ANY 3 CHARACTERS
KEYED BY USER. ON PROGRAM REQUEST THE USER KEYS IN THE 3
CHARACTERS TO BE TYPED, FOLLOWED BY A DELETE CODE IF FULL
SPEED TYPING IS DESIRED, OR BY ANY OTHER CHARACTER IF RANDOM
STALLS AFTER EACH CHARACTER ARE DESIRED.

8,11

```
PRG12 - TAPE GENERATOR - BINARY COUNT PATTERN
-----
```

PUNCHES BINARY COUNT PATTERN TEST TAPE.

```

/PDP-8/E TELETYPE CONTROL TEST, MAINDEC=08-DHKLD=A=L
/COPYRIGHT 1971,1972 DIGITAL EQUIPMENT CORPORATION, MAYNARD, MASS, 01754
/PROGRAMMER: ED FORTMILLER
/
/PRG0=BASIC OUTPUT CONTROL LOGIC TEST (CURRENT AND EIA)
/PRG1=BASIC OUTPUT AND INPUT LOGIC TEST (EIA - LOOP AROUND)
/PRG2=BASIC INPUT CONTROL LOGIC TEST - (USES READER)
/PRG3=READER TEST
/PRG4=PRINTER TEST
/PRG5=PUNCH TEST
/PRG6=KEYBOARD TEST
/PRG7=COMBINED TEST
/PRG10=READER EXERCISER; BINARY COUNT PATTERN.
/PRG11=PRINTER EXERCISER.
/PRG12=TAPE GENERATOR; BINARY COUNT PATTERN.
/
/

```

```

/*****
/BIT TIME TABLE
/0110 BAUD 11 BITS @ 9.09 MSEC = 100 MSEC
/0150 BAUD 10 BITS @ 6.67 MSEC = 66.7 MSEC
/0300 BAUD 10 BITS @ 3.33 MSEC = 33.33 MSEC
/0600 BAUD 10 BITS @ 1.67 MSEC = 16.67 MSEC
/1200 BAUD 10 BITS @ .833 MSEC = 8.33 MSEC
/2400 BAUD 10 BITS @ .416 MSEC = 4.167 MSEC
/*****

```

```

6001 ION=6001 /TURN INTRRRUPT ON;
6002 IOF=6002 /TURN INTRRRUPT OFF;
6003 SRQ=6003 /SKIP IF INTRRRUPT REQUEST;
6004 GTF=6004 /GET INTRRRUPT FLAGS
6005 RTF=6005 /RESTORE INTRRRUPT FLAGS AND TURN INTRRRUPT ON
6007 CAF=6007 /CLEAR ALL FLAGS, AC, LINK, AND ENABLE TTY INTRRRUPT
6030 KCR=6030 /CLEAR KBRD FLAG BUT DO NOT SET RDR RUN
6031 KSF=6031 /SKIP IF KEYBOARD/READER FLAG = 1;
6032 KCG=6032 /CLEAR AC AND KBRD/READER FLAG; SET READER RUN;
6034 KRS=6034 /READ KEYBOARD/READER BUFFER STATIC
6035 KIE=6035 /ENABLE TTY INTRRRUPT WHEN AC11 EQUALS 1
6036 KRB=6036 /CLEAR AC, READ KEYBOARD BUFFER, CLEAR
        /KEYBOARD FLAGS.
6040 SPF=6040 /SET PRINTER FLAG
6041 TSF=6041 /SKIP IF TELEPRINTER/PUNCH FLAG = 1;
6042 TCF=6042 /CLEAR TELEPRINTER/PUNCH FLAG;
6044 TPC=6044 /LOAD TELEPRINTER/PUNCH BUFFER
        /SELECT AND PRINT.
6045 SPI=6045 /SKIP IF TTY INTRRRUPT
6046 TLS=6046 /LOAD TELEPRINTER/PUNCH BUFFER,
        /SELECT AND PRINT AND CLEAR
        /TELEPRINTER/PUNCH FLAG.

```

```

7002 BSW=7002 /SWAP BYTES IN AC;
7200 CLA=7200
7402 HLT=7402

```

1644

7604 LAS=7604
7421 MQL=7421
7621 CAM=7621
7701 ACL=7701

0000 OPEN=0
4577 SETLOC=JMS I [STCTR
4576 DELAY=JMS I [DLYMS
4575 CRLF=JMS I [CRLF
4574 MOVE=JMS I [MOVE
4573 TYPE=JMS I [TYPSTG
6117 MTON=6117
6127 MTRS=6127
6115 MINT=6115
4572 UKSF=JMS I [XKSF
4571 UKCC=JMS I [XKCC
4570 UKRS=JMS I [XKRS
4567 UKRB=JMS I [XKRB
4566 UTSF=JMS I [XTSF
4565 UTCF=JMS I [XTCF
4564 UTPC=JMS I [XTPC
4563 UTLS=JMS I [XTLS
4562 UKCR=JMS I [XKCR
4561 UKIE=JMS I [XKIE
4560 USPF=JMS I [XSPF
4557 USPI=JMS I [XSPI
4556 STALL=JMS I [STAL
4555 CKSR37=JMS I [CK37
4554 CKSR33=JMS I [CK33
4553 CKSR35=JMS I [CK35
6577 BLOCKABEND
6601 BLOCK1=BLOCKA+2
6711 BLOCK8=BLOCKA+112
6722 BLK8B=BLOCKA+123
6713 BLOCK2=BLOCKA+114
6724 BLK2=BLOCKA+125
7023 BLOCKC=BLOCKA+224
7034 BLKCC=BLOCKA+235
7577 DBLK=BLOCKA+1000
7631 M147=147
0304 RRP=0304

/LOAD MQ FROM AC THEN CLEAR AC;
/CLEAR AC AND MQ;
/LOAD MQ INTO AC;

/PROGRAM MODIFIABLE;

/DC02; MULTIPLE TTY ON;
/DC02; MULTIPLE TTY READ STATUS;
/DC02; MULTIPLE TTY INTERRUPT CONTROL;

/-103 DECIMAL;

0000	*0	0000	JMP 1
0001		5001	2
0002		0002	3
0003		0003	
0005	*5	0005	JMP 1 2
5402		5402	0
0006		0006	
0016	*16	0016	OPEN
0000		0000	
0020	*20	0020	
0000		0000	TTYTYP, OPEN
0021		0021	TTYIOT, RRP
0304		0304	

/AUTO INDEX.

/CONSTANT TO DETERMINE IOT CODE

```

/RESET FOR 03 READER AND 04 PUNCH;
/TO CHANGE IOT CODE SET THIS LOCATION
/TO "RRPP" WHERE RP IS FOR
/THE READER AND PP IS FOR THE PUNCH,
/CONSTANT TO DETERMINE DELAY
/RESET FOR 110 BAUD;
/TO SELECT BAUD RATE DEPOSIT THE FOLLOWING:
/0110 FOR 110 BAUD;
/0150 FOR 150 BAUD;
/0300 FOR 300 BAUD;
/0600 FOR 600 BAUD;
/1200 FOR 1200 BAUD;
/2400 FOR 2400 BAUD;
/**THE ABOVE ARE THE ONLY LEGAL BAUD RATES**

```

```

/USER PROGRAM START;
/CHAIN RTN ENTRY;

```

0022 0110 BAUDRT, 110

0023	0000	KSTART, OPEN
0024	0000	DELAYM, OPEN
0025	0263	CHAIN, CHAIN
0026	1365	KFLAG, KFLAG
0027	0474	DLCNT1, DLCNT
0030	2012	S100, S100
0031	2000	S4000, S4000
0032	2005	S200, S200
0033	2126	TLCALL, TLCALL
0034	2134	TLC37, TLC37
0035	2144	FBI, FBI
0036	0000	PRGNUM, OPEN
0037	2200	PRGTAB, PRG0
0040	3000	PRG1
0041	3503	PRG2
0042	4307	PRG3
0043	4434	PRG4
0044	5274	PRG5
0045	5340	PRG6
0046	5465	PRG7
0047	5651	PRG10
0048	5722	PRG11
0049	5764	PRG12
0050	0000	OPEN
0051	0000	TEMP, OPEN
0052	0000	CURTST, OPEN
0053	0000	RTNNO, OPEN
0054	0000	NXTST, OPEN
0055	0000	MSCTR, OPEN
0056	0000	MILCTR, OPEN
0057	0000	MILI, OPEN
0062	0000	CTRA, OPEN
0063	0000	CTRB, OPEN
0064	0000	STLID, OPEN
0065	0530	SYNK, SYNK

```

/WORK
/LOCATIONS
/FCR CURRENT TEST ADDRESS
/FCR CURRENT TEST NUMBER
/FCR NEXT TEST ADDRESS
/MILLISECONDS COUNTER

```

```

/7372 FOR 110 BAUD;
/7522 FOR 150 BAUD;
/7652 FOR 300 BAUD;
/7726 FOR 600 BAUD;
/7754 FOR 1200 BAUD;
/7767 FOR 2400 BAUD;
/COUNTER A;
/COUNTER B;

```

/ENTRY TO SYNC TAPE RTN.

INPAT, IBIN
GETPT, GTBIN
CHECK, CHCK
PFLAG, 0
UOUT, OUT
UTPLN3, TYPLN3
UPUNCH, PUNCH
UMOVE, MOVE
RBSY, 0
AC, 0
LINK, 0
BLKCN, 0
DELAYS, 0
ERRCR, 0
UTEMP, 0
UTEMP1, 0
UTEMP2, 0
CR, 215
LF, 212
DLYMSK, 277
WTS6A, OPEN

/CARRIAGE RETURN
/LINE FEED

/CONTROL ROUTINE

*200
START, SKP CLA
HLT
CAM
JMS SEYRND
JMS STBAUD
BDR, LAS
AND C17
TAD C=12
SMA SZA
JMP START+1
LAS
AND C17
DCA PRGNUM
TAD PRGNUM
TAD CPGTAB
DCA TEMP
TAD I TEMP
DCA PRGADR
JMS DVCSEL
SLDC02, LAS
AND C7760
MTON
CLA IAC
MINT
JMS I UMOVE
5
1
-2
JMP I, +1
PRGADR, OPEN

/INCORRECT PROGRAM NUMBER
/CLEAR AC AND MQ
/SET UP RANDOM NUMBERS
/SET UP LOC M111 FOR SELECTED BAUD RATE.
/READ SR
/PROGRAM MASK = 17
/PROGRAM LIMIT = 12
/VALID PROGRAM NUMBER?
/NO
/YES, READ SR,
/SAVE PROGRAM NUMBER;
/DEVELOP PROGRAM START
/ADDRESS AND STORE AT
/PRGADR,
/PERFORM IOT SELECTION
/SELECT DC02 UNIT
/ENABLE DC02 INTERRUPT
/INITIALIZE
/INTERRUPT,
/AREA,

0236	7602	SRSET, HLT CLA	
0237	7200	GETRDY, CLA	
0240	1023	TAD KSTART	/SET ADDRESS OF 1ST ROUTINE
0241	3056	DCA NXTST	/STORE AT NXTST
0242	4302	JMS FORWD	
0243	7604	LAS	/READ SR
0244	7004	RAL	
0245	7700	SMA CLA	/ROUTINE SELECT? (SR1)
0246	5454	JMP I CURTST	/NO. START WITH 1ST RTN
0247	7604	LAS	/YES
0250	0146	AND C77	/SR 6=11 ENABLE MASK;
0251	7041	CIA	
0252	1055	TAD RTNNO	
0253	7650	SNA CLA	/IS IT THIS RTN?
0254	5454	JMP I CURTST	/YES, GO DO IT
0255	1056	TAD NXTST	/NO
0256	7001	IAC	/IS THIS LAST TRN?
0257	7640	SZA CLA	/NO
0260	5242	JMP GETRDY+3	
0261	7402	INCRTN, HLT GETRDY	/YES, INCORRECT ROUTINE NO.
0262	5237	JMP GETRDY	
0263	4317	CHAINN, JMS SHALT	/HALT? (SR0)
0264	7604	LAS	/READ SR
0265	7006	RTL	
0266	7630	SZL CLA	/SELECT ROUTINE? (SR1)
0267	5237	JMP GETRDY	/YES
0270	1056	TAD NXTST	
0271	7001	IAC	
0272	7640	SZA CLA	/LAST ROUTINE?
0273	5242	JMP GETRDY+3	/NO,
0274	7604	LAS	
0275	7006	RTL	
0276	7710	SPA CLA	/LOOP PROGRAM? (SR2)
0277	5237	JMP GETRDY	/YES
0300	7402	PRGEND, HLT	/END OF PROGRAM HALT
0301	5263	JMP CHAINN	
0302	0000	FORWD, 0	
0303	7300	CLA CLL	
0304	1456	TAD I NXTST	/GET NEXT RTN NO
0305	3055	DCA RTNNO	/STORE AT RTNNO
0306	2056	ISZ NXTST	
0307	1056	TAD NXTST	/SET CURRENT
0310	3052	DCA TEMP	/RTN NUMBER
0311	2056	ISZ NXTST	
0312	1056	TAD NXTST	/SET CURRENT
0313	3054	DCA CURTST	/RTN ADDR,
0314	1452	TAD I TEMP	/SET NEXT
0315	3056	DCA NXTST	/RTN ADDR,
0316	5702	JMP I FORWD	/EXIT
0317	0000	SHALT, 0	
0320	7604	LAS	/READ SR
0321	7700	SMA CLA	/HALT? (SR0)

/UNCONDITIONAL HALT (SR0 = 1)
/EXIT,

/GET CIR ADDR
/AND SAVE AT TEMP
/GET COUNT AND
/STORE PER C(TEMP)
/EXIT

/GET MS COUNT
/STORE IN MSCTR
/GET CONSTANT
/STORE IN MILCTR
/DELAY FINISHED?

/DONE DELAYING
/EXIT
/SUB TO CHECK FOR 33 TTY
/GET TTY TYPE
/33?
/YES,

/SUB TO CHECK FOR 35 TTY
/GET TTY TYPE
/35?
/YES,

/SUB TO CHECK FOR 37 TTY
/2
/GET TTY TYPE,
/37?
/YES,

PAGE

PAGE
OPEN
CLA CLL

/RANDOM NUMBER SUB A;

JMP I SHALT
TAD RTNCC
HLT
JMP I SHALT

STCTR,
0
CLA
TAD I STCTR
DCA TEMP
ISZ STCTR
TAD I STCTR
DCA I TEMP
ISZ STCTR
JMP I STCTR

DLYMS,
0
CLA CLL
TAD DELAYM
DCA MSCTR
TAD MIL1
DCA MILCTR
ISZ MILCTR
JMP I 1
ISZ MSCTR
JMP I 5
JMP I DLYMS
OPEN
CLA
TAD TTYTYP
SNA CLA
ISZ CK33
JMP I CK33

CK33,

CK35,

CK37,

0322 5717
0323 1055
0324 7402
0325 5717

0326 0000
0327 7200
0330 1726
0331 3052
0332 2326
0333 1726
0334 3452
0335 2326
0336 5726

0337 0000
0340 7300
0341 1024
0342 3057
0343 1061
0344 3060
0345 2060
0346 5345
0347 2057
0350 5343
0351 5737
0352 0000
0353 7200
0354 1020
0355 7650
0356 2352
0357 5752

0360 0000
0361 7240
0362 1020
0363 7650
0364 2360
0365 5760

0366 0000
0367 7344
0370 1020
0371 7650
0372 2366
0373 5766
0375 6000
0376 1513
0377 1740
0400

0400
0400
0401 7300

RGNA,

PAGE
OPEN
CLA CLL


```

0462 5257 JMP CHRCNT+1
0463 7041 CIA
0464 3273 DCA SCNT
0465 1656 TAD I CHRCNT
0466 3052 DCA TEMP
0467 1273 TAD SCNT
0470 3452 DCA I TEMP
0471 2256 ISZ CHRCNT
0472 5656 JMP I CHRCNT
0473 0000 SCNT, OPEN

```

```

/SUBROUTINE TO GENERATE RANDOM DELAY COUNT (NOT MORE THAN 3777(8));
DLCNT, 0
JMS RGNA
AND DLYMSK
SNA
JMP DLCLNT+1
CIA
DCA DELAYM
JMP I DLCLNT

```

/TABLE FOR BAUD RATES:

```

BAUTAB, -110
-150
-300
-600
-1200
-2400
OPEN

```

/FUTURE.

```

/SUBROUTINE TO COMPARE C(AC) TO CONTENTS STORED AT CALL+1
CHKC, 0

```

```

0513 0000 DCA WCHK
0514 3327 TAD I CHCK
0515 1713 CIA
0516 7041 TAD WCHK
0517 1327 ISZ CHCK
0520 2313 SEA CLA
0521 7640 JMP +3
0522 5325 ISZ CHCK
0523 2313 JMP I CHCK
0524 5713 TAD WCHK
0525 1327 JMP I CHCK
0526 5713 WCHK, 0
0527 0000

```

/SYNC ON TAPE SUBROUTINE

```

SYNK, 0
SETLOC
CTSK
-400
UKCC
UKSF
JMP +1
UKRS

```

```

/SET COUNT OF
/=256 (DEC) IN
/CTSK
/CLEAR AC AND FLAG
/READY?
/NO, TEST AGAIN
/YES, READ

```

```

0540 1144 TAD I=377
0541 7640 SZA CLA /377?
0542 7410 SKP
0543 5730 JMP I SYNK /YES, EXIT
0544 2350 ISZ CTSK /BUMP CHAR CTR +1
0545 5334 JMP SYNKA /GO READ AGAIN
0546 7402 HLT /256 CHARS READ, CAN'T SYNC
0547 5331 JMP SYNK+1 /GO TO SRST

0550 0000 CTSK, /CHAR COUNTER
0551 0000 STAL,
0552 7200 CLA
0553 1064 TAD STLID
0554 7700 SHA CLA /STALL?
0555 5751 JMP I STAL /NO, EXIT
0556 4274 JMS DLCNT /YES SET STALL COUNT
0557 4576 DELAY /STALL
0558 5751 JMP I STAL /EXIT
0559 0000 CRCTR,
0560 0000 CRALF,
0561 0000 OPEN
0562 0000 OPEN
0563 7200 CLA
0564 1762 TAD I CRALF
0565 3361 DCA CRCTR
0566 2362 ISZ CRALF
0567 4573 TYPE
0570 6250 CARLF
0571 2361 ISZ CRCTR
0572 5367 JMP I=3
0573 5762 JMP I CRALF
0600 PAGE

```

```

0600 PAGE
/SUBROUTINE TO MOVE VARIABLE LENGTH DATA FIELDS
MOVE, 0
0600 0000 CLA
0601 7200 TAD I MOVVE /GET "FROM ADDR" AND
0602 1600 DCA FADDR /STORE AT FADDR
0603 3223 ISZ MOVVE
0604 2200 TAD I MOVVE /GET "TO ADDR" AND
0605 1600 DCA TADDR /STORE AT TADDR,
0606 3224 ISZ MOVVE
0607 2200 TAD I MOVVE /GET "MOVE COUNT" AND
0610 1600 DCA MCTR /STORE AT MCTR,
0611 3225 ISZ MOVVE /SET UP FOR EXIT,
0612 2200 MOVEA, CLA
0613 7200 TAD I FADDR /GET "FROM" WORD
0614 1623 DCA I TADDR /STORE AT "TO" LOCATION
0615 3624 ISZ FADDR /+1 TO "FROM" ADDR
0616 2223 ISZ TADDR /+1 TO "TO" ADDR,
0617 2224 ISZ MCTR /ALL WORDS MOVED?
0620 2225 JMP MOVEA /NO, GO MOVE AGAIN
0621 5213 JMP I MOVVE /YES, EXIT
0622 5600 FADDR, 0
0623 0000

```

```

0626 0000 /TYPE CHARACTER STRING SUBROUTINE
0627 7200 TYPSTG, 0
0630 1626 CLA
0631 3314 TAD I TYPSTG
0632 3316 DCA TEMQ
0633 2226 DCA FLAG
0634 1714 ISZ TYPSTG
0635 7002 TAD I TEMQ
0636 4243 BSW
0637 1714 JMS TSC2
0640 4243 TAD I TEMQ
0641 2314 JMS TSC2
0642 5234 ISZ TEMQ
0643 0000 JMP TSC1
0644 0146 0
0645 3315 AND I77
0646 1316 DCA TEMR
0647 7640 TAD FLAG
0650 5260 SEA CLA
0651 1315 JMP TYPSP
0652 7450 TAD TEMR
0653 5256 SNA
0654 4271 JMP I+3
0655 5643 JMS PRINT
0656 2316 JMP I TSC2
0657 5643 ISZ FLAG
0660 3316 JMP I TSC2
0661 1315 DCA FLAG
0662 7450 TAD TEMR
0663 5643 SNA
0664 1377 JMP I TSC2
0665 7650 TAD (=97
0666 5626 SNA CLA
0667 1315 JMP I TYPSTG
0670 5254 TAD TEMR
    JMP TYPAT

PRINT,
0671 0000 OPEN
0672 1376 TAD (=45
0673 7640 SEA CLA
0674 5300 JMP I+4
0675 1107 TAD CR
0676 4474 JMS I UPUNCH
0677 5671 JMP I PRINT
0678 1315 TAD TEMR
0679 1375 TAD (=43
0680 7640 SEA CLA
0681 5306 JMP I+3
0682 1110 TAD LF
0683 5276 JMP PRINT+5
0684 1315 TAD TEMR
0685 1374 TAD (=40
    
```

/GET AND STORE
 /INITIAL ADDRESS
 /CLEAR FLAG;
 /SET UP EXIT
 /PICK UP DATA

/GO TYPE 1ST CHARACTER
 /PICK UP DATE
 /GO TYPE 2ND CHARACTER
 /EVEN STRING ADDRESS
 /GO BACK FOR MORE

/MASK OFF 6 BITS
 /SAVE CHARACTER
 /TEST "SPECIAL" FLAG;

/SET TYPE SPECIAL
 /NO. REGULAR CHARACTER
 /ZERO?
 /YES, SET FLAG;
 /NO, PRINT IT;
 /RETURN;
 /SET "SPECIAL" FLAG;
 /EXIT

/CLEAR FLAG;
 /TEST FOR 0;
 /0?
 /IGNORE IT;

/???
 /YES, EXIT CODE.

/IS IT 45?
 /NO.
 /YES, PRINT CR

/IS IT 43?
 /NO.
 /YES, TYPE LF

0710	7510	SPA			
0711	1143	TAD C100			
0712	1142	TAD C240			
0713	5276	JMP PRINT+5			
0714	0000	OPEN			
0715	0000	OPEN			
0716	0000	OPEN			
0717	0000	XKSF,		/SUB TO ISSUE KSF,	
0720	6031	KSF		/KSF	
0721	5717	JMP I XKSF		/NO SKIP	
0722	2317	ISZ XKSF		/SKIP	
0723	5717	JMP I XKSF			
0724	0000	XKCC,		/SUB TO ISSUE KCC,	
0725	6032	KCC			
0726	5724	JMP I XKCC		/EXIT	
0727	7402	HLT		/KCC SKIPPED,	
0730	0000	XKRS,		/SUB TO ISSUE KRS,	
0731	6034	KRS			
0732	5730	JMP I XKRS		/EXIT	
0733	7402	HLT		/KRS SKIPPED,	
0734	0000	XKRB,		/SUB TO ISSUE KRB,	
0735	6036	KRB			
0736	5734	JMP I XKRB		/EXIT	
0737	7402	HLT		/KRB SKIPPED,	
0740	0000	XTSF,		/SUB TO ISSUE ISF,	
0741	6041	TSF		/TSF	
0742	5740	JMP I XTSF		/NO SKIP,	
0743	2340	ISZ XTSF		/SKIP,	
0744	5740	JMP I XTSF			
0745	0000	XTCF,		/SUB TO ISSUE TCF,	
0746	6042	TCF			
0747	5745	JMP I XTCF		/EXIT	
0750	7402	HLT		/TCF SKIPPED,	
0751	0000	XTLS,		/SUB TO ISSUE TLS	
0752	6046	TLS			
0753	5751	JMP I XTLS		/EXIT	
0754	7402	HLT		/TLS SKIPPED,	
0755	0000	XKCR,		/SUB TO ISSUE KCR,	
0756	6030	KCR			
0757	5755	JMP I XKCR		/EXIT	
0760	7402	HLT		/KCR SKIPPED,	
0761	0000	XKIE,		/SUB TO ISSUE KIE,	
0762	6035	KIE			
0763	5761	JMP I XKIE		/EXIT,	
0764	7402	HLT		/KIE SKIPPED,	

2765 0000 XSPI, OPEN /SUB TO ISSUE SPI,
 2766 6045 SPI /
 2767 5765 JMP I XSPI /NO SKIP
 2770 2365 ISZ XSPI
 2771 5765 JMP I XSPI /EXIT

2774 7740
 2775 7735
 2776 7733
 2777 7701
 1000

PAGE

1000 PAGE
 1000 STBF,
 1001 4574 OPEN
 1002 0107 MOVE
 1003 6577 CR
 1004 7776 BLOCKA
 1005 4555 -2
 1006 5220 CKSR37
 1007 4574 JMP ST33B
 1010 0107 MOVE
 1011 6722 CR
 1012 7776 BLKBB
 1013 4574 -2
 1014 0107 MOVE
 1015 7034 CR
 1016 7776 BLKCC
 1017 5600 -2
 1020 4574 JMP I STBF
 1021 0107 ST33B,
 1022 6711 MOVE
 1023 7776 CR
 1024 4574 BLOCKB
 1025 0107 -2
 1026 7023 MOVE
 1027 7776 CR
 1030 5600 BLOCKC
 1031 0000 -2
 1032 7200 JMP I STBF
 1033 1631 FBF3,
 1034 3237 OPEN
 1035 2231 CLA
 1036 4574 TAD I FBF3
 1037 0000 DCA I+3
 1040 6601 ISZ FBF3
 1041 7775 MOVE
 1042 4555 OPEN
 1043 5255 BLOCK1
 1044 4574 -3
 1045 6601 CKSR37
 /37?
 /NO,
 /YES,
 /SUB TO SET UP BUFFER AREA;
 /CRLF TO BLOCKA,
 /KSR37?
 /NO,
 /CRLF TO BLKBB
 /CRLF TO BLKCC,
 /EXIT STBF
 /CRLF TO BLOCKB,
 /CRLF TO BLOCKC,
 /EXIT STBF,
 /SUB TO FILL CHAR BUFFER WITH
 /3 CHARACTERS SPECIFIED AT CALL+1;

1046	6604	BLOCK1*3	
1047	7662	-116	
1050	4574	MOVE	
1051	6601	BLOCK1	
1052	6724	BLK2	
1053	7657	-121	
1054	5631	JMP I FBF3	/EXIT FBF3,
1055	4574	MOVE	
1056	6601	BLOCK1	
1057	6604	BLOCK1*3	
1060	7673	-105	
1061	4574	MOVE	
1062	6601	BLOCK1	
1063	6713	BLOCK2	
1064	7670	-110	
1065	5631	JMP I FBF3	/EXIT FBF3,
1066	0000	OPEN	/FILL BUFFER WITH ALL CHARACTERS
1067	4555	CKSR37	/KSR37?
1070	5302	JMP FBA33	/NO,
1071	4574	MOVE	/YES,
1072	6107	A	
1073	6601	BLOCK1	
1074	7657	-121	
1075	4574	MOVE	
1076	6601	BLOCK1	
1077	6724	BLK2	
1100	7657	-121	
1101	5666	JMP I FBALL	/EXIT FBALL,
1102	4574	MOVE	
1103	6107	A	
1104	6601	BLOCK1	
1105	7701	-77	
1106	4574	MOVE	
1107	6107	A	
1110	6700	BLOCK1*77	
1111	7767	-11	
1112	4574	MOVE	
1113	6601	BLOCK1	
1114	6713	BLOCK2	
1115	7670	-110	
1116	5666	JMP I FBALL	/EXIT FBALL
1117	0000	FW336,	
1120	4574	MOVE	/MOVE 6 CHARACTERS ARS33 PRINTER
1121	6005	A33WP6	/WORST CASE PATTERN TO
1122	6601	BLOCK1	/BLOCK1
1123	7772	-6	
1124	4574	MOVE	/FILL BLOCKS WITH PATTERN
1125	6601	BLOCK1	
1126	6607	BLOCK1*6	
1127	7676	-102	
1130	4574	MOVE	
1131	6601	BLOCK1	
1132	6713	BLOCK2	

1133	7670	-110			
1134	5717	JMP I FW336	/EXIT		
1135	0000	FW356,			
1136	4574	0			
1137	6073	MOVE			
1140	6601	A35WP6	/MOVE 6 CHARACTER ASR35 PRINTER		
1141	7772	BLOCK1	/WORST CASE PATTERN TO BLOCK1		
1142	4574	-6			
1143	6601	MOVE			
1144	6607	BLOCK1	/FILL BUFFER WITH PATTERN		
1145	7676	BLOCK1+6			
1146	4574	-102			
1147	6601	MOVE			
1150	6713	BLOCK1			
1151	7670	BLOCK2			
1152	5735	-110			
		JMP I FW356	/EXIT		
1153	0000	FW376,			
1154	4574	OPEN	/MOVE 6 CHARACTER KSR37 PRINTER		
1155	6101	MOVE	/WORST CASE PATTERN TO BLOCK1		
1156	6601	A37WP6			
1157	7772	BLOCK1			
1160	4574	-6			
1161	6601	MOVE	/FILL BUFFER WITH PATTERN		
1162	6607	BLOCK1			
1163	7665	BLOCK1+6			
1164	5753	-113			
		JMP I FW376	/EXIT		
1165	0000	XSPF,			
1166	6040	OPEN	/SUB TO ISSUE SPF		
1167	5765	SPF			
1170	7402	JMP I XSPF	/EXIT		
		HLT	/SPF SKIPPED,		
1171	0000	XTPC,			
1172	6044	OPEN	/SUB TO ISSUE IPC		
1173	5771	TPC			
1174	7402	JMP I XTPC	/EXIT		
		HLT	/IPC SKIPPED,		
	1200	PAGE			
1200	0000	PAGE			
1201	4577	/PUNCH 70 (CODE 376) CHARACTERS SUBROUTINE			
1202	1211	PLTLR, 0			
1203	7672	SETLOC	/SET P70CTR TO -70		
1204	1377	P70CTR			
1205	4474	-106			
1206	2211	TAD (376)	/GET 376 CODE		
1207	5204	JMS I UPUNCH	/GO PUNCH IT		
1210	5600	ISZ P70CTR	/ALL CHARACTERS PUNCHED?		
1211	0000	JMP -3	/NO, REPEAT,		
		JMP I PLTLR	/YES, EXIT,		
		P70CTR, 0			

```

1212 0000 /PUNCH SYNC CHARACTER SUBROUTINE (RUBOUT)
1213 7240 PSYNC, 0
1214 4474 CLA CMA /SET AC TO 7777
1215 5612 JMS I UPUNCH /PUNCH A RUBOUT
JMP I PSYNC /EXIT,

/SYNC READER SUBROUTINE
RSYNC, 0 SETLOC /SET RSCTR TO -145
RSCTR -221
JMS RRDY /WAIT FOR READER NOT BUSY
CLA CMA /READER NOT BUSY,
DCA RBUSY /SET READER BUSY, INDICATOR
SETLOC /SET READER INTERRUPT
VCTR /SERVICE RETURN ADDRESS,
RSSERV /ENABLE INTERRUPT
ION /EXIT
JMP I RSYNC

RSCTR, 0 /READ
RSSERV, KRB /ADD MINUS RUBOUT
TAD [-377 /IS IT A RUBOUT?
SZA CLA /NO,
JMP +7 /YES, CLEAR READER BUSY,
DCA RBUSY
CLA CLL
TAD LINK
RAL
TAD AC
JMP I 0 /RESTORE LINK
ISZ RSCTR /RESTORE AC
JMP I UOUT /RETURN
HLT CLA /145 CHARACTER READ?
SETLOC /NO,
RSCTR /YES, NO SYNC,
-221 /SET RSCTR TO -145
JMP I UOUT /RETURN

INTSVC, DCA AC /SAVE AC
RAR /SAVE LINK
DCA LINK /PUNCH/PRINTER?
TSF /NO,
JMP +4 /YES, CLEAR FLAG,
TCF /CLEAR PFLAG
DCA PFLAG /RETURN
JMP OUT /READER/KYBD?
KSF /NO ERROR,
JMP +3 /GO SERVICE READER
JMP I +1 /UNEXPECTED INTERRUPT
VCTR, 0
OUT, HLT
CLA CLL
TAD LINK

```

1273 7004 RAL /RESTORE LINK
1274 1077 TAD AC /RESTORE AC,
1275 6001 ION /ENABLE INTERRUPT
1276 5400 JMP I 0 /RETURN

1277 0000 PSTUP, 0 /PUNCH SETUP
1300 4577 SETLOC /SET DATA ADDR
1301 1342 PADDR
1302 6577 BLOCKA
1303 4574 MOVE
1304 0101 BLKCNT
1305 1341 PCTR
1306 7777 -1
1307 5677 JMP I PSTUP /EXIT

1310 0000 PDCR, 0 /PUNCH DATA CHAR SUB.
1311 7200 CLA
1312 1742 TAD I PADDR
1313 2342 ISZ PADDR
1314 4474 JMS I UPUNCH
1315 5710 JMP I PDCR /GET DATA
/UPDATE PADDR,
/GO PUNCH/PRINT DATA
/EXIT

1316 0000 PBLK, 0 /PUNCH DATA BLOCK FULL SPEED
1317 4277 JMS PSTUP
1320 4310 JMS PDCR
1321 2341 ISZ PCTR
1322 5320 JMP I=2
1323 5716 JMP I PBLK /GO PUNCH CHARACTER
/ALL CHARS PUNCHED?
/NO, REPEAT
/YES, EXIT

1324 0000 PBLKR, 0 /PUNCH DATA BLOCK RANDOM STALLS.
1325 4277 JMS PSTUP /GO DO SET UP
1326 4776 JMS RGNB /GET A RANDOM NUMBER
1327 0111 AND DLYMSK /REMOVE EXCESS BITS
1330 7450 SNA /ZERO?
1331 5326 JMP I=3 /YES, GET ANOTHER NUMBER
1332 7041 CIA /NO, 2'S COMPLEMENT IT,
1333 3024 DCA DELAYM /PUT NUMBER IN DELAY
1334 4576 DELAY /DELAY,
1335 4310 JMS PDCR /GO PUNCH CHARACTER
1336 2341 ISZ PCTR /ALL CHARS PUNCHED?
1337 5326 JMP PBLKR+2 /NO, REPEAT
1340 5724 JMP I PBLKR /YES, EXIT,
1341 0000 PCTR, 0
1342 0000 PADDR, 0

1343 0000 RRDY, 0 /WAIT FOR RDR NOT BUSY SUB.
1344 7200 CLA
1345 1076 TAD RBUSY /FETCH RBUSY,
1346 7640 SZA CLA /READER BUSY?
1347 5345 JMP I=2 /YES, TRY AGAIN
1350 5743 JMP I RRDY /NO, EXIT

1351 0000 RSTUP, 0
1352 4343 JMS RRDY /WAIT FOR RDR NOT BUSY
1353 2076 ISZ RBUSY /SET RBUSY INDICATOR

```

1354 4577 SETLOC /SET DATA ADDR
1355 1416 RADDR
1356 6577 BLOCKA /SET DATA BLOCK LENGTH
1357 4574 MOVE
1360 0101 BLKCNT
1361 1417 RBCTR
1362 7777 -1
1363 3775 DCA ERRCTR /CLEAR ERROR COUNTER
1364 5751 JMP I RSTUP /EXIT;

/Routine TO SET KEYBOARD FLAG,
KFLAG, OPEN
1365 0000 UKCC
1366 4571 UKSF
1367 4572 UKSF
1370 5367 JMP I KFLAG
1371 5765 JMP I KFLAG /EXIT WITH KEYBOARD FLAG SET;

```

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```

1400 0000 /READ DATA BLOCK; FULL SPEED
1401 4777 /GO DO SETUP
1402 4577 /SET READER SERVICE
1403 1267 SETLOC
1404 1430 VCTR
1405 6001 RDRSRV
1406 5600 ION
1407 0000 JMP I RDBLK
1408 4777 /READ DATA BLOCK; RANDOM STALLS
1409 4577 /GO DO SETUP
1410 4577 /SET READER SERVICE
1411 1267 SETLOC
1412 1420 VCTR
1413 1420 RDRSRV
1414 6001 ION
1415 5607 JMP I RDBLK
1416 0000 RADDR;
1417 0000 RBCTR;

/READER SERVICE ROUTINES
RDRSRV, CLA
1420 7200 /GET A RANDOM NUMBER
1421 4776 JMS RGNA
1422 0111 AND DLYMSK
1423 7450 SNA
1424 5221 JMP I=3
1425 7041 CIA
1426 3102 DCA DELAYS
1427 4274 JMS DLMSR
1430 1616 RDRSRV, TAD I RADDR
1431 3235 DCA SB

```

/GET A RANDOM NUMBER
/REMOVE EXCESS BITS
/ZERO?
/YES, GET ANOTHER NUMBER
/NO, 2'S COMPLEMENT IT;
/STORE RANDOM NUMBER IN DELAYS;
/STALL;
/GET EXPECTED CHARACTER
/STORE AT SB


```

1514 1141 TAD I=6
1515 3346 DCA NTST
1516 1374 TAD (BAUTAB-1
1517 3010 DCA 10
1520 1022 TAD I 10
1521 1410 TAD I 10
1522 7650 SNA CLA
1523 5330 JMP I=5
1524 2346 ISZ NTST
1525 5320 JMP I=5
1526 7602 HLT CLA

1527 5326 JMP I=1
1530 1346 TAD NTST
1531 7040 CMA
1532 1373 TAD (BAUCON
1533 3061 DCA MIL1
1534 1461 TAD I MIL1
1535 3061 DCA MIL1
1536 5713 JMP I STBAUD

/GET USER DEFINED BAUD RATE.
/GET A RATE FROM TABLE.
/DO THEY MATCH?
/YES, GO SET DELAY ROUTINE.
/NO, KEEP A COUNT.
/NOW TRY NEXT ONE.
/ILLEGAL BAUD RATE: RESET BAUDRT AND
/RESTART PROGRAM AT 0200.
/NO CONTINUE ALLOWED.
/GET COUNT AND.
/COMPLEMENT IT.
/ADD TABLE ADDRESS
/
/
/EXIT,

```

/TABLE OF CONSTANTS FOR THE BAUD RATE.

```

1537 7767 BAUCON, 7767 /2400
1540 7754 7754 /1200
1541 7726 7726 /600
1542 7652 7652 /300
1543 7522 7522 /150
1544 7372 7372 /110
1545 0000 OPEN /FUTURE

```

/PUNCH TEST NORMAL TEST SEQUENCE ROUTINE

```

1546 0000 NTST, 0
1547 7200 CLA /CLEAR RBUSY
1550 3076 DCA RBUSY /SELECT PUNCH MODE
1551 1746 TAD I NTST
1552 3355 DCA NTSTA
1553 4772 JMS PLTLR /PUNCH LEADER
1554 4771 JMS PSYNC /PUNCH SYNC CHARACTER
1555 0000 NTSTA, 0
1556 4770 JMS RSYNC /SYNC READER
1557 4200 JMS ROBLK /READ DATA BLOCK
1560 4772 JMS PLTLR /PUNCH TRAILER
1561 4767 JMS RRDY /WAIT FOR RDR NOT BUSY
1562 5425 JMP I CHAIN /CHAIN

```

RM33A, TEXT /-----I@?/

```

1563 5555
1564 5555
1565 1100
1566 7700

```

```

1567 1343
1570 1216

```

PAGE

```

1600
1600 0000 /COMBINED TEST NORMAL TEST SEQUENCE
1601 7200 CNTST, 0
1602 3076 CLA
1603 4777 /CLEAR RBUSY
1604 4776 JMS PLTLR
1605 4775 JMS PSYNC
1606 4774 JMS PBLK
1607 4773 JMS RSYNC
1608 4772 JMS RDBLKR
1609 4771 JMS PBLKR
1610 4770 JMS RDBLK
1611 4769 JMS PLTLR
1612 4768 JMS RRDY
1613 4767 JMS RRDR
1614 5425 JMP I CHAIN
1615 0000 /PUNCH LEADER
1616 7200 /PUNCH SYNC CHARACTER
1617 3064 /PUNCH DATA BLOCK (NO STALLS)
1618 1615 /PUNCH DATA BLOCK (NO STALLS)
1619 3224 /PUNCH DATA BLOCK (STALLS)
1620 1615 /PUNCH DATA BLOCK (STALLS)
1621 3224 /PUNCH TRAILER
1622 2215 /WAIT FOR READER NOT BUSY
1623 4767 /CHAIN
1624 0000
1625 4227
1626 5615

```

```

1615 0000 /TYPE LINE OF 3 CHARACTERS (NO DELAY)
1616 7200 TYPLN3, 0
1617 3064 CLA
1618 1615 DCA STLD
1619 3224 TAD I TYPLN3
1620 1615 DCA +3
1621 3224 ISZ TYPLN3
1622 2215 JMS FBF3
1623 4767 0
1624 0000 JMS TYPLN
1625 4227 JMP I TYPLN3
1626 5615 /EXIT

```

```

1627 0000 /TYPE LINE OF ASCII PRINTABLE CHARACTERS
1628 0000 TYPLN, 0
1629 4555 CKSR37
1630 1140 TAD I11
1631 1137 TAD I=125
1632 3247 DCA TCTR
1633 4577 SETLOC
1634 1646 FETCH
1635 1646 BLOCKA
1636 6577 STALL
1637 4556 TAD I FETCH
1638 1646 JMS I UPUNCH
1639 4474 ISZ FETCH
1640 2246 ISZ TCTR
1641 2247 JMP TYPEA
1642 5237 JMP I TYPLN
1643 5627
1644 5627
1645 5627

```

```

/YES, SET CHARACTER
/GO PRINT CHARACTER
/SET UP FOR NEXT CHARACTER
/DONE?
/NO, REPEAT
/YES, EXIT,

```

1646	0000	FETCH,	0
1647	0000	TCTR,	0
1650	0000	ASCCN,	0
1651	1650		TAD I ASCCN
1652	3304		DCA WASC
1653	2250		ISZ ASCCN
1654	1650		TAD I ASCCN
1655	3305		DCA SASC
1656	2250		ISZ ASCCN
1657	1366		TAD (7700
1660	0704		AND I WASC
1661	7102		BSW CLL
1662	4271		JMS CNV
1663	2305		ISZ SASC
1664	1366		TAD (7700
1665	7040		CMA
1666	0704		AND I WASC
1667	4271		JMS CNV
1670	5650		JMP I ASCCN
1671	0000	CNV,	0
1672	3306		DCA ASCT
1673	1306		TAD ASCT
1674	7006		RTL
1675	7004		RAL
1676	0365		AND (707
1677	1306		TAD ASCT
1700	0365		AND (707
1701	1364		TAD (6060
1702	3705		DCA I SASC
1703	5671		JMP I CNV
1704	0000	WASC,	0
1705	0000	SASC,	0
1706	0000	ASCT,	0
1707	0000	SINPT,	OPEN
1710	7200		CLA
1711	3314		DCA SPT0
1712	3316		DCA SPIND
1713	5707		JMP I SINPT
1714	0000	SPT0,	OPEN
1715	0000	SPT1,	OPEN
1716	0000	SPIND,	OPEN
1717	0000	SGET,	OPEN
1720	7320		CLA STL
1721	2316		ISZ SPIND
1722	7340		CLA CMA CLL
1723	3316		DCA SPIND
1724	1314		TAD SPT0
1725	7420		SNL
1726	5331		JMP :+3
1727	7041		CIA
1730	7410		SKP
1731	7040		CMA
1732	3314		DCA SPT0

/SUB TO INITIALIZE SGCT SUB.

/ZERO SPT0
/ZERO SPIND
/EXIT

/"SPECIAL" BINARY COUNT
/PATTERN SUBROUTINE.

```

1733 1145      TAD C377
1734 0314      AND SPT0
1735 3315      DCA SPT1
1736 1315      TAD SPT1
1737 5717      JMP I SGET
                        /EXIT SGET SUB;

```

/SUBROUTINE TO INITIALIZE RANDOM NUMBER GENERATORS;

```

1740 0000      SETRND, OPEN
1741 1363      TAD C1233
1742 3762      DCA RP1A
1743 1363      TAD C1233
1744 3761      DCA RP1B
1745 1360      TAD C7622
1746 3757      DCA RP2A
1747 1360      TAD C7622
1750 3756      DCA RP2B
1751 5740      JMP I SETRND
                        /EXIT, AC=0

```

```

1756 0435
1757 0416
1760 7622
1761 0434
1762 0415
1763 1233
1764 6060
1765 0707
1766 7700
1767 1031
1770 1343
1771 1400
1772 1324
1773 1407
1774 1216
1775 1316
1776 1212
1777 1200
2000

```

PAGE

/ROUTINE TO SET CTRA EQUAL TO -7640 (-4000 DECIMAL);

```

2000 0000      S4000I, OPEN
2001 4577      SETLOC
2002 0062      CTRA
2003 0140      -7640
2004 5600      JMP I S4000I
                        /SET COUNT OF
                        /-4000 DECIMAL
                        /IN CTRA,
                        /EXIT, AC=0,

```

/ROUTINE TO SET DELAYM TO -310, (-200 DECIMAL);

```

2005 0000      S200I, OPEN
2006 4577      SETLOC
2007 0024      DELAYM
2010 7470      -310
2011 5605      JMP I S200I
                        /SET COUNT OF
                        /-200 DECIMAL
                        /IN DELAYM,
                        /EXIT WITH AC=0.

```

/ROUTINE TO SET CTRA EQUAL TO -144 (-100 DECIMAL);

2012	0000	S1001,	OPEN	
2013	4577	SETLOC		/SET COUNT OF
2014	0062	CTRA		/-100 DECIMAL
2015	7634	-144		/IN CTRA,
2016	5612	JMP I S1001		/EXIT, AC=0.

/ROUTINE TO DETERMINE DEVICE CAUSING UNEXPECTED INTERRUPT;

2017	0000	INTFND,	OPEN		
2020	7200	CLA		/KEYBOARD/READER?	
2021	6031	INTKSF,	KSF	/NO,	
2022	7410	SKP		/GO BOLT AND DISPLAY IOT	
2023	4276	JMS		TTY PRINTER/PUNCH?	
2024	6041	TSF		/NO,	
2025	7410	SKP		/GO BOLT AND DISPLAY IOT	
2026	4276	JMS		/H.S. READER?	
2027	6011	RSF		/NO,	
2030	7410	SKP		/HALT AND DISPLAY IOT	
2031	4276	JMS		/H.S. PUNCH?	
2032	6021	PSF		/NO,	
2033	7410	SKP		/HALT AND DISPLAY IOT	
2034	4276	JMS		/PT08/LT08 UNIT 1 IN?	
2035	6401	6401		/NO,	
2036	7410	SKP		/HALT AND DISPLAY IOT	
2037	4276	JMS		/PT08/LT08 UNIT 1 OUT?	
2040	6411	6411		/NO,	
2041	7410	SKP		/HALT AND DISPLAY IOT	
2042	4276	JMS		/PT08/LT08 UNIT 2 IN?	
2043	6421	6421		/NO,	
2044	7410	SKP		/HALT AND DISPLAY IOT	
2045	4276	JMS		/PT08/LT08 UNIT 2 OUT?	
2046	6431	6431		/NO,	
2047	7410	SKP		/HALT AND DISPLAY IOT	
2050	4276	JMS		/PT08/LT08 UNIT 3 IN?	
2051	6441	6441		/NO,	
2052	7410	SKP		/HALT AND DISPLAY IOT	
2053	4276	JMS		/PT08/LT08 UNIT 3 OUT?	
2054	6451	6451		/NO,	
2055	7410	SKP		/HALT AND DISPLAY IOT	
2056	4276	JMS		/PT08/LT08 UNIT 4 IN?	
2057	6461	6461		/NO,	
2060	7410	SKP		/HALT AND DISPLAY IOT	
2061	4276	JMS		/PT08/LT08 UNIT 4 OUT?	
2062	6471	6471		/NO,	
2063	7410	SKP		/HALT AND DISPLAY IOT	
2064	4276	JMS		/PT08/LT08 UNIT 5 OR DC02 IN?	
2065	6111	6111		/NO,	
2066	7410	SKP		/HALT AND DISPLAY IOT	
2067	4276	JMS		/PT08/LT08 UNIT 5 OR DC02 OUT?	
2070	6121	6121			

```

2071 5275 JMP I+4
2072 4276 JMS HLTD
2073 7777 7777
2074 7777 7777
2075 4276 JMS HLTD
2076 0000 OPEN
2077 1276 TAD HLTD
2100 1311 TAD M3
2101 3276 DCA HLTD
2102 1676 TAD I HLTD
2103 7402 HLT
2104 7001 IAC
2105 3306 DCA I+1
2106 0000 OPEN
2107 7200 CLA
2110 5617 JMP I INTFND
2111 7775 M3, -3 /EXIT

PUNCH, OPEN /SET PFLAG,
OUT0, TLS /PUNCH/PRINT,
CLA /GET C(PFLAG),
TAD PFLAG /FLAG RESET?
SNA CLA /YES
JMS OUT2 /NO, FLAG UP?
OUT1, TSF JMP I+4 /NO,
TCF /YES, CLEAR PRINTER FLAG,
DCA PFLAG /CLEAR PFLAG,
JMP I PUNCH /EXIT, AC=0,

M3, -3 /EXIT

```

```

2126 0000 /ROUTINE TO CONTROL THE CHARACTERS TO BE TYPED ON ALL TTYS,
2127 1726 TLCALI, OPEN
2130 3332 TAD I TLCALI
2131 4473 DCA I+2 /GET FIRST LETTER TO BE TYPED
2132 0000 JMS I UTPLN3 /SAVE IT,
2133 5425 OPEN /GO TYPE SAVED LETTER + NEXT 2;
JMP I CHAIN /FIRST LETTER TO BE TYPED,
/CHAIN

```

```

2134 0000 /ROUTINE TO CONTROL THE CHARACTER TO BE TYPED ON A "37",
2135 1734 TLC37I, OPEN
2136 3342 TAD I TLC37I
2137 4555 DCA I+4 /GET FIRST LETTER TO BE TYPED
2140 5425 CKSR37 /SAVE IT,
2141 4473 JMP I CHAIN /IS IT A "37"?
2142 0000 JMS I UTPLN3 /NO, CHAIN
2143 5425 OPEN /YES, GO TYPE LETTER + NEXT 2
JMP I CHAIN /FIRST LETTER TO BE TYPED,
/CHAIN

```

/CONTROL ROUTINE TO FILL A BUFFER WITH CHARACTERS;

```

2144 0000 FBFI, OPEN
2145 7300 CLA CLL
2146 1744 TAD I FBFI
2147 3351 DCA I+2
2150 4777 JMS FBFI
2151 0000 OPEN
2152 4776 JMS CNTST
/SAVE IT
/GO FILL A BUFFER
/WITH THIS +NEXT 2 CHAR
/GO TO COMBINED TEST SEQUENCE

```

/ROUTINE TO CONTROL TYPING A LINE WITHOUT STALLS
/AND THEN ONE WITH STALLS;

```

2153 0000 WOSWS, OPEN
2154 3064 DCA STLID
2155 4775 JMS TYPLN
2156 7240 CLA CMA
2157 3064 DCA STLID
2160 4775 JMS TYPLN
2161 5753 JMP I WOSWS
/ZERO STALL INDICATOR,
/TYPE LINE WITHOUT STALLS
/7777
/SET STALL INDICATOR
/TYPE LINE WITH STALLS
/EXIT,

```

/SUBROUTINE TO MARK TAB POSITIONS,

```

2162 0000 MTABP, OPEN
2163 3062 DCA CTRA
2164 4573 TYPE
2165 6300 TBMRK
2166 4573 TYPE
2167 6306 TBMRK1
2170 2062 ISZ CTRA
2171 5366 JMP I+3
2172 5762 JMP I MTABP
/MARK TAB POSITIONS
/EXIT,

```

```

2175 1627
2176 1600
2177 1031
2200

```

PAGE

/PROGRAM 0, BASIC TEST OF THE OUTPUT LOGIC;
/THE INSTRUCTIONS TESTED ARE:
/ SPF SET PRINTER FLAG,
/ TSF SKIP IF PRINTER FLAG IS SET,
/ TCF CLEAR PRINTER FLAG,
/ CAF CLEAR FLAGS, AC, LINK, AND ENABLE TTY INTERRUPT;
/ TPC CHECK THAT PRINTER FLAG WILL SET;
/ TLS CHECK THAT IT CLEARS PRINTER FLAG AND SETS PRINTER FLAG,

```

2200 4577 PRG0, SETLOC
2201 0023 KSTART
2202 2205 POTS0
2203 5604 JMP I I+1
2204 0236 SRSET
/SET KSTART TO INITIAL
/ROUTINE ADDRESS,
/GO START TEST

```

/TEST 0 CHECKS THE ABILITY OF

/SPF TO SET THE PRINTER FLAG, SET,
/TSF TO SKIP ON PRINTER FLAG, SET,
/CAF TO CLEAR PRINTER FLAG, AC, AND LINK,
/TCF TO CLEAR PRINTER FLAG,
/TSF TO NOT SKIP ON PRINTER FLAG EQUAL TO ZERO;

2205 0000
2206 2270
2207 4432

P0TS0, 0
P0TS1
JMS I S200

/SET DELAYM TO DELAY TWICE
/10 BIT TIMES FOR AN NON 110
/BAUD DEVICE AND TWICE 11 BIT
/TIMES FOR AN 110 BAUD DEVICE,
/SEE BIT TIME TABLE AT BEGINNING
/OF PROGRAM.

2210 4430 JMS I S100
2211 4560 P0TS0A, USPF
2212 4566 USPF
2213 5237 JMP P0E0A
2214 7360 CLA CMA CLL CML
2215 6007 P0TS0B, CAF
2216 4576 DELAY
2217 4566 USPF
2220 7410 SKP
2221 5244 JMP P0E0B
2222 7420 SNL
2223 7440 SZA
2224 5253 JMP P0E0C
2225 4560 USPF
2226 4566 USPF
2227 5237 JMP P0E0A
2230 4565 P0TS0E, UTCF
2231 4566 USPF
2232 7610 SKP CLA
2233 5262 JMP
2234 2062 ISZ
2235 5211 JMP P0TS0A
2236 5425 JMP I CHAIN

/SET UP TO DO TEST 100 TIMES,
/SET PRINTER FLAG
/FLAG SET?
/NO, SPF OR TSF FAILED
/AC AND LINK & 1
/YES, NOW CLEAR IT,
/GO DELAY
/FLAG SET?
/NO, CONTINUE TEST
/YES, CAF OR TSF FAILED
/LINK SET?
/NO, AC SET?
/YES, CAF FAILED TO CLEAR AC AND/OR LINK
/SET PRINTER FLAG
/PRINTER FLAG SET?
/NO, SPF OR TSF FAILED
/YES, CLEAR PRINTER FLAG
/PRINTER FLAG SET?
/NO, OK,
/YES, TCF FAILED TO CLEAR PRINTER FLAG,
/DONE TEST 100 TIMES?
/NO, REPEAT TEST
/YES, CHAIN NOW

/ERROR HLTS FOR P0TS0.

2237 7402 P0E0A, HLT
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
USPF
2240 4560
2241 4566
2242 5240
2243 5240 JMP
P0E0A+1
P0E0A+1

/SPF FAILED TO SET PRINTER FLAG
/OR TSF FAILED TO SKIP,
/SET PRINTER FLAG
/IS IT SET?
/NO, REPEAT,
/YES, REPEAT,

2244 7402 P0E0B, HLT
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
USPF
2245 4560
2246 6007 CAF

/CAF FAILED TO CLEAR PRINTER FLAG
/OR TSF SKIPPED,
/SET PRINTER FLAG
/CLEAR FLAGS

```

2247 4576      DELAY
2250 4566      /UTSF
2251 5245      JMP      P0E0B+1
2252 5245      JMP      P0E0B+1
2253 7402      P0E0C;
                /SCOPE LOOP, PRESS CONTINUE TO ENTER,
2254 7360      CLA CMA CLL CML
2255 6007      CAF
2256 7420      SNL
2257 7440      SZA
2260 5254      JMP      P0E0C+1
2261 5254      JMP      P0E0C+1
2262 7402      P0E0E; HLT
                /SCOPE LOOP; PRESS CONTINUE TO ENTER,
2263 4560      /USPF
2264 4565      /UTCFF
2265 4566      /UTSF
2266 5263      JMP      P0E0E+1
2267 5263      JMP      P0E0E+1
                /CAF FAILED TO CLEAR AC AND/OR LINK
                /LINK AND AC SET
                /CLEAR
                /LINK SET?
                /AC CLEAR
                /AC OR LINK SET, REPEAT
                /REPEAT,
                /TCF FAILED TO CLEAR PRINTER FLAG
                /OR TSF SKIPPED,
                /SET PRINTER FLAG
                /CLEAR PRINTER FLAG
                /FLAG SET?
                /NO, REPEAT;
                /YES, REPEAT;

```

/THIS ROUTINE CHECKS THAT NO OTHER DEVICE CAN CAUSE AN INTERRUPT
/AND THEN CHECKS THAT THE PUNCH/PRINTER FLAG CAN CAUSE AN INTERRUPT;

```

2270 0001      P0TS1; 1
2271 2327      P0TS2
2272 4577      SETLOC
2273 0002      2
2274 2304      P0E1A
2275 6007      CAF
2276 4560      /USPF
2277 4565      /UTCFF
2300 6001      /ION
2301 7000      /NOP
2302 6002      /IOF
2303 5306      /JMP
2304 4777;     /JMS
2305 5275      /INTFND
2306 4431      /P0TS1A
2307 4577      /JMS I S4000
2310 0002      SETLOC
2311 2324      2
2312 4560      P0TS1C
2313 4560      /USPF
2314 6001      P0TS1B; /ION
2315 7402      P0E1B; /NOP
                /HLT
                /SET PRINTER FLAG
                /ENABLE INTERRUPT
                /NO INTERRUPT
                /PRINTER FLAG FAILED TO INTERRUPT
                /OR INTERRUPT MALFUNCTION
                /SET INTERRUPT RETURN
                / TO P0TS1C-1
                /SET INTERRUPT RETURN
                / TO P0E1A;
                /ATTEMPT TO CLEAR ALL FLAGS
                /SET PRINTER FLAG
                /CLEAR PRINTER FLAG
                /ENABLE INTERRUPT
                /
                /DISABLE INTERRUPT
                /UNEXPECTED INTERRUPT
                /TRY AGAIN
                /SET UP TO DO TEST 4000 TIMES,
                /SET INTERRUPT RETURN
                / TO P0TS1C
                /SET PRINTER FLAG
                /ENABLE INTERRUPT
                /NO INTERRUPT
                /PRINTER FLAG FAILED TO INTERRUPT
                /OR INTERRUPT MALFUNCTION
                /SET INTERRUPT RETURN
                / TO P0TS1C-1
                /SETLOC
                /2
                /P0TS1C-1
                /SETLOC
                /2
                /P0TS1C-1

```



```

2404 7300 POTS2G, CLA CLL
2405 4561 UKIE
2406 6007 CAF
2407 4560 USPF
2410 4557 USPI
2411 5274 JMP P0E2G
2412 2062 ISZ CTRA
2413 5777 JMP POTS2A
2414 5425 JMP I CHAIN
/AC + LINK = 0
/DISABLE TTY INTERRUPT ENABLE F,F;
/CLEAR EVERYTHING AND ENABLE TTY INTERRUPT F,F;
/SET PRINTER FLAG;
/SKIP IF INTERRUPT REQUEST
/CAF FAILED TO ENABLE TTY INTERRUPT ENABLE F,F;
/DONE 4000 TIMES?
/NO, REPEAT TEST
/CHAIN

```

/ERROR HLTS FOR POTS2;

```

2415 7402 P0E2A, HLT
/SCOPE LOOP, PRESS CONTINUE TO ENTER;
SETLOC
2
P0E2A+1
CAF
2421 6007 UKIE
2422 4561 USPF
2423 4560 ION
2424 6001 NOP
2425 7000 JMP P0E2A+1
2426 5216 /REPEAT TEST;

```

```

2427 7602 P0E2B, HLT CLA
/SCOPE LOOP, PRESS CONTINUE TO ENTER;
UKIE
2430 4561 USPF
2431 4560 USPI
2432 4557 JMP P0E2B+1
2433 5230 JMP P0E2B+1
2434 5230 /REPEAT;

```

```

2435 7602 P0E2C, HLT CLA
/SCOPE LOOP, PRESS CONTINUE TO ENTER;
UKIE
2436 4561 USPF
2437 4560 SRQ
2440 6003 JMP P0E2C+1
2441 5236 JMP P0E2C+1
2442 5236 /REPEAT;

```

```

2443 7402 P0E2D, HLT
/SCOPE LOOP, PRESS CONTINUE TO ENTER;
SETLOC
2
P0E2D+4
UKIE
2446 2447 CLA IAC
2447 4561 UKIE
2450 7201 USPF
2451 4561 ION
2452 4560 NOP
2453 6001
2454 7000
/KIE FAILED TO ENABLE TTY INTERRUPT F,F;
/ENTER;
/SET INTERRUPT RETURN
/TO P0E2D+4
/DISABLE TTY
/ACII = 1
/ENABLE TTY
/SET PRINTER FLAG
/TURN INTERRUPT ON

```

```

2455 5247      JMP      P0E2D+4      /REPEAT
2456 7402      P0E2E; HLT          /SPI FAILED TO SKIP,
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
CLA IAC      /AC11 = 1
2457 7201      UKIE          /ENABLE TTY
2460 4561      USPF          /SET PRINTER FLAG
2461 4560      USPI          /SKIP IF INTERRUPT REQUEST
2462 4557      JMP          P0E2E+1  /REPEAT,
2463 5257      JMP          P0E2E+1  /REPEAT,
2464 5257

2465 7402      P0E2F; HLT          /SRQ FAILED TO SKIP,
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
CLA IAC      /AC11 = 1
2466 7201      UKIE          /ENABLE TTY
2467 4561      USPF          /SET PRINTER FLAG
2470 4560      SRQ          /SKIP IF INTERRUPT REQUEST
2471 6003      JMP          P0E2F+1  /REPEAT,
2472 5266      JMP          P0E2F+1  /REPEAT,
2473 5266

2474 7402      P0E2G; HLT          /CAF FAILED TO ENABLE TTY INTERRUPT
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
CLA CLL      /CLEAR
2475 7300      UKIE          /DISABLE TTY,
2476 4561      CAF          /ENABLE TTY INTERRUPT ENABLE F.F.,
2477 6007      USPF          /SET PRINTER FLAG
2500 4560      USPI          /TTY INTERRUPT REQUEST?
2501 4557      JMP          P0E2G+1  /NO, REPEAT,
2502 5275      JMP          P0E2G+1  /YES, REPEAT,
2503 5275

2504 0003      POTS3; 3
2505 2544      POTS4
2506 4430      JMS I S100
2507 4432      JMS I S200

/TEST 3 CHECKS THE ABILITY OF:
/TPC TO SET THE PRINTER FLAG,
/UTS TO CLEAR PRINTER FLAG,
/UTS TO SET PRINTER FLAG,
POTS3; 3
POTS4
JMS I S100
JMS I S200

/SET UP TO DO TEST 100 TIMES,
/SET DELAY TO DELAY TWICE
/10 BIT TIMES FOR AN NON 110
/BAUD DEVICE AND TWICE 11 BIT
/TIMES FOR AN 110 BAUD DEVICE,
/SEE BIT TIME TABLE AT BEGINNING
/OF PROGRAM,
/CLEAR PRINTER FLAG
/PRINT
/DELAY TWICE MAX TIME
/FLAG SET, IT SHOULD BE,
/FLAG NOT SET,
/CLEAR + SET PRINTER FLAG,
/FLAG SET?
/NO, OK

2510 4565      POTS3A; UTCF
2511 4564      UTPC
2512 4576      DELAY
2513 4566      UTSF
2514 5327      JMP P0E3A
2515 4563      POTS3B; UTLS
2516 4560      UTSF
2517 7610      SKP CLA

```

ADDRESS	INSTR	OPERAND	COMMENT
2520	JMP	P0E3B	/YES
2521	DELAY		/DELAY TWICE BAUD RATE;
2522	UTSF		/FLAG SET?
2523	JMP	P0E3C	/NO
2524	ISZ	CYRA	/YES, DONE 100 TIMES
2525	JMP	P0TS3A	/NO, DO TEST AGAIN
2526	JMP	1	/EXIT,
2527	HLT	CLA	/TPC FAILED TO SET PRINTER FLAG,
2530	LOOP,		/SCOPE
2531	UTCF		/CLEAR PRINTER FLAG
2532	UTPC		/SET FLAG BY BEGINNING OF 10TH BIT
2533	DELAY		/WAIT
2534	JMP	1=3	/REPEAT,
2535	HLT	CLA	/TLS FAILED TO CLEAR PRINTER FLAG,
2536	LOOP,		/SCOPE
2537	USPF		/SET PRINTER FLAG
2538	UTLS		/CLEAR PRINTER FLAG AT TP3;
2539	JMP	1=2	
2540	HLT	CLA	/TLS FAILED TO SET PRINTER FLAG,
2541	LOOP,		/SCOPE
2542	UTLS		/PRINT
2543	DELAY		/DELAY TWICE BAUD RATE;
2544	JMP	P0E3C+1	/REPEAT,
2545	PUNCH,	4	/PUNCH, 4
2546	P0TS4,		
2547	JMS	1 S100	/SET UP TO DO TEST 100 TIMES,
2548	SETLOC		/SET DELAYM
2549	DELAYM		/TO -81 (DEC)
2550	-121		
2551	TAD	BAUDRT	/GET BAUD RATE;
2552	TAD	1=110	/ADD A -110 TO IT,
2553	SNA	CLA	/IS IT 110 BAUD WE'RE WORKING WITH?
2554	JMP	1=3	/YES, LEAVE DELAYM ALONE;
2555	TAD	1=130	/NO, CHANGE DELAYM TO -88 (DEC);
2556	DCA	DELAYM	/DELAYM NOW SET TO -88 DECIMAL;
2557	UTLS		/PRINT
2558	DELAY		/DELAY A LITTLE LESS THAN 9 BIT TIMES
2559	UTSF		/FLAG SET
2560	SKP		/NO, OK
2561	JMP	P0E4A	/YES,
2562	SETLOC		/SET DELAYM
2563	DELAYM		/TO -7 (DEC)
2564	-7		
2565	DELAY		/DELAY SO WE'RE PAST THE 9.5 BIT TIME POINT
2566	UTSF		/FLAG SET?
2567	JMP	P0E4B	/NO
2568	JMP	P0TS4C	/CROSS PAGE
2569	2574	2600	

PAGE

2575 2614
2576 2607
2577 2332
2600

2600 4577
2601 0024
2602 7761
2603 4576
2604 2062
2605 5777
2606 5425

/DELAY SO WE'RE PAST THE END.
/DONE 100 TIMES?
/NO, DO TEST AGAIN
/CHAIN

CTRA
POTS4A
CHAIN

2607 7602 POTS4A: HLT CLA

/PROCESSOR TIMING TOO SLOW OR FLAG
/SETTING TOO SOON, (IS THE SLOW CYCLE
/JUMPER REMOVED FROM THE PROCESSOR
/TIMING MODULE? IS THE WRONG BAUD RATE SELECTED?

/SCOPE LOOP: PRESS CONTINUE TO ENTER,

UTLS
UTSF
JMP
JMP
I=1
I=3

/START PRINTER
/FLAG SET
/NO, CHECK AGAIN
/REPEAT

2610 4563
2611 4566
2612 5211
2613 5210

2614 7602
2615 5210

POTS4B: HLT CLA
/SCOPE LOOP: PRESS CONTINUE TO ENTER,
JMP POTS4A+1
/FLAG NOT SETTING IN REQUIRED TIME;
/GO TO SCOPE LOOP;

/TEST TO CHECK THAT THE PUNCH/PRINTER FLAG SETS AT THE PROPER TIME;

2616 0005
2617 2660
2620 4430
2621 4563
2622 4566
2623 5222
2624 4577
2625 0024
2626 7636
2627 1022
2630 7104
2631 7710
2632 2024
2633 4563
2634 4576
2635 4566
2636 7610
2637 5254
2640 4577
2641 0024
2642 7774
2643 4576
2644 4566
2645 5256

POTS5: 5
POTS6
JMS I S100
UTLS
UTSF
JMP I=1
SETLOC
DELAY
-142
TAD
RAL CLL
SPA CLA
ISZ
UTLS
DELAY
UTSF
SKP CLA
JMP POTS5A

POTS5A: SETLOC
DELAY
-142
TAD
RAL CLL
SPA CLA
ISZ
UTLS
DELAY
UTSF
SKP CLA
JMP POTS5B

BAUDRT
/GET BAUD RATE;
/MOVE INTO POSITION TO DETERMINE IF 2400;
/IS IT 2400?
/YES, INCREMENT DELAYM SO AS TO DELAY LESS;
/PRINT
/DELAY
/FLAG SET?
/NO, OK,
/YES, ERROR,
/SET DELAYM TO
/4 DECIMAL,
/DELAY
/FLAG NOW SET?
/NO, ERROR,

POTS5B: SETLOC
DELAY
-4
TAD
RAL CLL
SPA CLA
ISZ
UTLS
DELAY
UTSF
SKP CLA
JMP POTS5B

2646	4576	DELAY	
2647	4576	DELAY	
2650	4576	DELAY	
2651	2062	ISZ CTRA	/TEST DONE?
2652	5221	JMP P0TS5A-3	/NO, REPEAT,
2653	5425	JMP I CHAIN	/YES, CHAIN,
2654	7402	P0E5A, HLT	/FLAG SETTING TO SOON,
2655	5210	/SCOPE LOOP, PRESS	CONTINUE TO ENTER,
		JMP P0E4A+1	
2656	7402	P0E5R, HLT	/FLAG NOT SETTING SOON ENOUGH,
2657	5255	/SCOPE LOOP, PRESS	CONTINUE TO ENTER,
		JMP P0E5A+1	
/TEST OF GTF, TEST IS DONE			
/4000 TIMES,			
2660	0006	P0T6, 6	
2661	2736	P0T7	
2662	4431	JMS I 94000	/SET UP TO DO TEST 4000 TIMES;
2663	6007	CAF	/CLEAR ALL FLAGS, AC, LINK, AND ENABLE TTY INTERRUPT;
2664	7040	CMA	
2665	6004	GTF	/GET INTERRUPT FLAGS
2666	0335	AND K5200	/MASK,
2667	7440	SZA	
2670	7402	HLT	/GTF FAILED,
2671	7360	CLA CMA CLL CML	/SET LINK AND AC,
2672	6004	GTF	/GET INTERRUPT FLAGS; (AC SHOULD EQUAL 4000);
2673	0335	AND K5200	/MASK,
2674	7420	SNL	
2675	7402	HLT	/GTF CLEARED LINK,
2676	7104	CLL RAL	/(AC SHOULD EQUAL ZERO, LINK SHOULD EQUAL 1);
2677	7430	SZL	
2700	7440	SZA	
2701	7402	HLT	/GTF DID NOT GET LINK,
2702	6007	CAF	/CLEAR ALL FLAGS, AC, LINK, AND ENABLE TTY INTERRUPT;
2703	4560	USPF	/SET PRINTER FLAG,
2704	6004	GTF	/GET INTERRUPT FLAGS;
2705	0335	AND K5200	/MASK,
2706	7006	RTL	/PUT INTERRUPT BUS - (AC SHOULD EQUAL 1000)
2707	7004	RAL	/-FLAG INTO LINK, (AC SHOULD EQUAL ZERO);
2710	7430	SZL	
2711	7440	SZA	/IS LINK 1?
2712	7402	HLT	/IS AC ZERO?
2713	4577	SETLOC	/GTF FAILED TO GET INTERRUPT BUS,
2714	0002	2	/SET INTERRUPT RETURN LOCATION
2715	2725	P0T6F	/TO P0T6F,
2716	6007	CAF	/CLEAR ALL FLAGS;
2717	6001	ION	/TURN INTERRUPT ON
2720	6004	GTF	/GET INTERRUPT FLAGS;
2721	0335	AND K5200	

```

2722 4560      /SET PRINTER FLAG,
2723 7000      /INTERRUPT),
2724 7402      /GTF CLEARED ION,
2725 7102      /PUT ION = (AC SHOULD EQUAL 0002);
2726 7012      /FLAG INTO LINK, (AC SHOULD EQUAL 0000).
2727 7430      /LINK 1?
2730 7440      /AC ZERO?
2731 7402      /GTF FAILED TO GET ION,
2732 2062      /TEST DONE?
2733 5263      /NO, REPEAT,
2734 5425      JMP I CHAIN
2735 5200      K5200, 5200

```

/TEST OF RTF, TEST IS DONE
/4000 TIMES,

```

2736 0007      /SET UP TO DO TEST 4000 TIMES;
2737 7777      /SET INTERRUPT RETURN
2740 4431      /TO P077C+3,
2741 4577      /AC EQUALS ZERO, LINK EQUALS 1;
2742 0002      /RESTORE FLAGS,
2743 2760      /LINK SET?
2744 7320      /AC ZERO?
2745 6005      /RTF FAILED TO RESTORE LINK,
2746 7420      /AC EQUALS 4000
2747 7440      /RESTORE FLAGS, (LINK),
2750 7402      /LINK RESTORED?
2751 7330      /RTF FAILED TO RESTORE LINK,
2752 6005      /SET PRINTER FLAG,
2753 7420      /INTERRUPT),
2754 7402      /RTF DID NOT SET ION,
2755 4560      /TEST DONE?
2756 7000      /NO, REPEAT,
2757 7402      JMP P077A
2760 2062      /NO, REPEAT,
2761 5344      JMP P077A
2762 5425      JMP I CHAIN

```

2777 2547
3000 PAGE

/PROGRAM 1, LOOP AROUND INPUT TEST, OUTPUT MUST
/BE CONNECTED TO INPUT;
/PROGRAM CHECKS INPUT AND OUTPUT IOT'S, INTERRUPT AND TIMING,

```

3000 4577      PRG1, SETLOC
3001 0023      KSTART
3002 3005      P1TS0

```

```

3003 5604      JMP I 1-1
3004 0236      SRSET

/ISSUE KCC WITH AC=7777, AC SHOULD GO TO 0.
/AC NOT 0 INDICATES KCC FAILURE, TEST IS
/DONE 4000 TIMES.

3005 0000      P1TS0, 0
3006 3010      P1TS1
3007 4777      JMS P2TS0A

/ISSUE TLS AND THEN KCC, WAIT TWICE 10 OR 11 BIT TIMES
/(SEE TABLE AT BEGINNING OF PROGRAM) FOR FLAG TO SET.
/SKIP ON FLAG, FAILURE TO SKIP INDICATES THE THE
/FLAG IS NOT SET, OR KSF FAILURE, TEST IS DONE 100
/TIMES.

P1TS1, 1
3010 0001      P1TS2
3011 3034      JMS I S100
3012 4430      JMS I S200
3013 4432

/SET UP TO DO TEST 100 TIMES.
/SET DELAYM TO DELAY TWICE
/10 BIT TIMES FOR AN NON 110
/BAUD DEVICE AND TWICE 11 BIT
/TIMES FOR AN 110 BAUD DEVICE.
/SEE BIT TIME TABLE AT BEGINNING
/OF PROGRAM.

P1TS1A, UKCC
3014 4571      UTLS
3015 4563      DELAY
3016 4576      UKSF
3017 4572      JMP P1E1A
3020 5225      ISZ CTRA
3021 2062      JMP P1TS1A
3022 5214      CAF
3023 6007      JMP I CHAIN
3024 5425

/CLER AC AND KBRD FLFLAG.
/SEND,
/DELAY TWICE 10 OR 11 BIT TIMES.
/FLAG SET?
/NO.
/YES, TEST DONE 100 TIMES?
/NO, REPEAT,
/CLER
/CHAIN.

P1E1A, HLT CLA
3025 7602      /SCOPE LOOP, PRESS CONTINUE TO ENTER,
3026 4571      UKCC
3027 4563      UTLS
3030 4576      DELAY
3031 4572      UKSF
3032 5226      JMP I=4
3033 5226      JMP I=5

/DELAY TWICE 10 OR 11 BIT TIMES
/FLAG SET?
/NO, REPEAT
/YES, REPEAT,

/ISSUE TLS AND THEN KCC, WAIT TWICE 10 OR 11 BIT TIMES
/(SEE TABLE AT BEGINNING OF PROGRAM) FOR FLAG TO SET.
/SKIP ON FLAG 4000 TIMES TO VERIFY CONSISTENT SKIPPING.

P1TS2, 2
3034 0002      P1TS3
3035 3066      JMS I S200
3036 4432

/SET DELAYM TO DELAY TWICE
/10 BIT TIMES FOR AN NON 110
/BAUD DEVICE AND TWICE 11 BIT

```

```

3037 4431      JMS I S4000
3040 4571      P1TS2A, UKCC
3041 4563      UTLS
3042 4576      DELAY
3043 4565      UTCF
3044 4572      UKSF
3045 5253      JMP P1E2A
3046 4572      UKSF
3047 5262      JMP P1E2B
3050 2062      ISZ CTRA
3051 5246      JMP P1TS2B
3052 5425      JMP I CHAIN

      /TIMES FOR AN 110 BAUD DEVICE,
      /SEE BIT TIME TABLE AT BEGINNING
      /OF PROGRAM,

      /SET UP TO DO TEST 4200 TIMES,
      /CLEAR AC AND KBRD FLAG,
      /SEND,
      /DELAY TWICE 10 OR 11 BIT TIMES,
      /CLEAR TELEPRINTER FLAG,
      /KEYBOARD FLAG SET,
      /NO,
      /YES, KEYBOARD FLAG SET?
      /NO,
      /YES, DONE 4000 TIMES?
      /NO, REPEAT
      /CHAIN

P1E2A, HLT CLA      /FLAG NOT SET OR KSF FAILED TO SKIP,
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
UKCC
UTLS
UKSF
DELAY
JMP
JMP
      :=4
      :=1

P1E2B, HLT CLA      /KSF FAILED TO SKIP,
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
UKSF
JMP
JMP
      :=1
      :=2

/ISSUE TLS AND THEN KCC, WAIT TWICE MAXIMUM BIT RATE FOR
/FLAG TO SET, RESET FLAG (TLS AND THEN KCC) AND SKIP ON FLAG
/500 TIMES TO VERIFY NO SKIP OCCURS WITH FLAG = 0,

P1TS3, 3
P1TS4
SETLOG
CTRA
-764
JMS I S200

      /SET COUNT OF
      /=500 (DEC)
      /IN CTRA,
      /SET DELAY TO DELAY TWICE
      /10 BIT TIMES FOR AN NON 110
      /BAUD DEVICE AND TWICE 11 BIT
      /TIMES FOR AN 110 BAUD DEVICE,
      /SEE BIT TIME TABLE AT BEGINNING
      /OF PROGRAM,
      /CLEAR AC AND KBRD FLAG,
      /SEND,
      /DELAY TWICE 10 OR 11 BIT TIMES
      /FLAG SET,

P1TS3A, UKCC
UTLS
DELAY
UKSF

```

```

3100 5313 JMP P1E3A /NO,
3101 4571 UKCC /CLEAR AND AND KBRD FLAG,
3102 4563 UTLS /YES, SEND DATA,
3103 4572 UKSF /FLAG SET
3104 5306 JMP ,+2 /NO, OK
3105 5322 JMP P1E3B /YES,
3106 4566 UTSF /PRINTER FLAG SET?
3107 5306 JMP ,+1 /NO, WAIT TO CONTINUE TEST,
3110 2062 ISZ CTRA /DONE 500 TIMES?
3111 5301 JMP ,+10 /NO REPEAT TEST
3112 5425 JMP ,CHAIN,

3113 7602 P1E3A, HLT CLA /FLAG NOT SET OR KSF FAILED,
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
UTLS /SEND
UKCC /CLEAR AC AND KBRD FLAG
DELAY
UKSF
JMP ,+4
JMP ,+1

3122 7602 P1E3B, HLT CLA /KSF SKIPPED ON NO FLAG,
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
UTLS
JMS I KBFLAG
JMP P1E3B+1

3123 4563
3124 4426
3125 5323

```

/THIS ROUTINE CHECKS THAT NO OTHER DEVICE CAN CAUSE AN INTERRUPT;
/AND THEN CHECKS THAT THE READER FLAG IS CAPABLE OF INTERRUPTING;

```

3126 0004 P1TS4, 4
3127 3200 P1TS5
3130 4577 SETLOC
3131 0002 2
3132 3143 P1E4A
3133 4563 P1TS4A, UTLS
3134 4426 JMS I KBFLAG
3135 4565 UTSF
3136 4571 UKCC
3137 6001 ION
3140 7000 NOP
3141 6002 IOF
3142 5345 JMP ,+3
3143 4776 P1E4A, JMS INTFND
3144 5333 JMP P1TS4A
3145 4577 SETLOC
3146 0062 CTRA
3147 6030 ,+1750
3150 4577 SETLOC
3151 0002 2
3152 3167 P1TS4C
3153 4563 UTLS
3154 4426 JMS I KBFLAG
3155 4565 UTSF
3156 6001 ION

/SET INTERRUPT RETURN
/TO P1E4A
/SEND
/CLEAR PRINTER FLAG,
/CLEAR READER FLAG
/TURN INTERRUPT ON,
/TURN INTERRUPT OFF,
/SKIP OVER,
/UNEXPECTED INTERRUPT,
/TRY AGAIN,
/SET COUNT OF
/+1000 (DEC)
/IN CTRA,
/SET INTERRUPT RETURN
/SEND
/CLEAR PRINTER FLAG,
/INTERRUPT ON,

```

```

3157 7000 NOP
3160 7402 HLT
3161 4577 SETLOC
3162 0002 2
3163 3166 P1TS4C=1
3164 6001 /SCOPE LOOP, PRESS CONTINUE TO ENTER;
3165 7000 ION
3166 5364 NOP
3167 2062 JMP ,=2
3170 5353 P1TS4C, ISZ CTRA
3171 6007 JMP P1TS4B
3172 5425 CAF
3176 2017 JMP I CHAIN
3177 3513
3200 PAGE

```

```

/TEST 5 CHECKS THE ABILITY OF:
/SRQ TO SKIP ON AN INTERRUPT REQUEST,
/SPI TO SKIP ON A TTY INTERRUPT REQUEST,
/CAF TO CLEAR KBRD/READER FLAG,
/SRQ TO NOT SKIP ON NO INTERRUPT REQUEST,
/SPI TO NOT SKIP ON NO TTY INTERRUPT REQUEST.

P1TS5, 5
3200 0005 P1TS6
3201 3271 JMS I S100
3202 4430 CAF
3203 6007 P1TS5A, UTLS
3204 4563 JMS I KBFLAG
3205 4426 UTCF
3206 4565 SRQ P1E5A
3207 6003 JMP P1E5A
3210 5230 P1TS5B, USPI
3211 4557 JMP P1E5B
3212 5235 P1TS5C, CAF
3213 6007 UKSF
3214 4572 SKP CLA
3215 7610 JMP P1E5C
3216 5242 SRQ CLA
3217 6003 SRQ P1E5D
3220 7610 SKP CLA
3221 5257 JMP P1E5D
3222 4557 P1TS5E, USPI
3223 7610 SKP CLA
3224 5264 JMP P1E5E
3225 2062 ISZ CTRA
3226 5204 JMP P1TS5A
3227 5425 JMP I CHAIN
3230 7602 P1E5A, HLT CLA
/SRQ FAILED TO SKIP ON KBRD, FLAG;

```

```

3231 4250
3232 6003
3233 5231
3234 5233

/SCOPE LOOP, PRESS CONTINUE TO ENTER,
JMS PIE5
SRQ
JMP I:=2
JMP I:=1

```

```

3235 7602      PIE5B; HLT CLA      /SPI FAILED TO SKIP ON KBRD FLAG,
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
3236 4250      JMS PIE5
3237 4557      USPI
3240 5236      JMP I:=2
3241 5240      JMP I:=1

```

```

3242 7602      PIE5C; HLT CLA      /CAF FAILED TO CLEAR KBRD FLAG;
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
3243 4250      JMS PIE5
3244 6007      CAF
3245 4566      UTSF
3246 5243      JMP PIE5C+1
3247 5243      JMP PIE5C+1

```

```

3250 0000      PIE5; OPEN          /ROUTINE TO SET KBRD FLAG;
3251 7201      CLA IAC
3252 4561      UKIE
3253 4563      UTLS
3254 4426      JMS I KBFLAG
3255 4565      UTCF
3256 5650      JMP I PIE5      /EXIT

```

```

3257 7602      PIE5D; HLT CLA      /SRQ SKIPPED WITH NO FLAG;
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
3260 6007      CAF
3261 6003      SRQ
3262 5260      JMP PIE5D+1
3263 5260      JMP PIE5D+1

```

```

3264 7602      PIE5E; HLT CLA      /SPI SKIPPED WITH NO FLAG;
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
3265 6007      CAF
3266 4557      USPI
3267 5265      JMP PIE5E+1
3270 5265      JMP PIE5E+1

```

```

/READER TIMING TEST, CHECKS THAT READER FLAG IS = 1 NO
/LATER THAN THE TIME FOR THE FLAG TO SET;

```

```

3271 0006      PITS6; 6
3272 3314      PITS7
3273 4430      JMS I S100
3274 4577      SETLOC
3275 0024      DELAYM

/SET UP TO DO TEST 100 TIMES;
/SET DELAYM
/TO -103 DECIMAL

```

```

3276 7631          M147
3277 4563          P1TS6A, UTLS
3300 4571          /SEND
3301 4576          /RECEIVE
3302 4565          /DELAY 10=11 BIT TIMES
3303 4572          /CLEAR TELEPRINTER FLAG
3304 5310          /KBRD FLAG SET?
3305 2062          /FLAG NOT SET
3306 5277          /DONE 100 TIMES YET?
3307 5425          /CHAIN,

3310 7602          P1E6A, HLT CLA /FLAG NOT SETTING IN REQUIRED TIME;
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
UTLS
JMS I KBFLAG
JMP P1E6A+1

/TEST OF KEYBOARD AND PUNCH BUFFER USING
/KRS AND KCC TO RECEIVE AND TPC AND TCF
/TO SEND, A SPECIAL BINARY COUNT PATTERN
/IS USED.

3314 0007          P1TS7, 7
3315 3325          P1T10
3316 4577          SETLOC
3317 0062          CTRA
3320 7000          -1000
3321 4777          JMS SINPT
3322 4776          JMS SGET
3323 4337          JMS TRDATA
3324 5322          JMP P1TS7A

/SET COUNT OF
/512 (DEC)
/IN CTRA,
/INITIALIZE SPECIAL BIN COUNT,
/GET A NUMBER
/TRANSFER DATA AND CHECK,
/REPEAT

3325 0010          P1T10, 10
3326 3400          P1T11
3327 4577          SETLOC
3330 0062          CTRA
3331 7000          -1000
3332 4775          JMS SETRND
3333 4774          JMS RGNB
3334 0145          AND I397
3335 4337          JMS TRDATA
3336 5333          JMP P1T10A

/SET COUNT OF
/512 (DEC)
/IN CTRA,
/INITIALIZE RANDOM NUMBER GENERATOR,
/GET A RANDOM NUMBER,
/MASK,
/TRANSFER DATA AND CHECK,
/REPEAT

/SUBROUTINE USED BY P1TS7 AND P1T10

3337 0000          TRDATA, OPEN
3340 3346          DCA HOLD1
3341 1346          TAD HOLD1
3342 7421          MQL
3343 7701          ACL
3344 4353          JMS SNOREC

/STORE GOOD DATA IN MQ,
/RELOAD AC WITH 'E GOOD
/TRANSMIT AND FIVE

```

```

3345 4470 JMS I CHECK /DID I RECEIVE WHAT I SENT?
3346 0000 HOLD1, OPEN /WHAT I SENT,
3347 5366 JMP PIE710 /RECEIVED NOT SAME AS SENT,
3350 2062 ISZ CTRA /DONE?
3351 5737 JMP I TRDATA /NO,
3352 5425 JMP I CHAIN /YES, CHAIN,

/ROUTINE TO SEND AND RECEIVE DATA,

SNDREC, OPEN
3353 0000
3354 4565 UTCF
3355 4564 UTPC
3356 4571 UKCC
3357 4572 UKSF
3360 5357 JMP I=1
3361 7200 CLA /JUST IN CASE
3362 4570 UKRS
3363 4566 UTSF
3364 5363 JMP I=1
3365 5753 JMP I SNDREC /EXIT WITH RECEIVED DATA IN AC;

/COMMON HLT FOR PIT57 AND PIT10,

3366 7402 PIE710, HLT /DATA RECEIVED DOES NOT
/AGREE WITH DATA SENT,
/MQ CONTAINS DATA THAT WAS SENT,
/AC CONTAINS DATA THAT WAS RECEIVED,

/SCOPE LOOP, PRESS CONTINUE TO ENTER,
ACL
JMS SNDREC
3367 7701 JMP PIE710+1 /STAY IN LOOP,
3370 4353
3371 5367

3374 0417
3375 1740
3376 1717
3377 1707 3400

PAGE

/TEST OF KRS TO DO AN "OR" BY READING
/RANDOM DATA FROM KBRD BUFFER INTO AC
/EQUAL TO 7777, TEST IS DONE 500 TIMES,

PIT11, 11
3400 0011
3401 3435 PIT12
3402 4577 SETLOC
3403 0062 CTRA
3404 7014 -764
3405 6007 PIT11A, CAF
3406 4777 JMS RGNB
3407 7421 MQL
/SET COUNT OF
/500 (DEC)
/IN CTRA,
/CLEAR THE WORLD,
/GET A RANDOM NUMBER
/STORE IT IN MQ

```

```

3410 7701 ACL /RELOAD AC
3411 4563 UTLS /
3412 4566 UTSF /FLAG SET YET?
3413 5212 JMP I=1 /NO. WAIT;
3414 7240 CLA CMA /7777 TO AC
3415 4570 UKRS /READ KBRD BUFFER;
3416 7040 CMA /AC SHOULD NOW EQUAL 0
3417 7440 SZA /DOES IT = 0?
3420 5224 JMP PIE11A /NO.
3421 2062 ISZ CTRA /DONE 500 TIMES YET?
3422 5205 JMP PIT11A /NO. REPEAT
3423 5425 JMP I CHAIN /YES CHAIN;

3424 7402 PIE11A, HLT /KRS FAILED TO "OR" KBRD WITH AC
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
CAF
3425 6007 ACL /MQ TO AC
3426 7701 UTLS
3427 4563 UTSF
3430 4566 JMP I=1
3431 5230 CLA CMA
3432 7240 UKRS
3433 4570 JMP PIE11A+1
3434 5225

/TEST OF KRB

PIT12, 12
3435 0012 JMS I S100
3436 7777 SETLOC
3437 4430 DELAYM
3440 4577 M147
3441 0024
3442 7631
3443 6007 PIT12A, CAF
3444 1134 TAD I252
3445 4563 UTLS
3446 4566 UTSF
3447 5246 JMP I=1
3450 7240 CLA CMA
3451 4567 UKRB
3452 7041 CMA IAC
3453 1134 TAD I252
3454 7440 SZA
3455 5264 JMP PIE12A
3456 4572 PIT12B, UKSF
3457 7610 SKP CLA
3460 5274 JMP PIE12B
3461 2062 ISZ CTRA
3462 5243 JMP PIT12A
3463 5425 JMP I CHAIN

3464 7402 PIE12A, HLT /KRB FAILED TO JAM READER BUFFER TO AC;
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
CAF
3465 6007 TAD I252
3466 1134 UTLS
3467 4563

```

```

3470 4566      UTSF
3471 5270      JMP ,=1
3472 4567      UKRB
3473 5265      JMP P1E12A+1

3474 7402      P1E12B, HLT      /KRB FAILED TO CLEAR READER FLAG,
/SCOPE LOOP.  PRESS CONTINUE TO ENTER,
CAF
3475 6007      UTL
3476 4563      UTL
3477 4566      UTSF
3500 5277      JMP ,=1
3501 4567      UKRB
3502 5275      JMP P1E12B+1

```

/PROGRAM 2, ASR 33/35 TELETYPE BASIC INPUT TESTS;
/PROGRAM CHECKS INPUT IOT'S, INTERRUPT, AND READER TIMING

```

3503 4577      PRG2,  SETLOC      /SET KSTART TO INITIAL
3504 0023      KSTART      /ROUTINE ADDRESS.
3505 3510      P2TS0
3506 5707      JMP I ,+1      /GO START TEST
3507 0236      SRSET

```

/ISSUE KCC WITH AC=7777, AC SHOULD GO TO 0.
/AC NOT 0 INDICATES KCC FAILURE, TEST IS DONE 1000 TIMES;

```

3510 0000      P2TS0,  0
3511 3530      P2TS1
3512 4313      JMS P2TS0A
3513 0000      P2TS0A, OPEN
3514 4431      JMS I S4000
3515 7240      CLA CMA
3516 4571      UKCC
3517 7440      SZA
3520 5324      JMP P2E0
3521 2062      ISZ CYRA
3522 5315      JMP I =5
3523 5425      JMP I CHAIN
3524 7402      P2E0,  HLT

3525 7240      CLA CMA
3526 4571      UKCC
3527 5325      JMP ,=2

```

/SET UP TO DO TEST 4000 TIMES;
/SET AC TO 7777
/CLEAR AC AND FLAG
/IS AC = 0?
/NO, ERROR, GO TO P2E0
/DONE? REPEAT
/NO, REPEAT
/CHAIN
/TST0 ERR HALT, KCC DID
/NOT RESULT IN AC = 0
/SET A TO 7777
/CLEAR AC AND FLAG
/RPEAT

```

3530 0001      P2TS1,  1
3531 3545      P2TS2
3532 4432      JMS I S200

/ISSUE KCC, WAIT TWICE 10=11 BIT TIMES FOR FLAG TO SET;
/SKIP ON FLAG, FAILURE TO SKIP INDICATES
/THAT FLAG IS NOT SET, OR KSF FAILURE,
/TEST IS DONE 100 TIMES.
/SET DELAYM TO DELAY TWICE
/10 BIT TIMES FOR AN NON 110
/BAUD DEVICE AND TWICE 11 BIT
/TIMES FOR AN 110 BAUD DEVICE,

```

/SEE BIT TIME TABLE AT BEGINNING
/OF PROGRAM.

```

3533 4430      P2TS1A, JMS I S100
3534 4571      P2TS1B, UKCC
3535 4576      DELAY
3536 4572      UKSF
3537 5343      JMP P2E1
3540 2062      ISZ CTRA
3541 5334      JMP P2TS1B
3542 5425      JMP I CHAIN
3543 7402      HLT
                P2E1,
3544 5333      JMP P2TS1A

```

/ISSUE KCC; WAIT TWICE 10-11 BIT TIMES FOR FLAG TO BE SET;
/SKIP ON FLAG 1000 TIMES TO VERIFY CONSISTENT SKIPPING.

```

3545 0002      P2TS2, 2
3546 3600      P2TS3
3547 4432      JMS I S200

```

/SET DELAYM TO DELAY TWICE
/10 BIT TIMES FOR AN NON 110
/BAUD DEVICE AND TWICE 11 BIT
/TIMES FOR AN 110 BAUD DEVICE;
/SEE BIT TIME TABLE AT BEGINNING
/OF PROGRAM.

```

3550 4431      JMS I S4000
3551 4571      UKCC
3552 4576      DELAY
3553 4572      UKSF
3554 5362      JMP P2E2A
3555 4572      UKSF
3556 5364      JMP P2E2B
3557 2062      ISZ CTRA
3560 5355      JMP I=3
3561 5425      JMP I CHAIN
3562 7402      HLT
                P2E2A,
3563 5351      JMP P2TS2A
3564 7402      HLT
                P2E2B,
3565 4572      UKSF
3566 5365      JMP I=-1
3567 5365      JMP I=-2

```

PAGE

/ISSUE KCC; WAIT TWICE 10-11 BIT TIMES FOR FLAG TO SET;
/VERIFY THAT FLAG IS SET; RESET FLAG (KCC) AND

/SKIP ON FLAG 500 TIMES TO VERIFY THAT NO
/SKIP OCCURS WITH FLAG = 0,

3600 2023
3601 3630
3602 4432

P2TS3, 3
P2TS4
JMS I S200

/SET DELAYM TO DELAY TWICE
/10 BIT TIMES FOR AN NON 110
/BAUD DEVICE AND TWICE 11 BIT
/TIMES FOR AN 110 BAUD DEVICE.
/SEE BIT TIME TABLE AT BEGINNING
/OF PROGRAM,

3603 4577 SETLOC
3604 0062 CTRA
3605 7014 -764
3606 4571 UKCC
3607 4576 DELAY
3610 4572 UKSF
3611 5221 JMP P2E3A
3612 4571 UKCC
3613 4572 UKSF
3614 5216 JMP :+2
3615 5223 JMP P2E3B
3616 2062 ISZ CTRA
3617 5213 JMP :+4
3620 5425 JMP I CHAIN
3621 7402 HLT

3622 5206 JMP P2TS3A
3623 7402 HLT

/SET COUNT OF
/500 (DEC) IN
/CTRA
/CLEAR FLAG
/GO DELAY
/READY?
/NO, ERROR
/YES, RESET FLAG
/READY?
/NO, OK
/YES, ERROR
/ALL DONE TESTING?
/NO, REPEAT
/YES, CHAIN
/TST3 ERR HALT A; FLAG
/NOT SET OR KSF FAILURE
/TRY AGAIN
/TST3 ERR HALT B; FLAG
/FAILED TO RESET; OR KSF
/SKIPPED ERRONEOUSLY.

/TURN OFF READER BEFORE ENTERING

/SCOPE LOOP;
UKCC
UKSF
JMP :+2
JMP :+3

/CLEAR FLAG AND AC
/SKIP ON FLAG = 1
/REPEAT
/REPEAT

3624 4571
3625 4572
3626 5224
3627 5224

/THIS ROUTINE CHECKS THAT NO OTHER DEVICE CAN CAUSE AN INTERRUPT;
/AND THEN CHECKS THAT THE READER FLAG IS CAPABLE OF INTERRUPTING;

3630 0004
3631 3671 P2TS5
3632 4577 SETLOC
3633 0002 2
3634 3644 P2E4A
3635 4565 JMS I KBFLAG
3636 4426 UKCC
3637 4571 ION
3640 6001 NOP
3641 7000 IOF
3642 6002 JMP :+3
3643 5246 JMS INTFND
3644 4777 P2E4A;

/SET INTERRUPT RETURN
/TO P2E4A;

/CLEAR PUNCH/PRINTER FLAG
/CLEAR READER FLAG
/ENABLE INTERRUPT
/TURN OFF INTERRUPT
/UNEXPECTED INTERRUPT

```

3645 5235 JMP P2TS4A /TRY AGAIN
3646 4431 JMS I S4000 /SET UP TO DO TEST 4000 TIMES;
3647 4577 SETLOC /SET INTERRUPT RETURN
3650 0002 2 /TO P2TS4C,
3651 3666 P2TS4C
3652 4571 UKCC
3653 4572 UKSF
3654 5253 JMP ,=1 /WAIT FOR READER FLAG
3655 6001 ION /TO SET
3656 7000 NOP /ENABLE INTERRUPT
3657 7402 P2E4B, HLT /READER FLAG FAILED TO INTERRUPT,
/OR INTERRUPT SYSTEM MALFUNCTION
P2TS4B, ION /SET INTERRUPT RETURN
/TO P2TS4C=1,
P2E4B, HLT
/SCOPE LOOP
3660 4577 SETLOC
3661 0002 2
3662 3665 P2TS4C=1
3663 6001 ION
3664 7000 NOP
3665 5263 JMP ,=2
/
3666 2062 P2TS4C, ISZ CTRA /DONE?
3667 5255 JMP P2TS4B /NO, REPEAT
3670 5425 JMP I CHAIN

```

/READER TIMING TEST, CHECKS THAT READER FLAG IS =1 NO
/LATER THAN 103 MILLISECONDS AFTER KCC INSTRUCTION IS ISSUED.

```

3671 0005 P2TS5, 5
3672 3711 P2TS6
3673 4577 SETLOC /SET DELAYM
3674 0024 DELAYM /TO =103
3675 7631 M147
3676 4430 JMS I S100 /SET UP TO DO TEST 100 TIMES;
3677 4571 UKCC /START READER, CLEAR PC FLAG
3700 4576 DELAY /GO DELAY 103 MILLISECONDS
3701 4572 UKSF
3702 5306 JMP P2E5
3703 2062 ISZ CTRA
3704 5277 JMP P2TS5A
3705 5425 JMP I CHAIN
3706 7402 P2E5, HLT
/TS5 ERR HALT, FLAG NOT=1
/103 MSECS AFTER KCC INSTRUCTION,
3707 4426 JMS I KBFLAG
3710 5305 JMP ,=3 /YES, REPEAT,

```

/READ 256 DIFFERENT CHARACTERS, EACH CHARACTER IS READ 1000 TIMES
/TO VERIFY CONSISTENCY OF READING FROM ITI;

```

3711 0006 P2TS6, 6
3712 3762 P2TS7
3713 4577 SETLOC /SET COUNT OF

```

```

3714 0062      CTRA      /0256(DEC)
3715 7400      -420      /IN CTRA
3716 4426      P2TS6A,   JMS I KBFLAG
3717 4570      UKRS      /READ CHARACTER,
3720 3112      DCA WTS6A /SAVE AT WTS6A,
3721 4577      SETLOC    /SET COUNT OF
3722 0063      CTRB      /CTRB
3723 6030      -1750     /01000 (DEC) IN
3724 7200      P2TS6B,   CLA      /READ CHARACTER,
3725 4570      UKRS      /STORE IN MQ,
3726 7421      MQL       /GET IT BACK INTO THE AC,
3727 7701      ACL       /2'S COMPLEMENT IT
3730 7041      CIA       /ADD EXPECTED CHAR,
3731 1112      TAD WTS6A /RESULT 0?
3732 7640      SZA CLA   /NO, ERROR, GO TO E6A,
3733 5346      JMP P2E6A  /READ CHARACTER
3734 7240      P2TS6C,   CLA CMA
3735 4570      UKRS      /AC STILL 7777
3736 7040      CMA       /NO, ERROR GO TO P2E6C,
3737 7440      SZA       /READ CHAR 1000 TIMES?
3740 5356      JMP P2E6C  /NO, GO READ IT AGAIN,
3741 2063      ISZ CTRB   /YES, READ 256 DIFF. CHAR?
3742 5324      JMP P2TS6B /NO,
3743 2062      ISZ CTRA   /YES, CHAIN
3744 5316      JMP P2TS6A /MO TO AC,
3745 5425      JMP I CHAIN /TST6 ERR HALT A, AC DISPLAYS
3746 7701      P2E6A,   ACL      /INCORRECTLY READ CHAR, DEPRESS
3747 7402      HLT       /KEY CONTINUE

3750 7200      CLA WTS6A
3751 1112      TAD WTS6A
3752 7402      HLT

3753 7200      CLA
3754 4570      UKRS      /TST6 ERR HALT B, AC DISPLAYS
3755 5353      JMP ,=2     /WHAT THE CORRECT CHAR SHOULD
                               /BE.
3756 7402      HLT       /READ CHARACTER
                               /LOOP BACK
3757 7200      /SCOPE LOOP, PRESS CONTINUE TO ENTER,
3760 4570      CLA CMA
3761 5357      UKRS      /KRS FAILED TO "OR" KBRD BUFFER WITH AC,
                               JMP P2E6C+1

3762 0007      /ISSUE KCC, WAIT FOR FLAG TO SET, ISSUE KCR WITH
3763 4030      /AC=7777 AND DELAY 200 MSECs, AC NOT 7777 OR KBRD
                               /FLAG SET INDICATES A KCR FAILURE, TEST IS DONE
                               /100 TIMES,
3767 0007      P2TS7,   7
3768 4030      P2T10

```

3764 4430 JMS I S100
3765 4432 JMS I S200
/SET UP TO DO TEST 100 TIMES;
/SET DELAY TO DELAY TWICE
/10 BIT TIMES FOR AN NON 110
/BAUD DEVICE AND TWICE 11 BIT
/TIMES FOR AN 110 BAUD DEVICE;
/SEE BIT TIME TABLE AT BEGINNING
/OF PROGRAM.

3766 5776 JMS I KBFLAG
JMP P2TS7A

3776 4000
3777 2017
4000
PAGE
P2TS7A, JMS I KBFLAG
CLA CMA
UKCR
CMA
SZA
JMP P2E7A
P2TS7B, DELAY
UKSF
SKP
JMP P2E7B
ISZ CTRA
JMP P2TS7A
JMP I CHAIN

/AC=7777,
/CLEAR READER FLAG;
/AC SHOULD EQUAL ZERO/NOW;
/RESULT 0?
/NO, ERROR, GO TO P2E7A;
/GO DELAY 200 MILLISECS;
/READER FLAG SET?
/NO,
/YES, READER FLAG SET; ERROR, GO TO P2E7B;
/TEST DONE?
/NO, REPEAT.

4015 7402
4016 7240
4017 4562
4020 5216
P2E7A, HLT
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
CLA CMA
UKCR
JMP I=2
/KCR CLEARED AC,
/AC=7777, (SCOPE LOOP);
/CLEAR READER RUN, SHOULD NOT CLEAR AC,
/REPEAT.

4021 7402
4022 4426
4023 4562
4024 4576
4025 4572
4026 5222
4027 5222
P2E7B, HLT
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
JMS I KBFLAG
UKCR
DELAY
UKSF
JMP P2E7B+1
JMP P2E7B+1
/KCR DID NOT CLEAR READER FLAG
/CLEAR READER RUN,
/GO DELAY 200 MILLISECS
/REPEAT;
/REPEAT;

/ISSUE KCC, WAIT FOR FLAG TO SET, ISSUE KIE WITH
/AC11=0 THEN TURN THE INTERRUPT ON, AN INTERRUPT AT THIS TIME
/INDICATES A KIE FAILURE, WITH THE FLAG STILL SET ISSUE
/SRQ AND SPI, A SKIP BY EITHER INDICATES A FAILURE,
/ISSUE KIE WITH AC11=1 AND THE INTERRUPT ON, NO INTERRUPT
/INDICATES A KIE FAILURE, ISSUE SRQ AND THEN SPI, FAILURE OF
/EITHER TO SKIP INDICATES A FAILURE, THIS TEST IS DONE 4000 TIMES;

4030 0010 P2T10, 10
4031 4153 P2T11
4032 4431 JMS I S4000
/SET UP TO DO TEST 4000 TIMES;

```

4033 4426 JMS I KBFLAG
4034 4577 P2T10A, SETLOC
4035 2002 2
4036 4073 P2E10A
4037 4572 UKSF
4040 5233 JMP P2T10A+1
4041 7200 CLA
4042 4561 UKIE
4043 6001 ION
4044 7000 NOP
4045 6002 P2T10B, IOF
4046 6003 SRQ
4047 7410 SKP
4050 5307 JMP P2E10B
4051 4557 P2T10C, USPI
4052 7410 SKP
4053 5315 JMP P2E10C
4054 4577 P2T10D, SETLOC
4055 0002 2
4056 4064 P2T10E
4057 7201 CLA IAC
4060 4561 UKIE
4061 6001 ION
4062 7000 NOP
4063 5323 JMP P2E10D
4064 6003 P2T10E, SRQ
4065 5335 JMP P2E10E
4066 4557 P2T10F, USPI
4067 5344 JMP P2E10F
4070 2062 ISZ CTRA
4071 5234 JMP P2T10A
4072 5425 JMP I CHAIN

4073 7402 P2E10A, HLT
/SCOPE LOOP, PRESS CONTINUE TO ENTER;
4074 4572 UKSF
4075 4777 JMS INTFND
4076 4577 SETLOC
4077 0002 2
4100 4074 P2E10A+1
4101 4426 JMS I KBFLAG
4102 7200 CLA
4103 4561 UKIE
4104 6001 ION
4105 7000 NOP
4106 5274 JMP P2E10A+1

4107 7602 P2E10B, HLT CLA
/SCOPE LOOP, PRESS CONTINUE TO ENTER;
4110 4426 JMS I KBFLAG
4111 4561 UKIE
4112 6003 SRQ
4113 5310 JMP P2E10B+1

/SET INTERRUPT RETURN LOCATION
/TO P2E10A;

/AC=0
/DISABLE TTY INTERRUPT
/TURN INTERRUPT ON

/TURN INTERRUPT OFF;
/SKIP IF INTERRUPT REQUEST;

/ERROR, SRQ FAILED, GO TO P2E10B;
/SKIP IF TTY INTERRUPT;

/ERROR, SPI FAILED, GO TO P2E10C;
/SET INTERRUPT RETURN LOCATION
/TO P2T10E

/AC11=1
/ENABLE TTY INTERRUPT;
/TURN INTERRUPT ON;
/(SHOULD INTERRUPT);
/ERROR, GO TO P2E10D;
/SKIP IF INTERRUPT REQUEST;
/ERROR, GO TO P2E10E;
/SKIP IF TTY INTERRUPT;
/ERROR, GO TO P2E10F;
/DONE?
/NO, REPEAT.

/KIE FAILED TO DISABLE TTY;
/IS READER FLAG SET?
/NO, UNEXPECTED INTERRUPT;
/SET INTERRUPT RETURN LOCATION
/TO P2E10A+1;

/(SCOPE LOOP);
/DISABLE TTY INTERRUPT;
/INTERRUPT ON;

/REPEAT,

/SRQ SKIPPED WITH TTY DISABLED;

/SKIP IF INTERRUPT, (AC11=0); REQUEST, (SHOULD NOT SKIP)
/REPEAT

```

```

4114 5310      JMP P2E10B+1      /REPEAT
4115 7602      P2E10C, HLT CLA      /SPI SKIPPED WITH TTY DISABLED;
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
JMS I KBFLAG
UKIE
USPI
JMP P2E10C+1
JMP P2E10C+1
4123 7402      P2E10D, HLT      /KIE FAILED TO ENABLE TTY INTERRUPT WITH AC11=1.
/SCOPE LOOP, PRESS CONTINUE TO ENTER;
SETLOC
2
P2E10D+4
CLA IAC
UKIE
JMS I KBFLAG
ION
NOP
JMP P2E10D+4      /REPEAT,

```

```

4135 7402      P2E10E, HLT      /SRQ FAILED TO SKIP.
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
CLA IAC
UKIE
JMS I KBFLAG
SRQ
JMP P2E10E+1
JMP P2E10E+1

```

```

4144 7402      P2E10F, HLT      /SPI FAILED TO SKIP.
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
CLA IAC
UKIE
JMS I KBFLAG
USPI
JMP P2E10F+1
JMP P2E10F+1

```

```

4153 0011      P2T11, 11
4154 4233      P2T12
4155 4430      JMS I S100
4156 4432      JMS I S200

/ISSUE KIE WITH AC11=0 TO DISABLE TTY
/ISSUE CAF WITH AC, LINK, AND READER FLAG SET;
/TTY NOT ENABLED, OR AC AND LINK NOT
/ZERO INDICATES A FAILURE, TEST IS DONE 100 TIMES;

/SET UP TO DO TEST 100 TIMES;
/SET DELAY TO DELAY TWICE
/10 BIT TIMES FOR AN NON 110
/BAUD DEVICE AND TWICE 11 BIT
/TIMES FOR AN 110 BAUD DEVICE;

```

/SEE BIT TIME TABLE AT BEGINNING
/OF PROGRAM;

```

4157 4561 P2T11A, UKIE /DISABLE TTY (AC 11=0);
4160 4426 JMS I KBFLAG
4161 7360 CLA CMA CLL CML /AC AND LINK SET;
4162 6007 CAF /CLEAR ALL FLAGS, AC, LINK, AND ENABLE TTY;
4163 7420 SNL
4164 7440 SZA
4165 5776 JMP P2E11A
4166 4576 P2T11B, DELAY /ERROR, GO TO P2E11A;
4167 4572 UKSF /GO DELAY 200 MILLI SEC.
4170 7610 SKP CLA /DID FLAG COME UP?
4171 5775 JMP P2E11B /YES, ERROR, GO TO P2E11B;
4172 5774 JMP P2T11C /CROSS PAGE

```

PAGE

```

4200 4426 P2T11C, JMS I KBFLAG /SKIP IF TTY INTERRUPT REQUEST;
4201 4557 USPI /ERROR, GO TO P2E11C;
4202 5224 JMP P2E11C

```

```

4203 2062 ISZ CTRA /TEST DONE?
4204 5777 JMP P2T11A /NO, REPEAT,
4205 5425 JMP I CHAIN

4206 7402 P2E11A, HLT /CAF FAILED TO CLEAR AC AND LINK;
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
4207 7360 CLA CMA CLL CML /SCOPE LOOP;
4210 6007 CAF /CLEAR ALL FLAGS, AC, LINK, AND ENABLE TTY;
4211 7420 SNL
4212 7440 SZA
4213 5207 JMP P2E11A+1 /REPEAT;
4214 5207 JMP P2E11A+1 /REPEAT;

```

```

4215 7402 P2E11B, HLT /CAF DID NOT CLEAR FLAG OR FLAG SET AFTER BEING CLEARED,
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
4216 4426 JMS I KBFLAG
4217 6007 CAF /CLEAR THE FLAG,
4220 4576 DELAY /GO DELAY 200 MILLISEC;
4221 4572 UKSF /FLAG SET?
4222 5216 JMP P2E11B+1 /REPEAT
4223 5216 JMP P2E11B+1 /REPEAT

```

```

4224 7602 P2E11C, HLT CLA /CAF FAILED TO ENABLE TTY;
/SCOPE LOOP, PRESS CONTINUE TO ENTER;
4225 4561 UKIE /DISABLE TTY; (AC11=0)
4226 6007 CAF /ENABLE TTY,
4227 4426 JMS I KBFLAG

```

```

4230 4557      USPI
4231 5225      JMP P2E11C+1
4232 5225      JMP P2E11C+1

/TEST OF KRB INSTRUCTION.
P2T12, 12
4233 0012      /SET COUNT OF
4234 7777      /-256 DECIMAL
4235 4577      /IN CTRA
4236 0062      SETLOC
4237 7400      -400
4240 4577      SETLOC
4241 0024      DELAYM
4242 7631      M147
4243 4426      JMS I KBFLAG
4244 4570      UKRS
4245 3112      DCA WTS6A
4246 4426      JMS I KBFLAG
4247 1145      P2T12A, TAD I377
4250 4567      UKRB
4251 4572      UKSF
4252 7410      SKP
4253 5276      JMP P2E12A
4254 3104      DCA UTEMP
4255 4576      DELAY
4256 4572      UKSF
4257 5303      JMP P2E12B
4260 1112      UKSF
4261 7421      P2T12B, TAD WTS6A
4262 7701      MQL
4263 7001      AGL
4264 3112      IAC
4265 1112      DCA WTS6A
4266 0145      TAD WTS6A
4267 7041      AND I377
4270 1104      CMA IAC
4271 7640      TAD UTEMP
4272 5305      SEA CLA
4273 2062      JMP P2E12C
4274 5247      ISZ CTRA
4275 5425      JMP P2T12A
          JMP I CHAIN

P2E12A, HLT
/SCOPE LOOP, PRESS CONTINUE TO CLEAR READER FLAG;
4276 7402      JMS I KBFLAG
4277 4426      UKRB
4280 4567      DELAY
4281 4576      JMP P2E12A+1

P2E12B, HLT
/SCOPE LOOP, PRESS CONTINUE TO SET FLAG.
4303 7402      JMS I KBFLAG
4304 5277      UKRB
4305 7402      DELAY
          JMP P2E12A+1

P2E12C, HLT
/SCOPE LOOP, PRESS CONTINUE TO READ CORRECT DATA.

```

/PRESS CONTINUE TO TRY TEST AGAIN,
JMP P2Y12+2 /TRY TEST AGAIN,

4306 5235

/PROGRAM 3, ASR33/35 TELETYPE READER TEST, CHECKS ABILITY OF READER
/TO CORRECTLY READ AT FULL SPEED AND WITH RANDOM STALLS.

4307 4577
4310 0023
4311 4314
4312 5713
4313 0236

PRG3, SETLOC /SET KSTART TO INITIAL
KSTART /ROUTINE ADDRESS,
P3TS0 /GO START TEST
JMP I, +1
SRSET

/READ 4095 CHARACTERS, AT FULL SPEED, MATCHING EACH CHARACTER
/READ AGAINST COUNT PATTERN

4314 0000
4315 4344
4316 4465
4317 4577
4320 0062
4321 0001
4322 4571
4323 4466
4324 4467
4325 3332
4326 4572
4327 5326
4330 4567
4331 4470
4332 0000
4333 5337
4334 2062
4335 5324
4336 5425
4337 7402

P3TS0, 0
P3TS1 /GO SYNC TAPE
JMS I SYNC /SET COUNT OF
SETLOC /-4095(DEC) IN
CTRA /CTRA
-7777 /START READER
UKCC /GO INITIALIZE PATTERN
JMS I INPATT /GET PATTERN CHARACTER
JMS I GETPT /STORE AT SB0
DCA SB0 /READY?
UKSF /NO, TEST AGAIN
JMP I, -1 /YES, READ CHARACTER
UKRB /GO CHECK FOR CORRECT MATCH
JMS I CHECK /CORRECT CHAR HERE
0 /ERROR, GO TO P3E0
JMP P3E0 /OK, ALL DONE?
ISZ CTRA /NO, REPEAT
JMP P3TS0A /YES, CHAIN
JMP I CHAIN /TST10 ERR HALT, AC CONTAINS
HLT /CHAR THAT DID NOT MATCH
/AGAINST PATTERN, EPRESS
/KEY CONTINUE

4340 7200
4341 1332
4342 7402
4343 5334

CLA /GET CORRECT CHARACTER
TAD SB0 /AC CONTAINS THE EXPECTED CHARACTER
HLT
JMP P3T0B

/READ 2000 CHARACTERS WITH RANDOM DELAY BETWEEN CHARACTERS;
/MATCH EACH CHARACTER READ AGAINST COUNT PATTERN

4344 0001
4345 4400
4346 4465
4347 4577
4350 0062

P3TSI, 1
P3TS2 /TO SYNC TAPE
JMS I SYNC /SET COUNT OF
SETLOC /-2000 (DEC) IN
CTRA

```

4351 4060 /CTRA
4352 4571 /START READER
4353 4466 /INITIALIZE PATTERN
4354 4467 /GET PATTERN CHARACTER
4355 3364 /STORE AT SB1
4356 4427 /GENERATE RANDOM DELAY
4357 4576 /DELAY
4360 4572 /READY?
4361 5360 /NO, TEST AGAIN
4362 4567 /YES, READ CHARACTER
4363 4470 /GO CHECK FOR CORRECT MATCH

P3TS1A,
JMS I INPATT
JMS I GETPT
DCA SB1
JMS I DLCONT1
DELAY
UKSF
JMP ,=1
UKRB
JMS I CHECK

SB1,
0
JMP P3E1
P3T1B,
ISZ CTRA
JMP P3TS1A
JMP I CHAIN
HLT

P3E1,
CLA
TAD SB1
HLT

4372 7200 /GET CORRECT CHARACTER
4373 1364 /AC CONTAINS THE EXPECTED
4374 7402 /CHARACTER

4375 5366 JMP P3T1B

4377 4157 PAGE
4400 4400

```

```

/READ WITH RANDOM STALL BETWEEN RANDOM CHARACTER GROUPS
/100 GROUPS READ,
/
P3TS2, 2
7777
JMS I SYNC
JMS I S100
UKCC
JMS I INPATT
JMS I DLCONT1
JMS CHRCNT
CTRB
JMS I GETPT
DCA SB2
DELAY
UKSF
JMP ,=1
UKRB
JMS I CHECK

SB2,
0
JMP P3E2
ISZ CTRB
JMP P3TS2B

4400 0002 /GO SYNC TAPE
4401 7777 /SET UP TO DO TEST 100 TIMES:
4402 4465 /START READER
4403 4430 /INITIALIZE PATTERN
4404 4571 /SET RANDOM DELAY
4405 4466 /SET RANDOM CHARACTER
4406 4427 /COUNT IN CTRB
4407 4777 /GET PATTERN CHARACTER
4410 0063 /AND STORE AT SB2
4411 4467 /GO DELAY NO OF
4412 3220 /READY?
4413 4576 /NO, TEST AGAIN
4414 4572 /READ CHARACTER
4415 5214 /CHECK FOR CORRECT MATCH
4416 4567 /AGAINST SB2 CONTENTS
4417 4470 /ERROR, GO TO P3E2
4420 0000 /OK, ALL CHARS FOR GROUP DONE?
4421 5227 /NO
4422 2063
4423 5211

```

```

4351 4060          -3720      /CTRA
4352 4571          UKCC       /START READER
4353 4466          JMS I INPAT /INITIALIZE PATTERN
4354 4467          P3TS1A,    /GET PATTERN CHARACTER
4355 3364          DCA SB1     /STORE AT SB1
4356 4427          JMS I DLN1  /GENERATE RANDOM DELAY
4357 4576          DELAY      /DELAY
4360 4572          UKSF       /READY?
4361 5360          JMP ,=1     /NO, TEST AGAIN
4362 4567          UKRB       /YES, READ CHARACTER
4363 4470          JMS I CHECK /GO CHECK FOR CORRECT MATCH

SB1, 0             /CORRECT CHAR HERE
      JMP P3E1      /ERROR, GO TO P3E1
P3T1B, 15Z CTRA    /OK, ALL DONE?
      JMP P3TS1A    /NO,
      JMP I CHAIN   /YES, CHAIN
P3E1, HLT          /TST1 ERR HALT, AC CONTAINS
                     /CHARACTER THAT DID NOT MATCH
                     /AGAINST PATTERN, DEPRESS
                     /KEYCONTINUE

CLA               /GET CORRECT CHARACTER
TAD SB1           /AC CONTAINS THE EXPECTED
HLT              /CHARACTER

JMP P3T1B

PAGE
4372 7200
4373 1364
4374 7402

4375 5366
4377 4157
4400 4400

```

/READ WITH RANDOM STALL BETWEEN RANDOM CHARACTER GROUPS
/100 GROUPS READ,

```

4400 0002          P3TS2, 2   /GO SYNC TAPE
4401 7777          7777      /SET UP TO DO TEST 100 TIMES;
4402 4465          JMS I SYNC /START READER
4403 4430          JMS I S100 /INITIALIZE PATTERN
4404 4571          UKCC       /SET RANDOM DELAY
4405 4466          JMS I INPAT /SET RANDOM CHARACTER
4406 4427          JMS I DLN1  /COUNT IN CTRB
4407 4777,        JMS CHRCNT /GET PATTERN CHARACTER
4410 0063          CTRB       /AND STORE AT SB2
4411 4467          JMS I GETPT /GO DELAY NO OF
4412 3220          DCA SB2    /READY?
4413 4576          DELAY      /NO, TEST AGAIN
4414 4572          UKSF       /READ CHARACTER
4415 5214          JMP ,=1     /CHECK FOR CORRECT MATCH
4416 4567          UKRB       /AGAINST SB2 CONTENTS
4417 4470          JMS I CHECK /ERROR, GO TO P3E2
4420 0000          0         /OK, ALL CHARS FOR GROUP DONE?
4421 5227          JMP P3E2   /NO
4422 2063          15Z CTRB
4423 5211          JMP P3TS2B

```

```

4424 2062 P3T2C, ISZ CTRA /YES, ALL GROUPS DONE?
4425 5206 JMP P3TS2A /NO
4426 5425 JMP I CHAIN /YES, CHAIN

4427 7402 P3E2, HLT /TST2 ERROR HALT, AC CONTAINS CHAR THAT
/ DID NOT MATCH AGAINST PATTERN, DEPRESS KEY
/CONTINUE

4430 7200 CLA /GET CORRECT CHARACTER
4431 1220 TAD SB2 /AC CONTAINS THE EXPECTED CHARACTER
4432 7402 HLT
4433 5224 JMP P3T2C

```

/PROGRAM 4,

```

4434 4776, PRG4, JMS STBF /SET UP BUFFER AREA
4435 4577 SETLOC /SET KSTART TO INITIAL
4436 0023 KSTART /ROUTINE ADDRESS
4437 4442 P4TS0 /GO START PROGRAM
4440 5641 JMP I ,+1
4441 0236 SRSET

/CARRIAGE RETURN TEST
P4TS0, 0
4442 0000 P4TS1
4443 4475 CKSR37 /KSR37?
4444 4555 TAD [11] /NO,
4445 1140 TAD (=122 /YES
4446 1375 MQL /STORE IN MQ,
4447 7421 TYPE /PRINT TEST TITLE
4450 4573 CRTST
4451 6327 TAD [334
4452 1133 JMS I UPUNCH /GET "\ " CODE
4453 4474 ACL /PRINT IT
4454 7701 DCA UTEMP /MQ TO AC,
4455 3104 DCA UTEMP /ALL DONE?
4456 2104 ISZ UTEMP /NO
4457 7410 SKP I CHAIN /YES, CHAIN
4460 5425 JMP I CHAIN
4461 1104 TAD UTEMP
4462 3105 DCA UTEMP1 /UTEMP TO UTEMP1
4463 1142 TAD [240 /GET "SPACE" CODE
4464 4474 JMS I UPUNCH /PRINT IT
4465 2105 ISZ UTEMP1 /SPACED NO, OF TIMES IN UTEMP1?
4466 5263 JMP ,=3 /NO, SO SPACE AGAIN
4467 1107 TAD CR /YES, GET "CR" CODE,
4470 4474 JMS I UPUNCH /PRINT IT,
4471 4474 JMS I UPUNCH /DUMMY CYCLE,
4472 1132 TAD [257 /SET "/" CODE
4473 4474 JMS I UPUNCH /PRINT IT
4474 5256 JMP CRTSTA /GO TO CRTSTA

```

/RIGHT MARGIN TEST

```

4475 0001 P4TS1,
4476 4525 P4TS2
4477 7200 CLA
4500 1131 TAD C=16
4501 7421 MQL
4502 1130 TAD CRM33B
4503 3323 DCA RMB
4504 4555 CKSR37
4505 5312 JMP I=5
4506 1127 TAD C=17
4507 7421 MQL
4510 1126 TAD CRM37A
4511 3323 DCA RMB
4512 4573 TYPE
4513 6337 RMTST
4514 7701 ACL
4515 3104 DCA UTEMP
4516 4573 TYPE
4517 1563 RMTSTA
4520 2104 ISZ UTEMP
4521 5316 JMP RMTSTA
4522 4573 TYPE
4523 0000 OPEN
4524 5425 JMP I CHAIN

```

/KSR37?
/NO.
/YES.

/PRINT TEST TITLE

/PRINT ---- I

/DONE TIMES?
/NO, SO DO IT AGAIN
/YES, PRINT -1-

/CHAIN

```

4525 0002 /SPACE TEST
4526 4600 P4TS2,
4527 4573 P4TS3
4530 6354 TYPE
4531 4555 SPTST
4532 1125 CKSR37
4533 1124 TAD C=5
4534 3104 TAD C=51
4535 4573 DCA UTEMP
4536 6324 TYPE
4537 2104 SPTSTC
4540 5335 ISZ UTEMP
4541 4555 JMP SPTSTA
4542 1123 CKSR37
4543 1122 TAD C=50
4544 3104 DCA UTEMP
4545 1374 TAD C=1
4546 3105 DCA UTEMP1
4547 1105 TAD UTEMP1
4550 3106 DCA UTEMP2
4551 1107 TAD CR
4552 4474 JMS I UPUNCH
4553 4474 JMS I UPUNCH
4554 1142 TAD C240
4555 4474 JMS I UPUNCH
4556 2106 ISZ UTEMP2
4557 5354 JMP I=3
4560 1132 TAD C257

```

/PRINT TEST TITLE

/KSR37?

/NO

/YES

/=36 TO UTEMP

/PRINT \, SPACE

/DONE 36 TIMES?
/NO, SO DO IT AGAIN.

/KSR37?

/NO

/YES

/=36 TO UTEMP

/GET =1

/AC TO UTEMP1

/UTEMP1

/TO UTEMP2

/GET "CR" CODE

/PRINT IT

/DUMMY CYCLE

/GET "SPACE" CODE

/PRINT IT

/DONE SPACING?

/NO.

/GET "/" CODE

4561 4474 JMS I UPUNCH /PRINT IT
 4562 2104 /DONE 36 TIMES?
 4563 7410 SKP /NO,
 4564 5425 JMP I CHAIN /YES, CHAIN
 4565 7344 CLA CLL CMA RAL /-2 TO AC
 4566 1105 TAD UTEMP1 /ADD C(UTEMP1)
 4567 5346 JMP SPTSTB /GO TO SPTSTB

4574 7777
 4575 7656
 4576 1000
 4577 7456
 4600

PAGE

/LINE FEED TEST
 P4TS3, 3
 4600 0003
 4601 5122 P4TS47
 4602 7240 CLA CMA /SET STALL
 4603 3064 DCA STUID /INDICATOR
 4604 4573 TYPE /PRINT TEST TITLE
 4605 6366 LFTST
 4606 4555 CKSR37 /KSR37?
 4607 1140 TAD C11 /NO,
 4610 1377 TAD (-121 /YES,

4611 3104 DCA UTEMP
 4612 1133 LFTSTA, TAD C334
 4613 4474 JMS I UPUNCH /GET "A" CODE
 4614 1110 TAD LF UPUNCH /PRINT IT
 4615 4474 JMS I UPUNCH /GET "LF" CODE
 4616 2104 /PRINT IT
 4617 7410 /DONE?
 4620 5425 SKP I CHAIN /NO,
 4621 4536 JMP I CHAIN /YES, CHAIN
 4622 5212 STALL /GO TO LFTSTA
 JMP LFTSTA

/TYPE LINE OF CHARACTERS ABC
 P4TS4, 4
 4623 0004
 4624 4631 P4TS5
 4625 4573 TYPE /PRINT TITLE
 4626 6376 CHRTST
 4627 4433 JMS I TLCALL /PRINT LINE
 4630 6107 A

/TYPE LINE OF CHARACTERS DEF
 P4TS5, 5
 4631 0005
 4632 4635 P4TS6
 4633 4433 JMS I TLCALL
 4634 6112 D

4635	0006	/TYPE LINE OF CHARACTERS GHI
4636	4641	P4TS6, 6
4637	4433	P4TS7
4640	6115	JMS I TLCALL
		G
4641	0007	/TYPE LINE OF CHARACTERS JKL
4642	4645	P4TS7, 7
4643	4433	P4TS10
4644	6120	JMS I TLCALL
		J
4645	0010	/TYPE LINE OF CHARACTERS MNO
4646	4651	P4TS10, 10
4647	4433	P4TS11
4650	6123	JMS I TLCALL
		M
4651	0011	/TYPE LINE OF CHARACTERS PQR
4652	4655	P4TS11, 11
4653	4433	P4TS12
4654	6126	JMS I TLCALL
		P
4655	0012	/TYPE LINE OF CHARACTERS STU
4656	4661	P4TS12, 12
4657	4433	P4TS13
4660	6131	JMS I TLCALL
		S
4661	0013	/TYPE LINE OF CHARACTERS VMX
4662	4665	P4TS13, 13
4663	4433	P4TS14
4664	6134	JMS I TLCALL
		V
4665	0014	/TYPE LINE OF CHARACTERS YZ0
4666	4671	P4TS14, 14
4667	4433	P4TS15
4670	6137	JMS I TLCALL
		Y
4671	0015	/TYPE LINE OF CHARACTERS 123
4672	4675	P4TS15, 15
4673	4433	P4TS16
4674	6142	JMS I TLCALL
		ONE
4675	0016	/TYPE LINE OF CHARACTERS 456
4676	4701	P4TS16, 16
4677	4433	P4TS17
4700	6145	JMS I TLCALL
		FOUR
4701	0017	/TYPE LINE OF CHARACTERS 789
4702	4705	P4TS17, 17
4703	4433	P4TS20
4704	6150	JMS I TLCALL
		SEVEN
4705	0020	/TYPE LINE OF CHARACTERS !"#
4706	4711	P4TS20, 20
4707	4433	P4TS21
4710	6153	JMS I TLCALL
		C241

```

4711 0021
4712 4715
4713 4433
4714 6156

/TYPE LINE OF CHARACTERS $X8
P4TS21, 21
P4TS22
JMS I TLCALL
C244

/TYPE LINE OF CHARACTERS '()'
P4TS22, 22
P4TS23
JMS I TLCALL
C247

/TYPE LINE OF CHARACTERS **,
P4TS23, 23
P4TS24
JMS I TLCALL
C252

/TYPE LINE OF CHARACTERS =, (
P4TS24, 24
P4TS25
JMS I TLCALL
C255

/TYPE LINE OF CHARACTERS !|K
P4TS25, 25
P4TS26
JMS I TLCALL
C272

/TYPE LINE OF CHARACTERS =>?
P4TS26, 26
P4TS27
JMS I TLCALL
C275

/TYPE LINE OF CHARACTERS @ [\
P4TS27, 27
P4TS30
JMS I TLCALL
C300

/TYPE LINE OF CHARACTERS ]* AND LEFT ARROW
P4TS30, 30
P4TS31
JMS I TLCALL
C335

/TYPE LINE OF SMALL A, B, AND C
P4TS31, 31
P4TS32
JMS I TLC37
SA

/TYPE LINE OF SMALL D, E, AND F
P4TS32, 32
P4TS33
JMS I TLC37
SD

/TYPE LINE OF SMALL G, H, AND I
P4TS33, 33
4761 0033
4751 0031
4752 4755
4753 4434
4754 6206

4755 0032
4756 4761
4757 4434
4760 6211

```

4762 5000 P4TS34
4763 4434 JMS I TLC37
4764 6214 SG

4777 7657 PAGE
5000

5000 0034 /TYPE LINE OF SMALL J, K, AND L
5001 5004 P4TS34, 34
5002 4434 P4TS35
5003 6217 JMS I TLC37
SJ

5004 0035 /TYPE LINE OF SMALL M, N, AND O
5005 5010 P4TS35, 35
5006 4434 P4TS36
5007 6222 JMS I TLC37
SM

5010 0036 /TYPE LINE OF SMALL P, Q, AND R
5011 5014 P4TS36, 36
5012 4434 P4TS37
5013 6225 JMS I TLC37
SP

5014 0037 /TYPE LINE OF SMALL S, T, AND U
5015 5020 P4TS37, 37
5016 4434 P4TS40
5017 6230 JMS I TLC37
SS

5020 0040 /TYPE LINE OF SMALL V, W, AND X
5021 5024 P4TS40, 40
5022 4434 P4TS41
5023 6233 JMS I TLC37
SV

5024 0041 /TYPE LINE OF SMALL Y, AND Z, AND CODE 340 CHARACTER;
5025 5030 P4TS41, 41
5026 4434 P4TS42
5027 6236 JMS I TLC37
SY

5030 0042 /TYPE LINE OF CHARACTERS WHOSE CODE IS 373, 374, 375, 376;
5031 5047 P4TS42, 42
5032 4555 P4TS43
5033 5425 CKSR37 /KSR37?
5034 4574 JMP I CHAIN /NO, BYPASS TEST
5035 6241 MOVE
5036 6601 C373
5037 7774 BLOCK1
5040 4574 -4
5041 6601 MOVE
5042 6605 BLOCK1
5043 7663 BLOCK1+4
5044 3064 -115
DCA STLD

5045 4777/
5046 5425
JMS TYPLN
JMP I CHAIN

5047 0043
5050 5054
5051 4776/
5052 4775/
5053 5425
/TYPE 2 LINES OF ALL CHARACTERS, 1ST LINE NO DELAY; 2ND LINE WITH STALLS.
PATS43, 43
PATS44
JMS FBALL
JMS WOSWS
JMP I CHAIN /CHAIN
/FILL BUFFER WITH ALL CHARS,

5054 0044
5055 5072
5056 4573
5057 6412
5060 4554
5061 5425
5062 4774/
5063 4577
5064 0062
5065 7772
5066 4775/
5067 2062
5070 5266
5071 5425
/TYPE 12 LINES OF ASR33 WORST CASE PATTERN; ALTERNATE LINES WITH STALLS.
PATS44, 44
PATS45
TYPE
WCTST
CKSR33
JMP I CHAIN
JMS FW336
SETLOC
CTRA
-6
PATS44A, JMS WOSWS
ISZ CTRA
JMP PAT44A
JMP I CHAIN
/PRINT TITLE
/33?
/NO
/PATTERN TO BUFFER
/6 TO CTRA
/NO, REPEAT
/YES, CHAIN

5072 0045
5073 5106
5074 4553
5075 5425
5076 4773/
5077 4577
5100 0062
5101 7772
5102 4775/
5103 2062
5104 5302
5105 5425
/TYPE 12 LINES OF ASR35 WORST CASE PATTERN; ALTERNATE LINES WITH STALLS.
PATS45, 45
PATS46
CKSR35
JMP I CHAIN
JMS FW356
SETLOC
CTRA
-6
PATS45A, JMS WOSWS
ISZ CTRA
JMP PAT45A
JMP I CHAIN
/35?
/NO
/PATTERN TO BUFFER
/6 TO CTRA
/ALL LINES TYPED?
/NO, REPEAT
/YES, CHAIN

5106 0046
5107 7777
5110 4555
5111 5425
5112 4772/
5113 4577
5114 0062
5115 7772
5116 4775/
5117 2062
5120 5316
5121 5425
/TYPE 12 LINES OF KSR37 WORST CASE PATTERN; ALTERNATE LINES WITH STALLS.
PATS46, 46
7777
CKSR37
JMP I CHAIN
JMS FW376
SETLOC
CTRA
-6
PATS46A, JMS WOSWS
ISZ CTRA
JMP PAT46A
JMP I CHAIN
/37?
/NO, BYPASS TEST.
/YES, PATTERN TO BUFFER
/6 TO CTRA
/ALL LINES TYPED?
/NO, REPEAT
/YES, CHAIN

```

5122 0047 /KSR37, KSR35, OR ASR35 TAB TEST
5123 5231 P4TS47, 47
5124 4555 P4TS50
5125 5346 CKSR37
5126 4573 JMP TBTB
5127 6267 TYPE
5130 1121 TBTST
5131 4771 TAD C=11
5132 1370 JMS MTABP
5133 3340 TAD C=12
5134 1367 DCA TBCNT
5135 3062 TAD C=7
5136 3361 DCA CTRA
5137 4766 DCA SPCNT
5140 0000 JMS TABP
5141 2062 OPEN
5142 7410 ISZ CTRA
5143 5425 SKP
5144 2361 JMP I CHAIN
5145 5337 ISZ SPCNT
5146 4553 JMP TBTB+3
5147 5425 CKSR35
5150 4573 JMP I CHAIN
5151 6267 TYPE
5152 1367 TBTST
5153 4771 TAD C=7
5154 4573 JMS MTABP
5155 6301 TYPE
5156 1121 TBMRK+1
5157 3340 TAD C=11
5160 5334 DCA TBCNT
5161 0000 JMP TBTB
5162 0000 OPEN
SPCNT, OPEN
SPCTR, OPEN

```

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```

5200 0000 TABCTR, OPEN
5201 0000 TAPP, OPEN
5202 1601 TAD I TABP
5203 3200 DCA TABCTR
5204 2201 ISZ TABP
5205 4575 CRLF
5206 7777 -1

```

```
5207 1777/ SPAC, TAD SPENT /GET SPACE COUNT
5210 7450 SNA /0?
5211 5220 JMP TABPA /YES, DON'T SPACE
5212 7041 CIA /NO, NEGATE COUNT
5213 3776/ DCA SPCTR /SPACE
5214 1142 TAD C240
5215 4474 JMS I UPUNCH
5216 2776/ ISZ SPCTR
5217 5214 JMP I=3 /DONE SPACING?
5220 1140 TAD C11 /NO, SPACE AGAIN
5221 4474 JMS I UPUNCH /GET TAB CODE
5222 4474 JMS I UPUNCH /OUTPUT TO TELEPRINTER
5223 4474 JMS I UPUNCH /DUMMY CYCLE.
5224 1132 TAD C257 /DUMMY CYCLE.
5225 4474 JMS I UPUNCH /GET "/" CODE
5226 2200 ISZ TABCTR /AND TYPE IT
5227 5207 JMP SPAC /DONE?
5230 5601 JMP I TABP /NO, REPEAT
/YES, EXIT
```

```
/KSR37 BACKSPACE TEST;
P4TS50, 50
5231 0050 P4TS4
5232 4623 CKSR37
5233 4555 JMP I CHAIN
5234 5425 TYPE
5235 4573 BKSPTR
5236 6253 TAD C=51
5237 1124 DCA CTRA
5240 3062 TYPE
5241 4573 BKSU
5242 6574 ISZ CTRA
5243 2062 JMP I=3
5244 5241 TAD C=47
5245 1375 DCA CTRA
5246 3062 JMS BKSPC
5247 4263 -2
5250 7776 TAD C252
5251 1774/ JMS I UPUNCH
5252 4474 JMS BKSPC
5253 4263 -3
5254 7775 TAD C252
5255 1774/ JMS I UPUNCH
5256 4474 ISZ CTRA
5257 2062 JMP I=5
5260 5253 JMP I CHAIN
5261 5425 BKSPCTR, OPEN
5262 0000 BKSPC; OPEN
5263 0000 TAD I BKSPC
5264 1663 DCA BKSPCTR
5265 3262 ISZ BKSPC
5266 2263 TAD C210
5267 1373 JMS I UPUNCH
5270 4474
```

```
/KSR37?
/NO
/YES, TYPE TITLE
/=41 TO CTRA
/TYPE ALTERNATE U/S;
/DONE?
/NO,
/=39 TO CTRA
/BACKSPACE TWICE
/TYPE "*"
/BACKSPACE THRICE
/TYPE "*"
/DONE 39 TIMES?
/NO,
/YES, CHAIN
/GET BACKSPACE COUNT
/AND STORE AT BKSPCTR
/SET UP EXIT
/GET BACKSPACE CODE
/OUTPUT TO TELEPRINTER
```

5271 2262
5272 5267
5273 5663

ISZ BKSCTR
JMP I, 3
JMP I BKSPC

/DONE BACKSPACING?
/NO, REPEAT
/YES, EXIT

/PROGRAM 5; PUNCH TEST
PRG5, SETLOC

5274 4577
5275 0002
5276 1254
5277 4577
5300 0101
5301 7000
5302 4571
5303 1372
5304 3104
5305 1371
5306 3062
5307 4770
5310 4767
5311 3504
5312 2104
5313 2062
5314 5310
5315 4572
5316 5315
5317 7200
5320 3076
5321 4766
5322 4765
5323 4764
5324 4766
5325 4763
5326 4762
5327 4761
5330 4766
5331 4765
5332 4760
5333 4766
5334 4763
5335 4762
5336 4761
5337 5317

/SET INTERRUPT SERVICE ADDRESS
/TO INTSVC

/SET DATA BLOCK
/LENGTH TO
/512

/SET UP ADDRESS TO
/STORE DATA,
/512 TO CTRA

/INITIALIZE SPECIAL COUNT PATTERN
/GET CHARACTER
/STORE IT
/INCREMENT POINTER,
/DONE 512 CHARACTERS?
/NO, REPEAT

/YES, CLEAR READY BUSY

/PUNCH LEADER
/PUNCH SYNC CHARACTER
/PUNCH DATA BLOCK FULL SPEED;
/PUNCH TRAILER
/SYNC READER
/READ DATA BLOCK
/WAIT FOR READER NOT BUSY
/PUNCH LEADER
/PUNCH SYNC CHARACTER
/PUNCH DATA BLOCK (WITH STALLS);
/PUNCH TRAILER
/SYNC READER
/READ DATA BLOCK
/WAIT FOR READER NOT BUSY
/REPEAT.

PRG5A,

JMP I, -1

/PROGRAM 6; KEYBOARD TEST
PRG6, SETLOC

5340 4577
5341 0023
5342 5400
5343 4573
5344 6432
5345 5746
5346 0236

5360 1324
5361 1343

KSTART
P6T0
TYPE
KMSG1
JMP I, +1
SRSET

/SET KSTART TO INITIAL
/ROUTINE ADDRESS
/PRINT

5362 1400
5363 1216
5364 1316
5365 1212
5366 1200
5367 1717
5370 1707
5371 7000
5372 6577
5373 0210
5374 6164
5375 7731
5376 5162
5377 5161
5400

PAGE

/CLEAR AC AND FLAG (KCC), WAIT FOR FLAG TO SET, WITH FLAG SET, SKIP
/ON FLAG 4000 TIMES, KSF SHOULD SKIP EVERY TIME.

5400 0000
5401 5421
5402 4431
5403 4571
5404 4573
5405 6443
5406 4572
5407 5206
5410 4572
5411 5215
5412 2062
5413 5210
5414 5425
5415 7602
5416 4572
5417 5216
5420 5216

P6T1
JMS I 54000
UKCC
TYPE
KMSG2
UKSF
JMP I=1
UKSF
JMP P6E0
ISE CTRA
JMP I=3
JMP I CHAIN
P6E0, HLT CLA
UKSF
JMP I=1
JMP I=2

/CLEAR AC AND FLAG

/READY?
/WAIT
/READY, SKIP ON FLAG
/NO SKIP, ERROR
/ALL DONE?
/NO, REPEAT
/YES, CHAIN

/KSF FAILURE
/SCOPE LOOP
/SKIPS ON FLAG
/CONTINUOUSLY

/ECHO TEST CHARACTER RECEIVED FROM KEYBOARD IS TYPED; THE
/CHARACTER TYPED SHOULD MATCH CHARACTER KEYED; RUBOUT CHARACTER
/ENDS ROUTINE;

5421 0001
5422 5440
5423 4571
5424 4573
5425 6454
5426 4572
5427 5226
5430 4567
5431 4563
5432 4566
5433 5232
5434 1144
5435 7440

P6T1, 1
P6T2
UKCC
TYPE
KMSG3
UKSF
JMP I=1
UKRB
UTLS
UTSF
JMP I=1
TAD C=377
SEA
P6T1A,
UKSF
JMP I=1
UKRB
UTLS
UTSF
JMP I=1
TAD C=377
SEA
/CLEAR AC AND FLAG
/READY?
/WAIT
/READ CHARACTER
/PRINT IT
/PRINTER READY?
/NO, WAIT
/IS IT RUBOUT?

5436 5226
5437 5425

JMP P6T1A /NO
JMP I CHAIN /YES, CHAIN

/OCTAL EQUIVALENT TEST; THE OCTAL EQUIVALENT OF ANY
/CHARACTER KEYED IS PRINTED, RUBOUT ENDS ROUTINE;
P6T2,
2

5440 0002
5441 7777
5442 4571
5443 4573
5444 6521
5445 4573
5446 6462
5447 4572
5450 5247
5451 4567
5452 3112
5453 4777
5454 0112
5455 6541
5456 4573
5457 6537
5460 1112
5461 1144
5462 7640
5463 5247
5464 5425

/CLEAR AC AND FLAG
/PRINT TITLE AND
/INSTRUCTION

P6T2A,

/FLAG 1?
/NO, WAIT
/YES, READ KEYBOARD
/STORE CHARACTER
/CONVERT CHARACTER
/TO PRINTABLE OCTAL;
/PRINT CHARACTER

/WAS IT A RUBOUT?
/NO,
/YES, CHAIN

/PROGRAM 7; COMBINED READER, PRINTER, PUNCH TEST;
PRG7, SETLOC
2
INTSVC
SETLOC
BLKCNT
-226
JMS I KBFLAG
JMS STBF
SETLOC
KSTART
P7T0
JMP I '+1
SRSET

5465 4577
5466 0002
5467 1254
5470 4577
5471 0101
5472 7552
5473 4426
5474 4776
5475 4577
5476 0023
5477 5502
5500 5701
5501 0236

/SET DATA BLOCK LENGTH
/TO -150

/SET UP BUFFER AREA
/SET KSTART TO INITIAL
/ROUTINE ADDRESS

/START PROGRAM

5502 0000
5503 5506
5504 4435
5505 6107
5506 0001
5507 5512
5510 4435
5511 6112

/DATAI ABC

/DATAI DEF

5512	0002	P7T2,	2		
5513	5516	P7T3	JMS I FBF	/DATAI GHI	
5514	4435		G		
5515	6115				
5516	0003	P7T3,	3		
5517	5522	P7T4	JMS I FBF	/DATAI JKL	
5520	4435		J		
5521	6120		4		
5522	0004	P7T4,			
5523	5526	P7T5	JMS I FBF	/DATAI MNO	
5524	4435		M		
5525	6123		5		
5526	0005	P7T5,	P7T6	JMS I FBF	/DATAI PQR
5527	5532		P		
5530	4435		6		
5531	6126	P7T6,			
5532	0006		P7T7	JMS I FBF	/DATAI STU
5533	5536		S		
5534	4435		7		
5535	6131	P7T7,	P7T10	JMS I FBF	/DATAI VMX
5536	0007		V		
5537	5542		10		
5540	4435		P7T11	JMS I FBF	/DATAI YZ0
5541	6134	P7T10,			
5542	0010		11		
5543	5546		P7T12	JMS I FBF	/DATAI 123
5544	4435		ONE		
5545	6137		12		
5546	0011	P7T11,	P7T13	JMS I FBF	/DATAI 456
5547	5552		FOUR		
5550	4435		13		
5551	6142		P7T14	JMS I FBF	/DATAI 789
5552	0012	P7T12,	SEVEN		
5553	5556		14		
5554	4435		P7T15	JMS I FBF	/DATAI !"#
5555	6145	P7T13,	C241		
5556	0013		15		
5557	5562		P7T16	JMS I FBF	/DATAI \$%&
5560	4435		C244		
5561	6150	P7T14,	16		
5562	0014		P7T17	JMS I FBF	/DATAI '()
5563	5566		C247		
5564	4435				
5565	6153	P7T15,			
5566	0015				
5567	5572				
5570	4435				
5571	6156	P7T16,			
5572	0016				
5573	5600				
5574	4435				
5575	6161				
5576	1000				
5577	1650				

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5600 0017 P7T17, 17
 5601 5604 P7T20
 5602 4435 JMS I FBF /DATAI **,
 5603 6164 C252
 5604 0020 P7T20, 20
 5605 5610 P7T21
 5606 4435 JMS I FBF /DATAI -./,
 5607 6167 C255

5610 0021 P7T21, 21
 5611 5614 P7T22
 5612 4435 JMS I FBF /DATAI IJK
 5613 6172 C272
 5614 0022 P7T22, 22
 5615 5620 P7T23
 5616 4435 JMS I FBF /DATAI =>?
 5617 6175 C275
 5620 0023 P7T23, 23
 5621 5624 P7T24
 5622 4435 JMS I FBF /DATAI 0C\
 5623 6200 C300
 5624 0024 P7T24, 24
 5625 5630 P7T25
 5626 4435 JMS I FBF /DATAI J* AND LEFT ARROW
 5627 6203 C335
 5630 0025 P7T25, 25
 5631 5634 P7T26
 5632 4777, JMS FBALL
 5633 4776, JMS CNTST

5634 0026 P7T26, 26
 5635 5640 P7T27
 5636 4775, JMS FW336 /DATAI ASR33 PRINTER WORST CASE
 5637 4776, JMS CNTST /PATTERN
 5640 0027 P7T27, 27
 5641 5644 P7T30
 5642 4774, JMS FW356 /DATAI ASR35 PRINTER WORST CASE
 5643 4776, JMS CNTST /PATTERN
 5644 0030 P7T30, 30
 5645 7777 JMS FBFP3 /DATAI 1'S AND 0'S
 5646 4773, C377
 5647 6245 JMS CNTST

5650 4776, /PROGRAM 10, READS COUNT PATTERN,
 5651 4465 PRG10, JMS I SYNC /SYNC TAPE
 5652 3321 DCA ERRCTR /CLEAR ERROR COUNTER
 5653 4466 JMS I INPATT /INITIALIZE PATTERN,
 5654 4571 UKCC /START READER
 5655 7604 SRT0A, LAS /READ SR

```

5656 0120 AND C400
5657 7650 SNA CLA
5660 7040 CMA
5661 3064 DCA STLD

5662 4467 SRT08, JMS I GETPT
5663 3273 DCA SBSP
5664 4556 STALL
5665 4572 UKSF
5666 5265 JMP I=1
5667 4567 UKRB
5670 3103 DCA ERROR
5671 1103 TAD ERROR
5672 4470 JMS I CHECK
5673 0000
5674 7410
5675 5313 SKP
5676 2321 JMP HLTST
5677 5302 ISZ ERRCTR
5700 7240 JMP I+3
5701 3321 CLA CMA
5702 7604 DCA ERRCTR
5703 0143 LAS
5704 7650 AND C100
5705 5313 SNA CLA
5706 1103 JMP HLTST
5707 7402 TAD ERROR
5710 7200 HLT
5711 1273 CLA
5712 7402 TAD SBSP
5713 7604 HLT
5714 7700 LAS
5715 5255 SNA CLA
5716 1321 JMP SRT0A
5717 7402 TAD ERRCTR
5720 5255 HLT
5721 0000 JMP SRT0A
ERRCTR, 0 /ERROR COUNTER

SBSP, /GO CHECK CHARACTER WORD;
/ERROR, NO MATCH; GO INC. ERRCNT
/OK, /INCREMENT ERROR COUNTER
/OFLOW, RESET TO 7777.
/READ SR,
/HALT ON ERROR? (SR5)
/NO,
/YES, GET BAD CHAR;
/GET GOOD CHARACTER
/READ SR
/HALT? (SR0)
/NO,
/GET ERROR COUNT
/HALT, ERROR COUNT IN AC
/ERROR COUNTER

```

/PROGRAM 11, PRINTER EXERCISER, TYPES LINES OF ANY 3 CHARACTERS
/WITH STALLS, OR FULL SPEED, KEYBOARD CONTROLLED.

```

5722 4772/ PRG11, JMS STBF
5723 4573 TYPE
5724 6546 P11MG1
5725 1371 TAD (BLOCKI-1)
5726 3016 DCA 16
5727 4573 TYPE
5730 6562 P11MG2
5731 4353 JMS GKBCR
5732 3416 DCA I 16
5733 4353 JMS GKBCR
5734 3416 DCA I 16
5735 4353 JMS GKBCR
5736 3416 DCA I 16
5737 4353 JMS GKBCR

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5740 1144 TAD C=377
5741 7640 SZA CLA
5742 7240 CLA CMA
5743 3064 DCA STLD
5744 4773 JMS FBF3
5745 6601 BLOCK1
5746 4770 JMS TYPLN
5747 7604 LAS
5750 7700 SZA CLA
5751 5346 JMP I=3
5752 5325 JMP PRG11A
5753 0000 OPEN
5754 4572 UKSF
5755 5354 JMP I=1
5756 4567 UKRB
5757 7421 MQL
5760 7701 ACL
5761 4474 JMS I UPUNCH
5762 7701 ACL
5763 5753 JMP I GKBCR

```

GKBCR,

```

/STALL?
/YES,
/NO,
/SET UP LINE,
/TYPE LINE OF CHARACTERS
/READ SR,
/CHANGE DATA? (SP0=1)
/NO,
/YES,
/SUB TO GET KEYBOARD CHARACTER,
/WAIT FOR FLAG,
/READ CHARACTER,
/STORE CHARACTER,
/GET IT BACK,
/ECHO IT,
/GET CHARACTER AGAIN,
/EXIT

```

/PROGRAM 12, PUNCHES BINARY COUNT PATTERN,

```

5764 4466 PRG12, JMS I INPAT
5765 4467 JMS I GETPT
5766 4474 JMS I UPUNCH
5767 5365 JMP I=2

```

```

/INITIALIZE BINARY COUNT PATTERN
/GET BINARY COUNT CHARACTER,
/PUNCH CHARACTER
/REPEAT,

```

PAGE

```

6000 0000 DVCSEL, OPEN
6001 1117 TAD CINTAB
6002 3052 DCA TEMP
6003 1021 TAD TTYIOT
6004 7012 RTR
6005 7010 RAR
6006 0116 AND C0770
6007 3104 DCA UTEMP
6010 4222 JMS DVCOM
6011 1115 TAD COUTTAB
6012 3052 DCA TEMP
6013 1021 TAD TTYIOT
6014 7006 RTL
6015 7004 RAL
6016 0116 AND C0770
6017 3104 DCA UTEMP

```

```

/DEVICE CODE SELECT ROUTINE,
/GET START ADDR OF INPUT IOT TABLE,
/AND SAVE AT TEMP,
/OBTAIN NEW INPUT IOT AND
/STORE AT UTEMP,

/PERFORM INPUT IOT SELECTION,
/GET START ADDR OF OUTPUT IOT TABLE,
/AND OBTAIN NEW OUTPUT IOT AND
/OBTAIN NEW OUTPUT IOT AND
/STORE AT UTEMP,

```


/SMALL Q
/BIG A
/SWUNG DASH
/BIG A
/SMALL Q

0361
0301
0376
0301
0361

6102 0361
6103 0301
6104 0376
6105 0301
6106 0361

6107 0301 A,
6110 0302
6111 0303
6112 0304 D,
6113 0305
6114 0306
6115 0307 G,
6116 0310
6117 0311
6120 0312 J,
6121 0313
6122 0314
6123 0315 M,
6124 0316
6125 0317
6126 0320 P,
6127 0321
6130 0322 S,
6131 0323
6132 0324
6133 0325 V,
6134 0326
6135 0327
6136 0330 Y,
6137 0331
6140 0332
6141 0260 ONE,
6142 0261
6143 0262
6144 0263
6145 0264 FOUR,
6146 0265
6147 0266

0301
0302
0303
0304
0305
0306
0307
0310
0311
0312
0313
0314
0315
0316
0317
0320
0321
0322
0323
0324
0325
0326
0327
0330
0331
0332
0260
0261
0262
0263
0264
0265
0266

267
270
271
241
242
243
244
245
246
247
250
251
252
253
254

SEVEN,

C241,

C244,

C247,

C252,

267
270
271
241
242
243
244
245
246
247
250
251
252
253
254

6167	0255	C255,	255
6170	0256		256
6171	0257		257
6172	0272	C272,	272
6173	0273		273
6174	0274		274
6175	0275	C275,	275
6176	0276		276
6177	0277		277
6200	0300	C300,	300
6201	0333		333
6202	0334		334
6203	0335	C335,	335
6204	0336		336
6205	0337		337
6206	0341	SA,	341
6207	0342		342
6210	0343		343
6211	0344	SD,	344
6212	0345		345
6213	0346		346
6214	0347	SG,	347
6215	0350		350
6216	0351		351
6217	0352	SJ,	352
6220	0353		353
6221	0354		354
6222	0355	SM,	355
6223	0356		356
6224	0357		357
6225	0360	SP,	360
6226	0361		361
6227	0362		362
6230	0363	SS,	363
6231	0364		364
6232	0365		365
6233	0366	SV,	366
6234	0367		367
6235	0370		370
6236	0371	SY,	371
6237	0372		372
6240	0340	C373,	340
6241	0373		373
6242	0374		374
6243	0375		375
6244	0376		376
6245	0377	C377,	377
6246	0000		000
6247	0377		377

CARLF, TEXT /%#0?/

6250	4543
6251	0077
6252	0000
6253	4543
6254	4302

BKSPT, TEXT /%#BACKSPACE TESTX##0?/

6255 0103
 6256 1323
 6257 2001
 6260 0305
 6261 4024
 6262 0523
 6263 2445
 6264 4343
 6265 0077
 6266 0000
 6267 4543
 6270 4324
 6271 0102
 6272 4024
 6273 0523
 6274 2445
 6275 4343
 6276 0077
 6277 0000
 6300 4040
 6301 4040
 6302 4040
 6303 4040
 6304 5700
 6305 7700
 6306 4040
 6307 4040
 6310 4040
 6311 4057
 6312 0077
 6313 0000
 6314 5511
 6315 5500
 6316 7700
 6317 5555
 6320 5555
 6321 1155
 6322 1100
 6323 7700
 6324 3440
 6325 0077
 6326 0000
 6327 4543
 6330 4303
 6331 2240
 6332 2405
 6333 2324
 6334 4543
 6335 4300
 6336 7700
 6337 4543
 6340 4322
 6341 1107
 6342 1024
 6343 4015

TBTST, TEXT /X#TAB TESTX#0?/

TBMRK, TEXT / /0?/

TBMRK1, TEXT / /0?/

RM33B, TEXT /-1-0?/

RM37A, TEXT /-1-1-10?/

SPTSIC, TEXT / \ 0?/

CRIST, TEXT /X#CR TESTX#0?/

RMTST, TEXT /X#RIGHT MARGIN TESTX#0?/

6344 0122
 6345 0711
 6346 1640
 6347 2405
 6350 2324
 6351 4543
 6352 4300
 6353 7700
 6354 4543
 6355 4323
 6356 2001
 6357 0305
 6360 4024
 6361 0523
 6362 2445
 6363 4343
 6364 0077
 6365 0000
 6366 4543
 6367 4314
 6370 0640
 6371 2405
 6372 2324
 6373 4543
 6374 4300
 6375 7700
 6376 4543
 6377 4303
 6400 1001
 6401 2201
 6402 0324
 6403 0522
 6404 4024
 6405 0523
 6406 2423
 6407 4543
 6410 4300
 6411 7700
 6412 4543
 6413 4327
 6414 1722
 6415 2324
 6416 4003
 6417 0123
 6420 0540
 6421 2001
 6422 2424
 6423 0522
 6424 1640
 6425 2405
 6426 2324
 6427 4543
 6430 4300
 6431 7700
 6432 4543

SPTST, TEXT /X##SPACE TESTX##0?/

LFTST, TEXT /X##LF TESTX##0?/

CHRTST, TEXT /X##CHARACTER TESTX##0?/

WCPTST, TEXT /X##WORST CASE PATTERN TESTX##0?/

KMSGI, TEXT /X##KYBD TESTX##0?/

6433 4313
 6434 3102
 6435 0440
 6436 2405
 6437 2324
 6440 4543
 6441 0077
 6442 0000
 6443 4543
 6444 2022
 6445 0523
 6446 2340
 6447 0140
 6450 1305
 6451 3145
 6452 4300
 6453 7700
 6454 4543
 6455 0503
 6456 1017
 6457 4024
 6460 0523
 6461 2400
 6462 4543
 6463 0310
 6464 0122
 6465 0103
 6466 2405
 6467 2240
 6470 1305
 6471 3105
 6472 0440
 6473 2711
 6474 1414
 6475 4002
 6476 0540
 6477 2431
 6500 2005
 6501 0456
 6502 0000
 6503 4543
 6504 2225
 6505 0217
 6506 2524
 6507 4005
 6510 1604
 6511 2340
 6512 2217
 6513 2524
 6514 1116
 6515 0556
 6516 4543
 6517 4300
 6520 7700
 6521 4543

KMSG2, TEXT /X#PRESS A KEY#07/

KMSG3, TEXT /X#ECHO TEST/

KMSG3A, TEXT /X#CHARACTER KEYED WILL BE TYPED./

TEXT /X#RUBOUT ENDS ROUTINE;X##07/

KMSG4, TEXT /X#OCTAL EQUIVALENT TEST07/

6522 4317
 6523 0324
 6524 0114
 6525 4005
 6526 2125
 6527 1126
 6528 0114
 6531 0516
 6532 2440
 6533 2405
 6534 2324
 6535 0077
 6536 0000
 6537 4543
 6540 0000
 6541 4040
 6542 4040
 6543 4543
 6544 0077
 6545 0000
 6546 4543
 6547 2022
 6550 1116
 6551 2405
 6552 2240
 6553 0530
 6554 0522
 6555 0311
 6556 2305
 6557 2245
 6560 4300
 6561 7700
 6562 4543
 6563 2431
 6564 2005
 6565 4011
 6566 1640
 6567 0401
 6570 2401
 6571 4072
 6572 0077
 6573 0000
 6574 2540
 6575 0077
 6576 0000
 6577 0000

KMSG, TEXT /X#/
 OCTEQV, TEXT / X#0?/

PL1MG1, TEXT /X#PRINTER EXERCISERX#0?/

PL1MG2, TEXT /X#TYPE IN DATA 10?/

BKSU, TEXT /U 0?/

END, 0 /BEG OF 1000 WORD BUFFER

\$

0114 7007
 0115 6050
 0116 0770
 0117 6035
 0120 0400
 0121 7767
 0122 7730

0123 0004
0124 7727
0125 0005
0126 6317
0127 7761
0130 6314
0131 7762
0132 0257
0133 0334
0134 0252
0135 7650
0136 7670
0137 7653
0140 0011
0141 7772
0142 0240
0143 0100
0144 7401
0145 0377
0146 0077
0147 7760
0150 0037
0151 7766
0152 0017
0153 0360
0154 0352
0155 0366
0156 0551
0157 0765
0160 1165
0161 0761
0162 0755
0163 0751
0164 1171
0165 0745
0166 0740
0167 0734
0170 0730
0171 0724
0172 0717
0173 0626
0174 0600
0175 0562
0176 0337
0177 0326

[illegible]

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A	6107	CLA	7200	HLTD	V141	MTON	6117
A33WP6	6065	CNTST	1600	HLTST	2076	MTRS	6127
A35WP6	6073	CNV	1671	HOLD1	5713	NTST	1546
A37WP6	6101	CR	0107	IBIN	3346	NTSTA	1555
AC	0077	CRALF	0562	IN0	0436	NXTST	0056
ACL	7701	CRCR	0561	INCRIN	1433	OCIEGV	6541
ASCCN	1650	CRLF	4575	INKSF	0261	ONE	6142
ASCT	1706	CRST	6327	INPATT	1264	OPEN	0000
BAUCON	1537	CRSTTA	4456	INTAB	6035	OUT	1271
BAUDRT	0022	CRSTTB	4461	INTCF	1261	OUT0	2114
BAUTAB	0504	CTRA	0062	INTFND	2017	OUT1	2121
BDRET	0205	CTRB	0063	INTKSF	2021	OUT2	2123
BKSCTR	5262	CTSK	0550	INTSF	1257	OUTTAB	6050
BKSPC	5263	CURTST	0054	INTSVC	1254	P	6126
BKSPT	6253	D	6112	INTTST	2024	P0E0A	2237
BKSU	6574	DBLK	7577	IOF	0002	P0E0B	2244
BLK2	6724	DELAY	4576	ION	6001	P0E0C	2253
BLK88	6722	DELAYM	0024	J	6120	P0E0E	2262
BLKCC	7034	DELAYS	0102	K5200	2735	P0E1A	2304
BLKCNT	0101	DLGNT	0474	K5ELAG	0026	P0E1B	2315
BLKCK1	6001	DLGNT1	0027	KCC	6032	P0E2A	2415
BLKCK2	6713	DLMSR	1474	KCR	6030	P0E2B	2427
BLCKA	6577	DLMS	0227	KFLAG	1365	P0E2C	2433
BLCKB	6711	DLMSK	0111	KFL	6035	P0E2D	2456
BLCKC	7023	DVCOM	6022	KMSG1	6432	P0E2E	2465
BSSH	7002	DVSEL	6000	KMSG2	6443	P0E2F	2474
C241	6153	END	6577	KMSG3	6454	P0E2G	2527
C244	6156	ERRCNT	5676	KMSG3A	6462	P0E3A	2534
C247	6161	ERRR	0103	KMSG4	6521	P0E3B	2540
C252	6164	ERRCTR	5721	KMSG5	6537	P0E4A	2607
C255	6167	ERRR	1440	KRB	6036	P0E4B	2614
C252	6172	FADR	0623	KRS	6034	P0E5A	2634
C273	6175	FBA33	1102	KSTRT	6031	P0E5B	2636
C330	6200	FBALL	1000	LAS	0023	P0E6A	2670
C335	6203	FBF	0033	LF	7604	P0E6B	2675
C373	6241	FBF3	1031	LFST	0110	P0E6C	2701
C377	6249	FBEI	1055	LFSTA	6366	P0E6D	2712
CAF	6007	FETCH	2144	LINK	4612	P0E6E	2724
CAM	7021	FLAG	1046	M	0100	P0E6F	2731
CALP	6250	FORND	0716	M147	6123	P0E7A	2750
CHAIN	0025	FOUR	0302	M3	7631	P0E7B	2754
CHAINN	0263	FOUR	6145	MCTR	2111	P0E7C	2757
CHCK	0513	FH336	1117	MILCTR	0625	P0T6	2660
CHECK	0070	FH356	1135	MILCTR	0061	P0T6A	2663
CHCNT	0456	FH376	1153	MINT	0000	P0T6B	2671
CHRTST	6376	G	6115	MOVE	6115	P0T6C	2676
CK33	0352	GETPT	0067	MOVEA	4574	P0T6D	2702
CK35	0360	GETRDR	0237	MOVEA	0613	P0T6E	2713
CK37	0366	GTBIN	5737	MOVE	0600	P0T6F	2725
CKSR33	4554	CTF	0444	MGL	7421	P0T7	2736
CKSR37	4555	HLT	6004	MTABP	0057	P0T7A	2744

P0T7B	2751	P1T11	3400	PAL10	V141	7 APR 72	10155
P0T7C	2755	P1T11A	3405	P2E7B	4021	P4T45A	5102
P0TS0	2205	P1T12	3435	P2T10	4030	P4T46A	5116
P0TS0A	2211	P1T12A	3443	P2T10A	4034	P4TS0	4442
P0TS0B	2214	P1T12B	3456	P2T10B	4045	P4TS1	4475
P0TS0C	2222	P1TS0	3005	P2T10C	4051	P4TS10	4645
P0TS0D	2225	P1TS1	3010	P2T10D	4054	P4TS11	4651
P0TS0E	2230	P1TS1A	3014	P2T10E	4064	P4TS12	4655
P0TS1	2270	P1TS2	3034	P2T10F	4066	P4TS13	4661
P0TS1A	2275	P1TS2A	3040	P2T11	4153	P4TS14	4665
P0TS1B	2313	P1TS2B	3046	P2T11A	4157	P4TS15	4671
P0TS1C	2324	P1TS3	3066	P2T11B	4166	P4TS16	4675
P0TS2	2327	P1TS3A	3074	P2T11C	4200	P4TS17	4701
P0TS2A	2332	P1TS4	3126	P2T12	4233	P4TS2	4525
P0TS2B	2342	P1TS4A	3133	P2T12A	4247	P4TS20	4705
P0TS2C	2345	P1TS4B	3153	P2T12B	4255	P4TS21	4711
P0TS2D	2350	P1TS4C	3167	P2T12C	4260	P4TS22	4715
P0TS2E	2400	P1TS5	3200	P2TS0	3510	P4TS23	4721
P0TS2F	2402	P1TS5A	3204	P2TS0A	3513	P4TS24	4725
P0TS2G	2404	P1TS5B	3211	P2TS1	3530	P4TS25	4731
P0TS3	2504	P1TS5C	3213	P2TS1A	3533	P4TS26	4735
P0TS3A	2510	P1TS5D	3217	P2TS1B	3534	P4TS27	4741
P0TS3B	2515	P1TS5E	3222	P2TS2	3545	P4TS30	4745
P0TS3C	2521	P1TS6	3271	P2TS2A	3551	P4TS31	4751
P0TS4	2547	P1TS6A	3314	P2TS3	3600	P4TS32	4755
P0TS4A	2547	P1TS7	3314	P2TS3A	3606	P4TS33	4761
P0TS4B	2565	P1TS7A	3322	P2TS4	3630	P4TS34	5004
P0TS4C	2600	P20	3524	P2TS4A	3635	P4TS35	5004
P0TS5	2616	P21	3543	P2TS4B	3666	P4TS36	5010
P0TS5A	2624	P2E10A	4073	P2TS4C	3671	P4TS37	5014
P11MG1	6546	P2E10B	4107	P2TS5	3677	P4TS4	4625
P11MG2	6562	P2E10C	4115	P2TS5A	3711	P4TS40	5020
P1E11A	3424	P2E10D	4123	P2TS5B	3716	P4TS41	5024
P1E12A	3464	P2E10E	4135	P2TS5A	3724	P4TS42	5030
P1E12B	3474	P2E10F	4144	P2TS5B	3734	P4TS43	5034
P1E1A	3025	P2E11A	4206	P2TS7	3762	P4TS44	5034
P1E2A	3053	P2E11B	4215	P2TS7A	4000	P4TS45	5072
P1E2B	3062	P2E11C	4224	P2TS7B	4006	P4TS46	5106
P1E3A	3113	P2E12A	4276	P3E0	4337	P4TS47	5122
P1E3B	3122	P2E12B	4303	P3E1	4371	P4TS48	4631
P1E4A	3143	P2E12C	4305	P3E2	4427	P4TS49	5231
P1E5	3250	P2E2A	3562	P3T0B	4334	P4TS50	4635
P1E5A	3230	P2E2B	3564	P3T1B	4366	P4TS7	4641
P1E5B	3235	P2E3A	3621	P3T2C	4424	P6E0	5415
P1E5C	3242	P2E3B	3623	P3TS0	4314	P6T0	5400
P1E5D	3257	P2E4A	3644	P3TS0A	4324	P6T1	5421
P1E5E	3264	P2E4B	3657	P3TS1	4344	P6T1A	5426
P1E6A	3310	P2E5	3706	P3TS1A	4354	P6T2	5440
P1E710	3366	P2E6A	3746	P3TS2	4400	P6T2A	5447
P1T10	3325	P2E6B	3752	P3TS2A	4406	P70CTR	1211
P1T10A	3333	P2E6C	3756	P3TS2B	4411	P7T0	5502
		P2E7A	4015	P4T44A	5066	P7T1	5506

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P7T10	5542	PUNCH	2112	SHALT	V141	TEMR	0715
P7T11	5546	RADDR	1416	SINPT	0317	TLC37	0034
P7T12	5552	R8CTR	1417	SJ	1707	TLC371	2134
P7T13	5556	RBUSY	0076	SLDC02	6217	TLCALI	2126
P7T14	5562	RCTRA	1511	SM	0223	TLCALL	0033
P7T15	5566	RCTRB	1512	SNDREC	6222	TLS	6046
P7T16	5572	RDBLK	1400	SP	3353	TPC	6044
P7T17	5600	RDBLKR	1407	SPAC	6225	TRDATA	3337
P7T2	5512	RDRSRV	1420	SPCNT	5207	TSC1	0634
P7T20	5604	RDSRV	1430	SPCTR	5161	TSC2	0643
P7T21	5610	RGNA	0400	SPF	5162	TSF	6041
P7T22	5614	RGNB	0417	SPI	6045	TTYIOT	0021
P7T23	5620	RM33A	1563	SPIND	1716	TTYTYP	0020
P7T24	5624	RM33B	6314	SPT0	1714	TYPAT	0634
P7T25	5630	RM37A	6317	SPT1	1715	TYPEA	4573
P7T26	5634	RMB	4523	SPTST	6334	TYPLN	1637
P7T27	5640	RMTST	4537	SPTSTA	4535	TYPLN3	1627
P7T3	5516	RP1A	0415	SPTSTB	4546	TYPSP	1615
P7T30	5644	RP1B	0434	SPTSTC	6324	TYRSTG	0660
P7T4	5522	RP2A	0416	SRG	6003	UKCC	0626
P7T5	5526	RP2B	0435	SRSET	0236	UKCR	4571
P7T6	5532	RRDY	1343	SRTOA	5655	UKIE	4562
P7T7	5536	RRPP	0304	SRTOB	5662	UKRB	4567
PADDR	1342	RSCTR	1232	SS	6230	UKRS	4570
PBLK	1316	RSSERV	1233	ST33B	1020	UKSF	4572
PBLKR	1324	RSTUP	1331	STAL	0551	UNOVE	0075
PCTR	1341	RSYNC	1216	STALL	4556	UOUT	0072
PDCR	1310	RTF	6005	STBAUD	0200	UPUNCH	0074
PFLAG	0071	RTNNO	0055	STBF	1513	USPF	4560
PLTLR	1200	RUDONE	1456	STCTR	1000	USPI	4537
PRG0	2200	S	6131	STLID	0326	UTCF	4565
PRG1	3000	S100	0030	SV	0044	UTEMP	0104
PRG10	5651	S1001	2012	SY	6233	UTEMP1	0103
PRG11	5722	S200	0032	SYNC	6236	UTEMP2	0106
PRG11A	5725	S2001	2005	SYNKA	0065	UTLS	4563
PRG12	5764	S2001	0031	SYNKB	0534	UTPC	4564
PRG2	3503	S4000	2000	SYNKA	5200	UTPLN3	0073
PRG3	4307	SA	6206	TABCTR	5201	V	4566
PRG4	4434	SASC	1705	TABP	5220	VCTR	6134
PRG5	5274	SB	1435	TABPA	0624	WASC	1267
PRG5A	5317	SB0	4332	TADDR	5140	WCHK	1704
PRG6	5340	SB1	4364	TBCNT	6300	WCPTST	0527
PRG7	5465	SB2	4420	TBMRK	6306	WCPTST	6412
PRGADR	0235	SBSP	5673	TBMRK1	5134	WOSNS	2153
PRGEND	0300	SCNT	0473	TBTA	5146	WTS6A	0112
PRGNUM	0056	SD	6211	TBTB	6267	XKCC	0724
PRGTAB	0037	SETLOC	1740	TBTST	6042	XKCR	0755
PRINT	0671	SETRND	6150	TCF	1647	XKIE	0761
PSTUP	1212	SEVEN	6214	TEMP	0052	XKRB	0734
PSYNC	0442	SG	1717	TEMP1	0053	XKRS	0730
PT0	0443	SGET		TEMQ	0714	XKSF	0717

XSPF 1165
XSPI 0765
XTCF 0745
XTLS 0751
XTPC 1171
XTSF 0740
Y 6137

ERRORS DETECTED: 0

LINKS GENERATED: 106

RUN-TIME: 32 SECONDS

3K CORE USED

